GESTURE CONTROLLED BOT FOR MATERIAL HANDLING

A Project Report

Submitted by

ANANTHALAKSHMI S NAIR(TVE22IE012)

ARATHI M ANAND(TVE22IE016)

K SREEKESH(TVE22IE041)

SRUTHY M(TVE22IE064)

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DECLARATION

We undersigned hereby declare that the project report "Gesture Controlled Bot for Material Handling" , submitted for fulfillment of the requirements for the award of degree of Bachelor of Technology of the APJ Abdul Kalam Technological University, Kerala is a bonafide work done by me under supervision of Prof. Praveen A. This submission represents my ideas in my own words and where ideas or words of others have been included, I have adequately and accurately cited and referenced the original sources. I also declare that I have adhered to ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in my submission. I understand that any violation of the above will be a cause for disciplinary action by the institute and/or the University and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been obtained. This report has not been previously formed the basis for the award of any degree, diploma or similar title of any other University.

Thiruvananthapuram

ANANTHALAKSHMI S NAIR
ARATHI M ANAND
K SREEKESH
SRUTHY M

DEPARTMENT OF MECHANICAL ENGINEERING

COLLEGE OF ENGINEERING TRIVANDRUM

TRIVANDRUM

2025 - 2026

CERTIFICATE

This is to certify that the report entitled 'Gesture Controlled Bot for Material Handling' submitted by 'Ms Anantha Lakshmi S Nair, Ms Arathi M Anand, Mr K Sreekesh, Ms Sruthy M' to the APJ Abdul Kalam Technological University in partial fulfillment of the requirements for the award of the Degree of Bachelor of Technology in B.Tech in Industrial Engineering is a bonafide record of the project work carried out by them under my/our guidance and supervision.. This report in any form has not been submitted to any other University or Institute for any purpose.

Dr. Karthika A S
Professor & Head
Department of Mechanical Engineering
College of Engineering Trivandrum

Dr. Rajesh N R
Professor & UG Coordinator
Department of Mechanical Engineering
College of Engineering Trivandrum

Prof. V B Suresh Kumar
Associate Professor & Project Coordinator
Department of Mechanical Engineering
College of Engineering Trivandrum

Prof. Praveen A
Assistant Professor & Project Guide
Deptartment of Mechanical Engineering
College of Engineering Trivandrum

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ABBREVIATIONS

Abbreviation	Full Form
ESP	Espressif
MPU	Motion Processing Unit
IoT	Internet of Things
IIoT	Industrial Internet of Things
AGV	Automated Guided Vehicle
MNC	Multi National Corporations
MSME	Micro Small and Medium Enterprises
I2C	Inter-Integrated Circuit
IR/US	Infrared/Ultra Sonic
AS/RS	Automated Storage and Retrieval System
ANOVA	Analysis of Variance
IMU	Inertial Measurement Unit
BLE	Bluetooth Low Energy
RSSI	Received Signal Strength Indicator
CV	Computer Vision
CNN-BiLSTM	Convolutional Neural Network and Bidirectional Long Short-Term Memory

Abstract

Under the Mechatronics domain, this project aims to build a Gesture Controlled Bot which is potentially capable of automating warehouse navigation. Industry automation is a growing sector with many requisites from innovation and technology. With easily available and flexible components like ESP 32 Module, MPU6050 sensor, 2 Wheel Smart Car Robot Chassis Kit and L298 Dual Motor driver board, we aim to build a mini prototype. The principle of operation of this device is the use of a gyro sensor to detect the orientation of the hand, and the data is transmitted to an ESP32 Bluetooth transmitter. The Bluetooth receiver then receives the hand data which is converted to corresponding motor movements on the Micro Mouse side. A wearable hand gesture device that includes a gyro sensor and microcontroller are used to convert gyro values to corresponding motor movements and send the data to an RC car via Bluetooth. This project will enable the user to control a robot car wirelessly with the ESP32 module. The hand movements will be captured by a wearable hand gesture device and the data will be sent to the Micro Mouse wirelessly. Thus the bot will move according to the hand movements. This prototype is named Bridge Model Warehousing to signify the gap between fully manual warehouses and fully automated warehouses(MNC: Amazon, etc). This project can be implemented in the lowest category of warehouses so as to enable them to derive the benefits of higher productivity, efficiency, cost reduction, better workplace ergonomics and social benefits like higher employment in labour and technical sectors. This is a scalable project that is targeted at automating Indian warehouses. Among the 2 widely seen extremes of industry automation in India, this prototype will serve as a bridge model which can effectively automate the warehouse navigation while having a low to none adverse impact on the social dimension of employment.

KEYWORDS: Mechatronics, Gesture Controlled Bot, Industrial Automation, ESP 32 Module, Gyro sensor, Bluetooth, Bridge Model Warehousing, Indian Warehouses.

Chapter 1

INTRODUCTION

In the modern industrial landscape, the convergence of multiple disciplines-mechanical engineering, electronics, computer science and control system has given rise to Mechatronics, a field that has revolutionized automation and intelligent system design. Mechatronics represents the synergistic integration of mechanical systems, electronic control and software-based intelligence to create adaptive and efficient electromechanical systems. Its applications span from precision robotics and automotive control systems to manufacturing automation and smart devices. The rapid evolution of the Internet of Things (IoT) has further accelerated the pace of automation and digitalization across industries. IoT allows physical devices, sensors and actuators to communicate and exchange data through the internet, enabling real-time decision-making and seamless interaction between the physical and digital worlds. When applied to industrial domains, this connectivity evolves into the Industrial Internet of Things (IIoT), forming the backbone of smart manufacturing and intelligent production ecosystems. In recent years, the industrial paradigm has shifted from Industry 4.0-which focused on automation, data exchange and cyber-physical systems-to Industry 5.0, a more human-centric framework emphasizing collaboration between humans and intelligent machines. Industry 5.0 integrates advanced technologies such as artificial intelligence (AI), IoT, robotics, edge computing and data analytics with human creativity, emotional intelligence and adaptability. The goal is to develop production systems that are not only efficient and autonomous but also sustainable, flexible and responsive to human needs. In this context, the development of a gesture-controlled robot for material handling presents a practical embodiment of these principles. By merging mechatronic design, IoT-enabled communication, and embedded system integration, this project aims to demonstrate how human-robot collaboration can be achieved through intuitive and contactless gesture-based control mechanisms.

Material handling constitutes a crucial component of any industrial or warehousing environment, involving the movement, storage and control of goods and raw materials throughout the production process. Traditional material handling systems rely heavily on manual labour, which can be time-consuming, physically demanding and prone to inefficiencies. Repetitive tasks, physical fatigue and human error contribute to reduced productivity and workplace hazards. Automation has long been employed to address these challenges, but conventional automated systems often require significant infrastructure investment, complex programming and lack adaptability for small and medium-scale industries (MSMEs). These limitations create an opportunity for cost-effective, flexible and semi-autonomous solutions that combine automation with human oversight. The concept of a gesture-controlled robot aligns perfectly with this need. Instead of relying on complex control interfaces or pre-programmed instructions, a robot that responds to human hand gestures enables natural and intuitive interaction. Such systems not only enhance operational efficiency but also improve worker safety by reducing direct human involvement in hazardous environments. By integrating IoT-based monitoring and IIoT connectivity, the robot's performance can be tracked, optimized and scaled to fit different industrial scenarios. This project thus bridges the gap between low-cost automation and intelligent industrial systems, promoting a new class of human–machine collaboration for material handling operations.

The development of the gesture-controlled robot is grounded in the principles of Mechatronics, which combine mechanical design, electronic control systems and embedded programming into a cohesive unit. The mechanical subsystem provides the structural foundation and mobility of the robot, enabling precise movement and load handling. The electronic subsystem, comprising sensors, actuators, motor drivers, and microcontrollers, provides the necessary control and feedback mechanisms. The software subsystem integrates these components through embedded control algorithms, enabling real-time gesture recognition, decision-making and motion control. The ESP32 microcontroller chosen for this project serves as the brain of the system. It integrates Wi-Fi and Bluetooth connectivity, facilitating communication between the gesture controller and the mobile platform. The gyroscopic sensor captures orientation data, allowing the system to interpret specific hand movements and translate them into directional commands. The L298 motor driver interfaces the control signals with the motors, enabling smooth and coordinated movement. This tight integration of mechanical, electrical, and computational components exemplifies the mechatronic approach—where the focus is not merely on individual subsystems but on their seamless interaction to achieve intelligent and adaptive behaviour.

The Internet of Things (IoT) forms the communication backbone of modern smart systems. By embedding sensors, actuators, and communication modules into physical devices, IoT allows real-time data acquisition, analysis, and control. In the context of the gesture-controlled robot, IoT connectivity enables: Wireless communication between the user's gesture interface and the robot through Bluetooth Low Energy (BLE) or Wi-Fi, Data logging and monitoring, allowing parameters such as position, movement speed, and battery status to be tracked remotely. Integration with cloud platforms, enabling predictive maintenance and performance analytics. Scalability and interoperability, where multiple robots or devices can be networked within a larger industrial framework. Using BLE technology through the ESP32 module ensures a low-power and long-range communication protocol, making the system energy-efficient while maintaining reliable performance. This approach not only enhances operational flexibility but also positions the robot within the broader IoT ecosystem, paving the way for future smart factory integration.

The Industrial Internet of Things (IIoT) extends IoT concepts to manufacturing and industrial environments, focusing on improving productivity, safety and system intelligence. IIoT integrates sensors, data analytics and machine-to-machine communication to create self-monitoring and self-optimizing industrial systems. In this project, the robot represents a node in an IIoT-enabled manufacturing environment. Its embedded sensors and communication modules collect and transmit data about operation status, task progress and environmental conditions. This information can be visualized and analysed to improve decision-making, reduce downtime and predict maintenance needs. By incorporating IIoT functionalities, the robot transitions from a stand-alone automation unit to an intelligent, connected system capable of real-time collaboration with other devices or systems within a factory. This level of connectivity enhances flexibility, scalability and responsiveness-key attributes of next-generation industrial automation. Industry 5.0 represents the next evolutionary stage in industrial development, emphasizing human-centered and collaborative automation. Unlike Industry 4.0, which focused primarily on machine efficiency and data integration, Industry 5.0 restores the human to the center of production systems. It promotes synergy between human creativity and robotic precision, aiming for sustainability, personalization and resilience. The gesture-controlled robot embodies the principles of Industry 5.0 by enabling intuitive human–robot interaction (HRI) through gesture recognition. Instead of programming complex control sequences, operators can direct the robot's movements through simple, natural gestures. This form of interaction not only simplifies operation but also makes automation more inclusive, allowing workers without specialized training to collaborate with robotic systems effectively. Furthermore, the robot's semi-autonomous nature aligns with Industry 5.0's focus on collaborative intelligence, where the robot performs repetitive or hazardous tasks while humans retain control and decision-making authority. The result is a system that is not only efficient and productive but also safe, ergonomic and adaptable to human preferences.

The success of this project lies in the seamless integration of embedded systems with mechatronic hardware. Embedded systems act as the control hub, processing sensor data, executing control algorithms and managing communication protocols in real time. The ESP32 microcontroller serves as the central processing unit, handling tasks such as gesture input interpretation, motor control and wireless data exchange. Gyroscopic sensors and accelerometers form the primary input interface, capturing hand movement data and translating it into digital control signals. The L298 motor driver module provides power and direction control for the robot's DC motors, ensuring precise movement and navigation.This embedded system is further enhanced with power management features and a backup power source, ensuring reliability in continuous operation. The implementation of Bluetooth-based tracking and RSSI (Received Signal Strength Indicator) measurements enables location estimation and real-time tracking, making the robot capable of intelligent navigation in confined spaces such as warehouses or production floors. By merging these elements, the system achieves a semi-autonomous, low-cost and scalable solution that can be adapted for a variety of industrial applications. The development of the gesture-controlled robot demonstrates the integration of multiple emerging technologies: Mechatronics-for the design and coordination of mechanical and electronic subsystems. Embedded Systems-for real-time control and data processing. IoT and IIoT-for connectivity, monitoring and system scalability. Human-Robot Interaction (HRI)-for intuitive, gesture-based control aligned with Industry 5.0 goals. Cyber-Physical Systems (CPS)-enabling synchronization between digital commands and physical responses. This multidisciplinary integration highlights the essence of modern engineering-creating systems that are intelligent, adaptive and capable of bridging the gap between human intention and machine execution.

The proposed gesture-controlled robot offers a range of advantages: Enhanced safety by minimizing direct human involvement in material transport. Improved efficiency and precision through sensor-based control. Low-cost implementation, making it feasible for MSMEs and small-scale industries. Scalable architecture, allowing future integration with IIoT networks. Human-centric operation, consistent with Industry 5.0 principles. Potential applications include warehouse automation, assembly line support, laboratory logistics and transport in hazardous or confined environments. With further development, the robot can be integrated into larger smart manufacturing ecosystems, contributing to the evolution of connected, intelligent and sustainable factories. The gesture-controlled robot for material handling embodies the evolution of automation from mechanization to intelligent collaboration. By integrating Mechatronics, IoT, IIoT and embedded systems, the project demonstrates how advanced technology can be used to design human-friendly and cost-effective automation solutions. As industries transition toward Industry 5.0, such systems will play a vital role in redefining the relationship between humans and machines-shifting from replacement to partnership. This project stands as a small yet significant step toward that vision: creating smart, connected and collaborative robots that enhance productivity while maintaining the human touch in automation.

This introduction is followed by a comprehensive literature review, problem definition from obtained research gap and the methodology of implementation. An effort to simulate the implementation in Simio was also undertaken during this phase, details of which are also included in this report. The developments in Design phase, addition of which complete the first phase works of this project.

Chapter 2

LITERATURE REVIEW

The paper (Zafari, S.,et al, 2024)[1] presents a gesture-based interactive system designed to facilitate collaboration between a human operator and a semi-autonomous forklift in material handling processes. It addresses the intersection of Human-Agent Interaction, Human-Robot Interaction, and Automation in Industrial Settings with particular focus on gesture-based control for automated material handling vehicles. The core concepts of gesture based implementations are heavily discussed in this study including its benefits and usability challenges with a focus on Industrial Human robot interaction. Gesture control offers a more intuitive, natural, and non-contact form of interaction, which is advantageous in industrial and outdoor settings when traditional controls are hard to reach. Technical issues encountered in this implementation were latency, stability and precision based factors. Human-centered factors also posed challenges in the form of intuitiveness, ergonomics, gesture recall, and reliability of the sensor technology faults. This study inculcates the application of complex commands like starting, defining pallet amounts, setting loading/unloading areas, etc for a semi autonomous forklift in material handling domain. The methodology of this study involves a comparison analysis as well. The gesture based system is compared against a touch based User Interface to explore qualitative and quantitative factors in real world settings. The design principles were focused on demands of industrial material handling context and Human-Centered Automation. The challenges of an outdoor setting were also taken into consideration by testing in an outdoor environment. The system has been designed for robust gesture recognition so as to prevent unintended movements and enhance overall safety by allowing the operator to maintain a safe distance from the semi-autonomous forklift. The aim of the interface is to optimize efficiency by providing an alternative to tedious physical controls in the work area. The gestures naturally integrate into the operator's existing material handling work process. The gesture vocabulary is designed to be intuitive and easily recallable for ease of use. Gestures are easily performable and sustainable over a period of time and they prevent operator fatigue as well. Continuous, informative feedback on the system state, gesture recognition status, and the vehicle's actions build operator trust and ensure accurate command execution. The gesture set is designed to support the core, high-level commands required for the material handling process, including: Starting the operation, Defining the amount of pallets to be transported, Defining the loading and unloading destinations, Initiating the transport action. The system employs an algorithm to localise the operator's body and classify the performed static pose or dynamic movement against the predefined gesture vocabulary. The output of this perception system is a confirmed, discrete control command. The recognized gesture command is sent to a central control unit which translates the high-level command into a sequence of low-level instructions for the semi-autonomous forklift. The vehicle executes the complex, multi-step command autonomously after receiving the high-level instruction from the operator's gesture. The system provides feedback to the operator to indicate state of system and apt recognition of gesture. The overall architecture of this system is an Industry 5.0 compatible format wherein the zero human intervention concept is undone by the central control of the system relying in human hands

The increasing complexity of supply chains, driven by just-in-time production, mass customization, and e-commerce, has necessitated the integration of automation in warehouses. Over the past three decades, research has focused extensively on developing flexible, automated systems that improve warehouse efficiency, adaptability, and sustainability. Automated Guided Vehicles (AGVs) have emerged as a key technology for material handling in smart warehouses. Early studies emphasized traditional navigation methods—such as wire, magnetic, and laser guidance—while recent works have explored advanced methods like computer vision, LiDAR, and SLAM (Simultaneous Localization and Mapping) for autonomous navigation. Modern AGVs are now part of intelligent and decentralized systems that enable real-time communication with other machines and management systems, aligning with Industry 4.0 principles. The literature highlights various warehouse automation technologies, including Automated Storage and Retrieval Systems (AS/RS), Robotic Mobile Fulfillment Systems (RMFS), and order picking strategies optimized through algorithms and artificial intelligence. Studies show that integrating data collection technologies (like RFID), robotic systems, and warehouse management systems (WMS) enhances efficiency, space utilization, and flexibility. A significant focus has been on optimization techniques—including path planning, time and cost optimization, and the use of AI-based decision systems. Researchers have adopted genetic algorithms, fuzzy logic, reinforcement learning, and deep learning to improve AGV navigation and warehouse operations. Recent advancements in hardware and software—such as AI-enabled processors, Li-ion batteries, sensor fusion, and machine learning—have accelerated the evolution of AGVs into Autonomous Intelligent Vehicles (AIVs). These AIVs support adaptive material flow and collaborative operations with humans, ensuring safe and flexible production environments. Furthermore, studies emphasize the importance of collaborative robots (cobots) for tasks like palletizing, picking, and packaging, especially in e-commerce and high-mix warehouses. Human–robot cooperation, flexible grippers, and predictive algorithms for human intention recognition are emerging areas of interest. Overall, the literature establishes that a flexible automated warehouse relies on the seamless integration of automated machinery, intelligent control systems, and real-time data analytics. The reviewed studies collectively guide the development of a conceptual framework for Industry 4.0–driven smart warehouses, where AGVs, AIVs, and collaborative robots operate autonomously to optimize logistics, productivity, and safety.

Automation in material handling has been an area of rapid technological advancement, aiming to enhance efficiency, safety, and cost-effectiveness in industrial environments. Traditional scissor lifts, commonly used in maintenance, construction, and warehousing, are robust but heavily dependent on manual operation. To overcome limitations such as high cost, labor dependency, and safety risks, researchers have explored semi-autonomous and fully autonomous lifting systems that integrate mechatronic principles and smart control technologies.Several studies have focused on improving scissor lift mechanisms through optimized structural design and material selection using finite element analysis (Dengiz et al., 2018). These improvements ensure load stability and minimize deformation under stress. The integration of sensors—including infrared (IR), ultrasonic, and gesture sensors—has become a significant focus in developing intelligent lift systems capable of object detection, navigation, and human interaction. These sensors provide safety, precision, and user-friendly control compared to purely mechanical systems.The use of open-loop control systems (Teja, 2023) simplifies operation by allowing discrete input commands for lifting and navigation. Studies on line-following and obstacle-avoidance robots (Bendimrad et al., 2020) support the use of IR and ultrasonic sensors for path tracking and collision prevention, forming the foundation for semi-autonomous navigation in constrained environments like warehouses.Compliance with ANSI/ITSDF B56.5 safety standards ensures that semi-autonomous industrial vehicles meet industry-grade safety and operational criteria. The paper's reference to ANSI and OSHA collaboration highlights the importance of standardized testing in ensuring the reliability and safety of robotic lift systems.In terms of cost analysis, previous works have shown that fully autonomous cargo robots, such as Piaggio's Gita series, are effective but expensive. The reviewed design merges affordability with automation by using low-cost components like plywood, steel, and Arduino-based control systems. Such approaches align with current trends toward cost-effective, small-scale industrial automation.Overall, existing literature demonstrates a clear progression toward intelligent, low-cost, semi-autonomous systems that enhance material handling. The presented work builds upon these foundations by combining mechanical strength, sensor-based control, and adherence to safety standards to produce an efficient and affordable robotic lift suitable for industrial applications.

Gesture-based robotic systems have become an important research area in human–machine interaction, offering intuitive and contactless control for automation and assistive technologies. Traditional robot control methods required manual programming or remote operation, which were less efficient and more complex for untrained users. The literature highlights a steady transition from mechanical to sensor-based robotic systems that can recognize and interpret human gestures for real-time motion control.Early research introduced accelerometer and gyroscope-based sensors to detect hand orientation and movement, converting these signals into control commands for robotic systems. Such sensors provided a simpler, low-cost alternative to camera-based gesture recognition systems. Studies have also explored electromyography (EMG) and sound-activated controls as alternative human–machine interfaces, but accelerometer-based systems remain preferred for their accuracy, affordability, and reduced computational load.Researchers have emphasized that effective gesture-controlled robots must incorporate modularity, flexibility, and fault tolerance while maintaining simplicity for practical use. The reviewed literature supports using Arduino-based microcontrollers and L293D motor drivers to control DC motors for movement in multiple directions. The use of the ADXL335 accelerometer sensor allows recognition of hand tilt and direction, enabling the robot to move forward, backward, left, or right depending on hand gestures.Gesture-controlled robots are widely applied in industrial automation, medical assistance, military robotics, and mobility aids for the physically challenged. Studies also note that such systems can improve worker safety and reduce human effort in repetitive or hazardous environments. Despite these advantages, challenges remain in ensuring stable wireless communication, reliable power supply, and accurate gesture recognition under varying conditions.Overall, existing research demonstrates that gesture-controlled robotic systems offer an efficient and user-friendly interface between humans and machines. The reviewed work contributes to this field by designing a cost-effective, Arduino-based robot that uses accelerometer sensors for intuitive gesture-based motion control. The findings highlight the feasibility of low-cost automation solutions that can be adapted for diverse real-world applications.

Manghisi, et al [5] presents a careful, application-driven review that frames gesture-based sensing as an enabling technology for human-centred smart factories — i.e., factories that augment rather than replace human workers. The paper's value lies less in novel algorithms and more in synthesizing interdisciplinary threads: sensing modalities (vision, depth sensors, IMUs/ wearables, EMG), data processing approaches (temporal models, sensor fusion), and the human factors and ergonomics necessary when deploying gesture systems on the shop floor. The authors explicitly position gestures as a bridge between the increasing automation of Industry 4.0 and the need to preserve operator safety, comfort and situational awareness; this human-first argument shapes how technical tradeoffs are discussed throughout the review. A central contribution is the structured mapping the authors create between sensing technologies and industrial use cases. For example, they note that vision and depth cameras are attractive for hands-free control and high-fidelity pose estimation but are vulnerable to occlusion and raise privacy concerns on the shop floor; IMUs and wearable sensors provide continuous motion streams and resilience to occlusion but need calibration and can suffer drift; EMG and force sensors can pick up intent earlier (muscle activation) but are intrusive and harder to normalize across workers. By pairing these technological constraints with desired industrial attributes (real-time responsiveness, low false positives, minimal worker training, robustness to dirt/lighting), the review gives practitioners a decision rubric for selecting or hybridizing sensors rather than only reporting raw recognition metrics. The review also pushes evaluation beyond classical machine-learning metrics. Manghisi et al. argue for evaluation protocols that include ergonomic outcomes (e.g., reductions in risky postures or cumulative muscle load), cognitive load measures, and integration-level tests (how a gesture control mode affects normal task flow and safety practices). This broadened evaluation lens is important because a high classification accuracy for a predefined gesture vocabulary may still disrupt work if gestures increase cognitive burden or encourage awkward postures; therefore the authors call for standardized industrial datasets and benchmarks that incorporate human factors. This emphasis on holistic evaluation is a distinguishing strength of the paper. On the algorithmic side, the review summarizes progress in temporal models and deep approaches for gesture recognition — recurrent networks, temporal convolutional networks and early transformer adaptations — but it stresses that “state-of-the-art” models must be balanced with constraints on compute, latency and interpretability in industrial deployments. The authors highlight sensor fusion as the research sweet spot: combining vision for contextual scene understanding with IMU/wearable signals for robust kinematics yields complementary strengths but raises systems engineering challenges (time synchronization, bandwidth, edge vs cloud processing). They also point to data scarcity for industrial gestures and the need for transfer learning and synthetic data augmentation tailored to factory contexts. Privacy, safety certification and deployment economics are recurring systems-level themes. Manghisi et al. underscore legal and organizational barriers (e.g., camera surveillance concerns, maintenance cycles for wearables, and the cost/benefit calculus for retrofitting legacy lines) and recommend design patterns: prioritize on-device/edge processing to reduce data leakage, anonymize video streams, and use minimalistic gesture sets that are ergonomically comfortable. The paper's concluding research agenda asks for multi-disciplinary datasets, protocols that evaluate ergonomic impact, and design guidelines to choose sensing modalities depending on task criticality and environmental constraints — a pragmatic roadmap that aligns research with industrial adoption needs. In short, this review is less about a single technological innovation and more about reframing gesture technologies around human outcomes in Industry 4.0: it is a practical, well-referenced guide for researchers and practitioners who must trade off accuracy, robustness, ergonomics and privacy when building gesture-enabled smart factories.

Gourob, et al [6] presents a compact, demonstrative system that maps visually detected hand gestures from a single camera to commands that control a robotic hand. The paper reads like an engineering proof-of-concept: it prioritizes accessibility and simplicity (monocular webcam, contour and fingertip heuristics, rule-based gesture classification) rather than cutting-edge deep learning. This focus makes the study useful as a reproducible baseline: the authors demonstrate how a low-cost vision pipeline can enable teleoperation tasks and educational prototypes without specialized hardware. Methodologically, the pipeline follows a classic CV flow: image capture → background removal/segmentation → contour and convex hull analysis → fingertip/defect detection → discrete gesture classification →

mapping to robot actuators. The authors implement simple rules to distinguish a small gesture vocabulary (open/close/grab/point), translate those into actuator commands for a robotic hand assembly, and measure end-to-end latency and recognition accuracy in controlled lighting. The paper reports that the system is sufficiently responsive for basic teleoperation and educational demonstrations, and it documents typical failure modes (illumination changes, complex backgrounds, occlusion and hand orientation variation). The experimental evaluation, while limited in scale, is informative about practical constraints. Tests include a handful of users and primarily lab conditions; recognition accuracy drops significantly under dynamic backgrounds and irregular illumination, and the monocular setup struggles when the hand rotates or when multiple hands enter the field. The authors candidly acknowledge these limitations and position their contribution as a low-barrier entry point: useful for labs, student projects and initial feasibility studies, but not yet robust for industrial deployment. This transparency is helpful for readers who want a clear baseline to improve upon. From a technical critique standpoint, the paper exposes the limits of classical heuristic pipelines. Subsequent literature and conference follow-ups suggest three clear upgrade paths: (1) incorporate depth sensing or multi-camera setups to handle occlusion and scale, (2) adopt light-weight CNNs or vision transformers for more robust feature extraction under variable conditions, and (3) integrate IMU or wearable data to provide redundancy and orientation awareness. Gourob et al. themselves point toward these hybridizations as future work. The paper therefore functions well as a “simple baseline + failure analysis” — an important role in engineering research because it clarifies what problems remain to be solved by more complex methods. Beyond the algorithmic aspects, the study is useful for its practical considerations: it documents latency budgets acceptable for the tested actuation tasks, explores tradeoffs between gesture vocabulary size and misclassification risk, and briefly discusses user training requirements. These pragmatic details are valuable for implementers building working prototypes in constrained environments (education, assistive tech, hobbyist robotics). The paper’s limited but explicit discussion of usability (simple gesture sets, minimal calibration) is aligned with the recurring theme in HRI: simpler, well-chosen gesture vocabularies often outperform larger, brittle ones in real tasks. In conclusion, Gourob et al. (2021) do not claim theoretical novelty; rather they provide a clear, reproducible, low-cost demonstration of vision-based gesture control for a robotic hand, coupled with a candid analysis of failure modes and directions for improving robustness — making the paper a practical baseline that subsequent research can readily extend.

Roda-Sanche, et al [7] presents an IoT-centric architecture for real-time gesture control of robotic arms using wearable IMU devices as the primary sensing modality. Unlike vision-centric works, the paper foregrounds the industrial requirements of latency, robustness to occlusion and environmental variability — constraints that naturally lead to choosing inertial/wearable sensors and an IoT middleware to distribute low-latency motion data across edge processing nodes and actuators. The paper therefore occupies a different, complementary design space to vision-first proposals and explicitly targets collaborative industrial tasks. Architecturally, the authors describe layered components: sensor nodes (IMUs attached to the operator), an IoT gateway for data aggregation and lightweight preprocessing, a pose/gesture interpretation module (running on an edge server), and a robot controller that translates interpreted commands to manipulator trajectories. Emphasis is placed on time synchronization, drift compensation in IMUs (complementary filters and periodic recalibration), and network reliability considerations (QoS and message prioritization for safety-critical commands). The modular architecture illustrates how existing IoT stacks can be repurposed for HRI while preserving industrial requirements for determinism and safety. The experimental work reported is practical and multi-dimensional: the authors evaluate latency (end-to-end), kinematic accuracy (IMU-derived pose vs ground truth), and human usability metrics (task completion time and perceived comfort). Results show the IMU/IoT approach achieves low latency suitable for direct teleoperation and remains resilient when visual line-of-sight is blocked — a decisive advantage on cluttered shop floors where cameras frequently suffer occlusion. However, IMU drift and the need for calibration across different operators are noted drawbacks; the authors propose sensor fusion with vision as a robust hybrid strategy for future systems. From a human-centred perspective, the paper gives attention to operator comfort and ergonomics: lightweight wearables reduce interference with tasks, and the gesture vocabulary is designed to avoid awkward or sustained postures. Importantly, the authors test operators performing collaborative tasks with a cobot and measure subjective workload and acceptance metrics — aligning with the broader Industry 4.0 goals of safe human–robot teaming rather than full automation. This experimental emphasis on human outcomes strengthens the practical impact of the work. Technically, the paper highlights three avenues for improvement that many later researchers pick up: (1) hybrid sensor fusion (IMU + vision) to combine continuous motion streams with contextual scene understanding; (2) adaptive calibration and personalization to handle different body sizes and motion styles; and (3) edge intelligence for lightweight ML models that can run near the sensor to minimize network round trips. The paper's combination of architecture, empirical results, and human usability tests creates a convincing case for IoT-driven wearable strategies in many industrial HRI scenarios. In summary, Roda-Sanchez et al. provide a mature, engineering-oriented demonstration that IoT and wearable IMUs can meet industrial latency and robustness needs for gesture-based robot control, while also foregrounding operator comfort and system modularity — an influential contribution that argues for hybridized, human-centred solutions in Industry 4.0.

A novel web-based hand and thumb gesture recognition model that revolutionizes human-robot interaction by eliminating the need for wearable sensors or specialized hardware. The system utilizes a custom-trained YOLOv5s model integrated with Raspberry Pi and Arduino hardware to create an omnidirectional autonomous vehicle controlled through natural hand gestures. The innovative approach employs a Finite State Machine (FSM) framework that maps five distinct gesture commands – forward, backward, left, right, and stop – to vehicle control signals with remarkable precision.The technical architecture demonstrates exceptional integration capabilities through a browser-based interface built using TensorFlow.js, Node.js, and WebSocket technologies, enabling real-time video streaming and device-agnostic control without external processing requirements. The system achieves impressive performance metrics with 94.2 percent. classification accuracy, stable frame rates of 12-18 FPS, and operational latency between 150-300 ms. Memory utilization remains optimized at approximately 280 MB during active inference and 160 MB during idle states, making it highly efficient for deployment on resource-constrained platforms. The research addresses critical limitations of existing gesture-controlled systems by providing a cost-effective solution that operates through standard web browsers without requiring specialized hardware installations.The system's scalability and accessibility make it particularly suitable for industrial material handling applications,healthcare logistics, warehouse automation, and assistive technologies. The comprehensive validation includes statistical analysis through confusion matrices and ANOVA testing, confirming the system's reliability across multiple gesture classes and environmental conditions.The practical implications of this work extend beyond laboratory demonstrations to real-world applications where contactless operation is essential. The elimination of wearable technology removes barriers to adoption and reduces maintenance requirements, while the web-based interface ensures compatibility across diverse computing platforms.Future developments proposed by the authors include integration of 3D vision capabilities and obstacle avoidance functionalities to enhance autonomous navigation in complex environments

The research also addresses the critical labor shortage in agriculture through the development of a sophisticated gesture controlled robotic arm system that combines data gloves with bending sensors and OptiTrack motion tracking systems.The work emerges from the urgent need to mechanize fruit harvesting, where labor requirements have decreased minimally compared to grain production mechanization. The system represents a hybrid approach that bridges the gap between fully automated systems and manual operations by providing intuitive gesture-based control.The technical implementation integrates ten flex sensors within a custom data glove connected via ESP32-S3 microcontroller with WiFi connectivity, while the OptiTrack system provides high-frequency spatial tracking at 120 Hz. The core innovation lies in the CNN-BiLSTM hybrid neural network architecture with 365,511 optimized parameters that processes both spatial and temporal gesture dynamics. This dual-sensor approach addresses fundamental limitations of single-modality systems by combining the fine motor control capabilities of bending sensors with the spatial precision of optical tracking.Experimental validation demonstrates exceptional precision with 96.43 percent gesture recognition accuracy, 0.0131 m meanEuclidean distance, and 0.0095 m Root Mean Square Error (RMSE) in trajectory replication. The system successfully replicates complex hand movements including individual finger flexion and coordinated grasping motions, enabling the myCobot robotic arm to perform delicate manipulation tasks with human-like dexterity. The research methodology includes comprehensive testing across various environmental conditions and user populations to ensure robustness and reliability.The practical applications extend beyond agricultural harvesting to industrial pick-and-place operations, precisionassembly tasks, and any application requiring fine motor control and spatial coordination. The system's modular design allows for adaptation to different robotic platforms and task requirements. The authors identify future research directions including multi-sensor fusion with IMU and camera data to enhance robustness under challenging environmental conditions such as occlusion and varying lighting conditions.The economic implications are significant, as the system offers potential solutions for labor-intensive agricultural operations while maintaining the adaptability and speed advantages of semi-automated systems. The combination of precision control and intuitive operation reduces the learning curve for operators while ensuring consistent performance across different users and environmental conditions.

The comprehensive research addresses the critical need for intuitive human-robot collaboration interfaces in manufacturing environments that require frequent robot path revisions. The study presents a distributed processing architecture utilizing MediaPipe hand landmark detection and Intel RealSense RGB-D cameras to create a naturalgesture-based interface for robot trajectory definition. The system differentiates itself from existing approaches by enabling direct waypoint definition through simple hand gestures, significantly reducing programming complexity and operator training requirements.The technical architecture employs multiple Jetson Nano units for distributed processing, with each unit handling data from a single camera to ensure real-time performance and system scalability. The MediaPipe solution localizes hand landmarks in RGB images, enabling precise gesture recognition and 3D spatial mapping for trajectory definition. The system supports comprehensive robot control functions including waypoint creation, confirmation, deletion, orientation editing, and path traversal validation. Experimental validation involved

20 volunteers in realistic industrial workspace conditions closely resembling intended applications. The comparative study demonstrated that gesture-based path definition was four times faster than conventional teach pendant methods, with NASA-TLX stress index scores below 50 percent and System Usability Scale (SUS) scores of 81.0, indicating superior usability and reduced cognitive load. The experiments collected both objective performance metrics and subjective user experience data to provide comprehensive evaluation of system effectiveness.The research methodology includes rigorous safety protocols and comprehensive user training to ensure reliable data collection under controlled conditions. The distributed architecture ensures fault tolerance and enables seamless scaling to larger workspace areas through additional camera-processing units. The system's ability to operate in collaborative environments where humans and robots share workspace makes it particularly valuable for flexible manufacturing applications.The practical implications extend to various manufacturing scenarios including assembly line reconfiguration, materialrepositioning tasks, and quality control applications where rapid path adjustment is essential. The system reduces downtime associated with robot reprogramming and enables non-expert operators to modify robot behavior without specialized training. Future research directions include integration with augmented reality systems for enhanced visualization and expansion to more complex gesture vocabularies for advanced robot control functions.

[11] This research presents the GestureMoRo algorithm, a sophisticated approach to mobile robot teleoperation that leverages LeapMotion sensor technology to create intuitive gesture-based control systems. The work addresses fundamental challenges in human-robot interaction by providing natural, equipment-free control methods that are accessible to users regardless of technical expertise. The system eliminates the burden of wearable devices while maintaining high precision and responsiveness in robot control applications. The technical implementation centers on palm-based gesture interpretation where pitch and roll movements correspond to linear and angular velocity commands respectively. The innovation lies in the integration of Gaussian filtering mechanisms that effectively mitigate tremor effects and environmental noise, ensuring stable and predictable robot movement.

The client-server architecture ensures robust data transmission between the LeapMotion sensor and the WATER2 mobile robot platform, providing reliable communication even in challenging wireless environments.Comprehensive experimental validation demonstrates exceptional performance with 98 percent gesture recognition accuracy and average response latency of 168.06 ms. The system maintains error rates below 1 percent across extended operation periods, proving its reliability for continuous use applications. User experience evaluation with 20 participants revealed minimal learning curves, with average navigation task completion times of 23 seconds, indicating the intuitive nature of the control interface.The algorithm's design philosophy prioritizes user comfort and natural interaction patterns, requiring only relaxed hand positioning over the LeapMotion sensor. The three-dimensional coordinate mapping is performed separately rather than through direct world coordinate transformation, effectively reducing cumulative errors and improving overall system accuracy. This approach makes the system particularly suitable for assistive applications where traditional control methods may be inaccessible or difficult to use. The practical applications span warehouse navigation, material transport vehicles, smart wheelchair control, and assistive robotics where contactless operation provides significant advantages. The system's ability to operate in confined spaces and dynamic environments makes it valuable for applications where traditional remote controls would be impractical. The research establishes a foundation for future developments in autonomous vehicle control and human machine interaction, with potential for integration into larger automated systems for industrial and healthcare applications

The introduction to reinforcement learning methodologies to hand gesture recognition systems,representing a paradigm shift from traditional supervised learning approaches to adaptive, experience-based control mechanisms. The work addresses fundamental limitations of conventional machine learning methods that require fully labeled datasets and cannot adapt to changing user patterns or environmental conditions during operation. The Deep Q Network (DQN) approach enables continuous learning and improvement through online experience, making it particularly suitable for dynamic robotic control applications.The technical architecture combines electromyography (EMG) and inertial measurement unit (IMU) signals from Myo Armband sensors with a CNN-based DQN agent that learns optimal gesture-to-action mappings through continuous interaction and feedback. The system processes EMG-IMU signal windows with 20 ms inference time, enabling real-time control applications. The reinforcement learning framework allows the system to adapt to individual user characteristics and compensate for signal drift over extended use periods.Experimental validation demonstrates superior performance with 97.45 percent classification accuracy and 88.05 percent recognition accuracy, outperforming traditional CNN and SVM approaches by 5-8 percent. The system was tested on two distinct robotic platforms: a three-degrees-of-freedom tandem helicopter test bench and a six-degree-of-freedom UR5 manipulator, proving its versatility across different control applications. Both platforms utilized PID controller schemes with gesture commands providing high-level control inputs. The adaptive capabilities of the DQN-based system represent a significant advancement in practical deployment scenarios where user fatigue, signal drift, and environmental changes can affect performance. The ability to learn from online experience eliminates the need for extensive pre-training datasets and enables personalized calibration for individual users. The system demonstrates robust performance across multiple users and extended operation periods,indicating its suitability for commercial applications.The research implications extend to various fields including assistive technologies, industrial automation, and human robot collaboration where adaptive control systems can provide significant advantages over static approaches. The reinforcement learning framework enables the system to optimize performance for specific tasks and user preferences,leading to improved efficiency and user satisfaction. Future developments may include integration with additional sensor modalities and expansion to more complex gesture vocabularies and robot control scenarios

The development of advanced anthropomorphic control system that achieves unprecedented precision in replicating human upper limb movements through multi-sensor fusion technologies. The work addresses the complex challenge of creating robotic systems that can mimic the natural dexterity and coordination of human arms, enabling applications in rehabilitation medicine, industrial automation, and virtual reality environments. The system integrates Kinect sensors, IMU units, and data gloves to capture comprehensive information about human movement patterns and translate them to robotic actuators.The technical innovation centers on Kalman filter-based sensor fusion that combines four upper limb joint angles from Kinect and IMU sensors with seven wrist and finger angles from data glove technology. This multimodal approach provides complete control over an 11-degree-of-freedom robotic arm, enabling complex anthropomorphic movements including shoulder flexion, abduction, rotation, elbow flexion, and fine finger movements. The Kalman filtering ensures sub-0.5° angular accuracy while maintaining real-time performance requirements.The system successfully demonstrates a variety of anthropomorphic movements with high fidelity reproduction of human motion patterns. Experimental validation includes curve smoothness analysis showing significant improvement in trajectory stability after sensor fusion implementation. The integration of tactile feedback from fingertip sensors enhances safety and precision during manipulation tasks, making the system suitable for applications requiring direct human-robot interaction.The anthropomorphic nature of the control system enables intuitive operation by users without extensive training, as the robotic arm mirrors natural human movements. This characteristic makes it particularly valuable for applications in collaborative manufacturing cells where humans and robots must work in close proximity. The system's precision and responsiveness enable delicate manipulation tasks that require human-like dexterity while maintaining the consistency and repeatability advantages of robotic systems. The practical applications span rehabilitation robotics where precise movement replication can assist in patient therapy, industrial automation where anthropomorphic control enables complex assembly operations, and virtual reality environments where immersive interaction requires natural movement mapping. The research establishes a foundation for next-generation human-machine interfaces that can bridge the gap between human capability and robotic precision,enabling new forms of collaborative automation and assistive technologies

2.1 Research Gap

The existing models have advanced functionalities that have a high potential for industrial applications and easing human intervention in material handling. The technology used in these models are siloed and provide a gap for integration. The literature presents singular interaction among existing human machine interaction technology. The reviewed models also pose a challenge of higher power consumption and lack of remote tracking. The lack of a backup power source also poses safety issues and operational inefficiencies. The design aspects of the scissor lift model can be integrated into the chassis for increasing the weight carrying capacity and to add a second degree of freedom to gesture controlled bots. The core focus of this project is to implement multimodal interaction via gesture control and bluetooth based tracking to increase the overall efficiency of existing models.

Chapter 3

PROBLEM DEFINITION

Traditional material handling utilises manual labour to move goods within warehouses. This is a time consuming, ergonomically concerning and unproductive process. This project proposes to automate material handling by deploying a bot that is controlled by hand gestures for safe and faster material movement within facilities. The proposed solution is an embedded system based implementation that can track and control the bot movements in real time. This approach is a faster, safer, and low cost alternative to manual material handling built on the foundation of Industry 5.0 and Mechatronics.

3.1 Objectives

Based on the literature review and research gap identified the following objectives are fixed for the study. To design and develop a prototype capable of gesture controlled navigation and multimodal interaction by incorporating the features of ESP32 module. Thus create a blueprint for an economical alternative to manual material handling so as to enhance overall efficiency of existing models. The application of this project will be analysed via a case study on a pharmaceutical material handling facility scenario to prove that this embbeded system based application serves as a low cost alternative to traditional material handling with a guaranteed boost in the productivity of the organization and its workers.

3.2 Project Plan

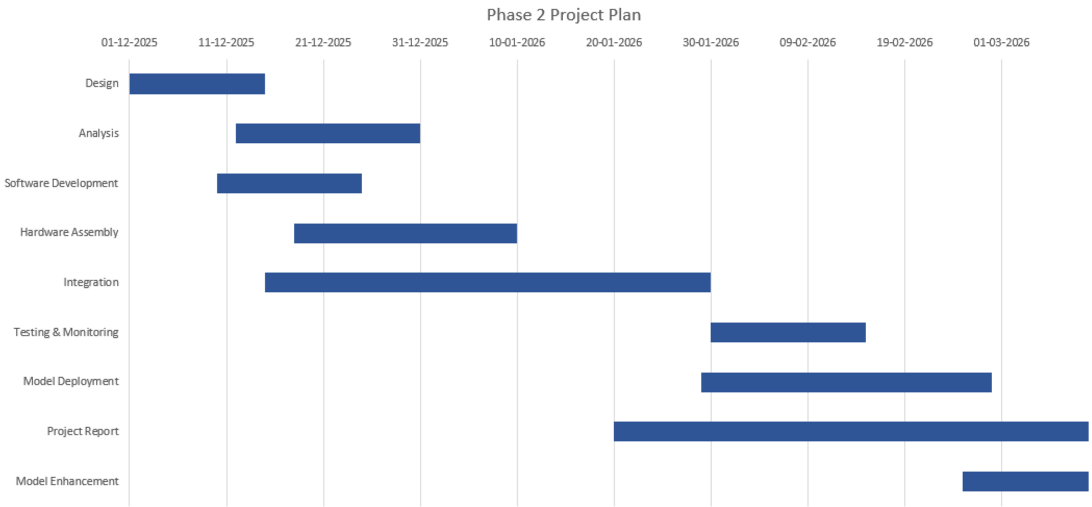


Figure 3.1: Project Plan

3.3 Scope

Unlike Multi National Corporations (MNC) where material handling is fully automated, this project targets Micro Small and Medium Enterprises (MSME). The warehouses of such industries have low to none mechanisations for material handling. Manual labour carries the inventory and this is a slower and ergonomically unproductive phase. This project aims for a low cost implementation using embedded systems that can carry out the material movements saving time and physical pain. Since this is a semi autonomous implementation the job displacements associated with its execution are also lesser than general automation attempts. The targeted industries include textiles, stationery, pharmaceuticals, food processing and agro based industries.

Chapter 4

METHODOLOGY

This chapter signifies the methodology of work that is to be carried out for the project.

The initial step involves the design of the prototype implementation including the circuit design. The component selection involves choosing the required hardware and software components for each function. Hardware assembly is a crucial phase wherein the circuitry is completed on a breadboard. This is followed by the integration of ESP32 into the system and calibration of the sensors. The penultimate phase involves the coding part in Embedded C language. Once the model deployment is complete, the bot is ready for testing and validation.

The model analysis is done in 2 phases prior to deployment as in Design and Simulation. A model design in Fusion 360 is intended for analysing its physical structure and behaviour in real world conditions. A pre deployment simulation for assessing its compatibility is done in Simio.

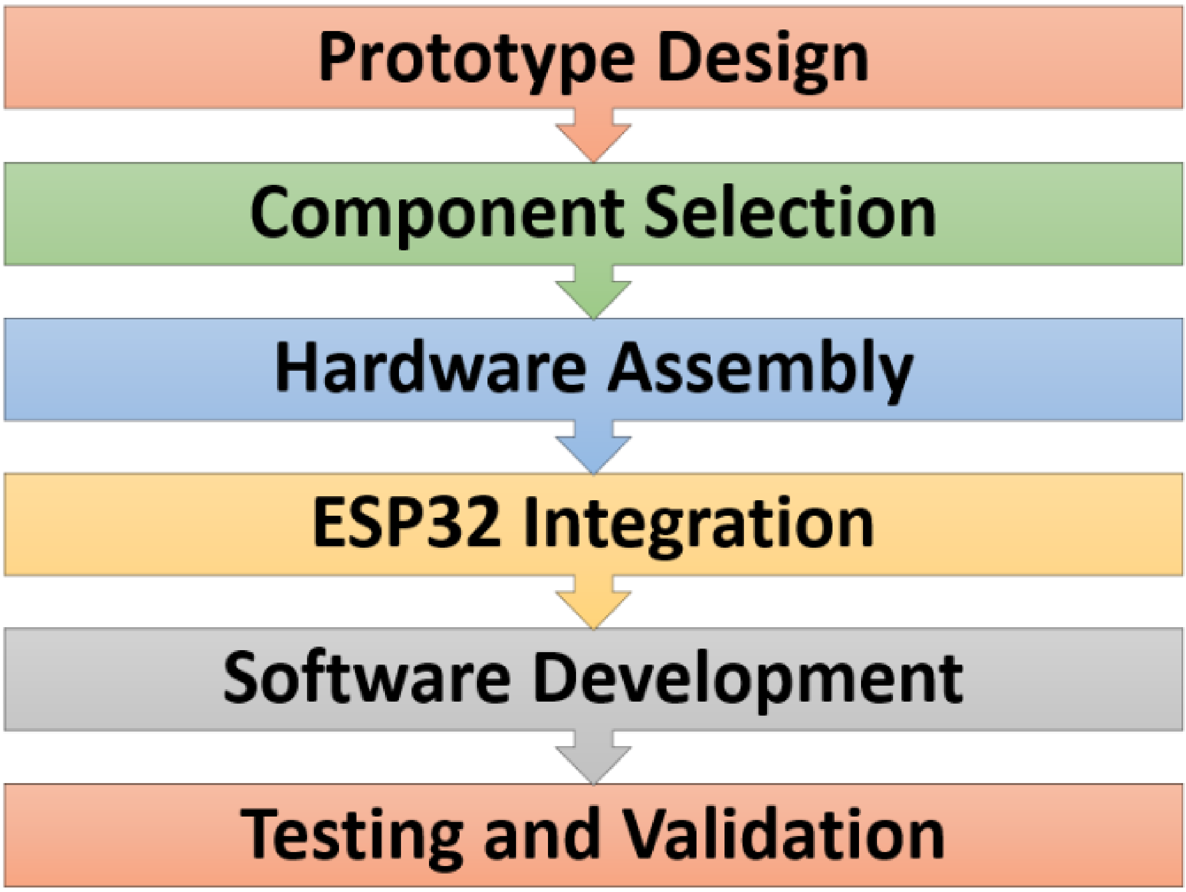


Figure 4.1: Workflow Design

4.1 Components

The hardware components used are: ESP 32 Module, MPU6050 sensor, Small breadboard (170 points), 2 Wheel Smart Car Robot Chassis Kit, Jumper wires, USB cables and L298 Dual Motor driver board. The software components include Arduino IDE, MPU6050 library, sensors library, Wire (I2C communication),and Wi-Fi library.

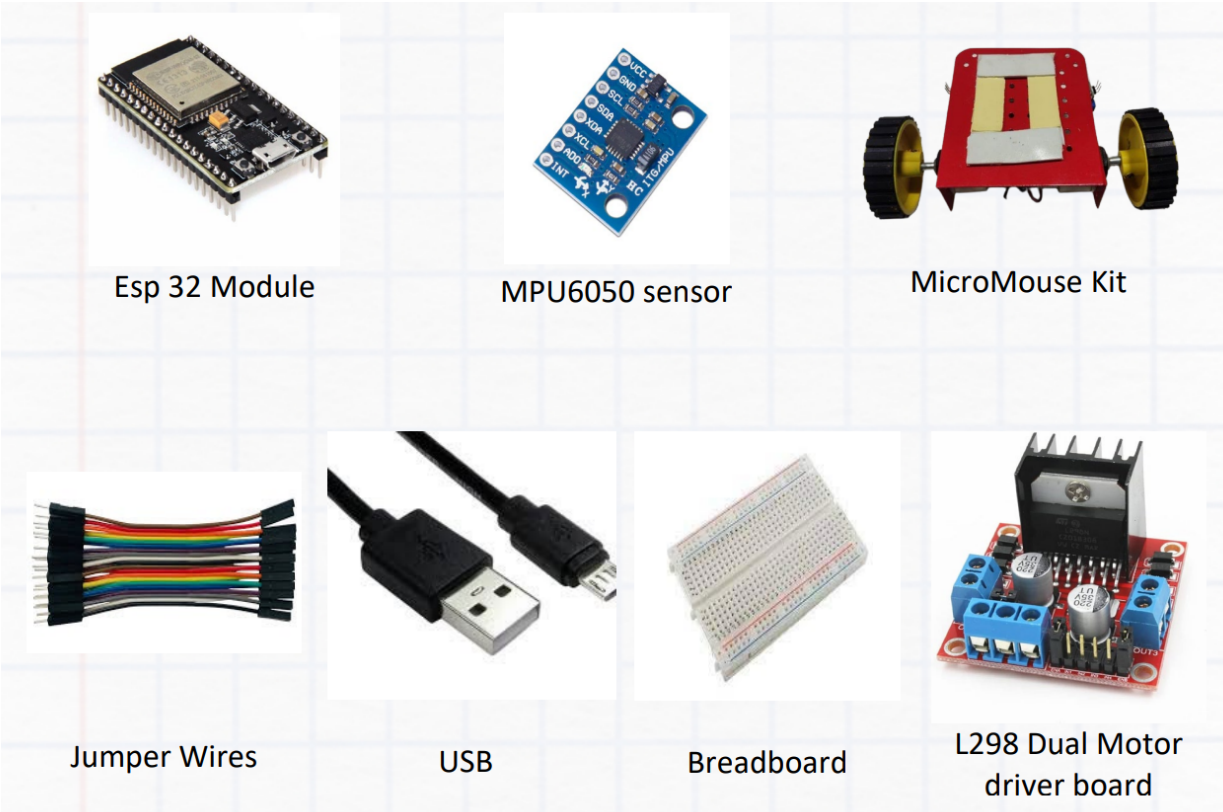


Figure 4.2: Hardware Components [15]

Each component has its own pivotal role to play in bringing the model to life. First and foremost MPU6050 Gyro sensor is the device used for gesture detection. It would be attached to a wearable which would capture the hand movements and that data will be sent to the Micro Mouse wirelessly. It is a complete 6-axis Motion Tracking Device which combines a 3-axis Gyroscope, 3-axis Accelerometer and Digital Motion Processor all in a small package. Also, the sensor has an additional feature of on-chip Temperature sensor. It has an I2C bus interface to communicate with the microcontrollers. It has Auxiliary I2C bus to communicate with other sensor devices like 3-axis Magnetometer, Pressure sensor etc. If the 3-axis Magnetometer is connected to the auxiliary I2C bus, then the MPU6050 can provide complete 9-axis Motion Fusion output. The MPU6050 consists of a 3-axis Gyroscope with Micro Electromechanical System technology which is used to detect rotational velocity along the X, Y, Z axes. When the gyros are rotated about any of the sensor axes, the Coriolis Effect causes a vibration that is detected by a MEM inside MPU6050. The resulting signal is amplified, demodulated, and filtered to produce a voltage that is proportional to the angular rate. This voltage is digitized using 16-bit ADC to sample each axis. The full-scale range of output are +/- 250, +/- 500, +/- 1000, +/- 2000. It measures the angular velocity along each axis in degrees per second unit. The MPU6050 consists of a 3-axis Accelerometer with Micro Electromechanical (MEMs) technology which detects the angle of tilt or inclination along the X, Y and Z axes. Acceleration along the axes deflects the movable mass. This displacement of moving plate (mass) unbalances the differential capacitor which results in sensor output. Output amplitude is proportional to acceleration. 16-bit ADC is used to get digitized output. The full-scale range of acceleration are +/- 2g, +/- 4g, +/- 8g, +/- 16g. It is measured in g (gravity force) units. When the device is placed on a flat surface it will measure 0g on the X and Y axis and +1g on the Z axis.

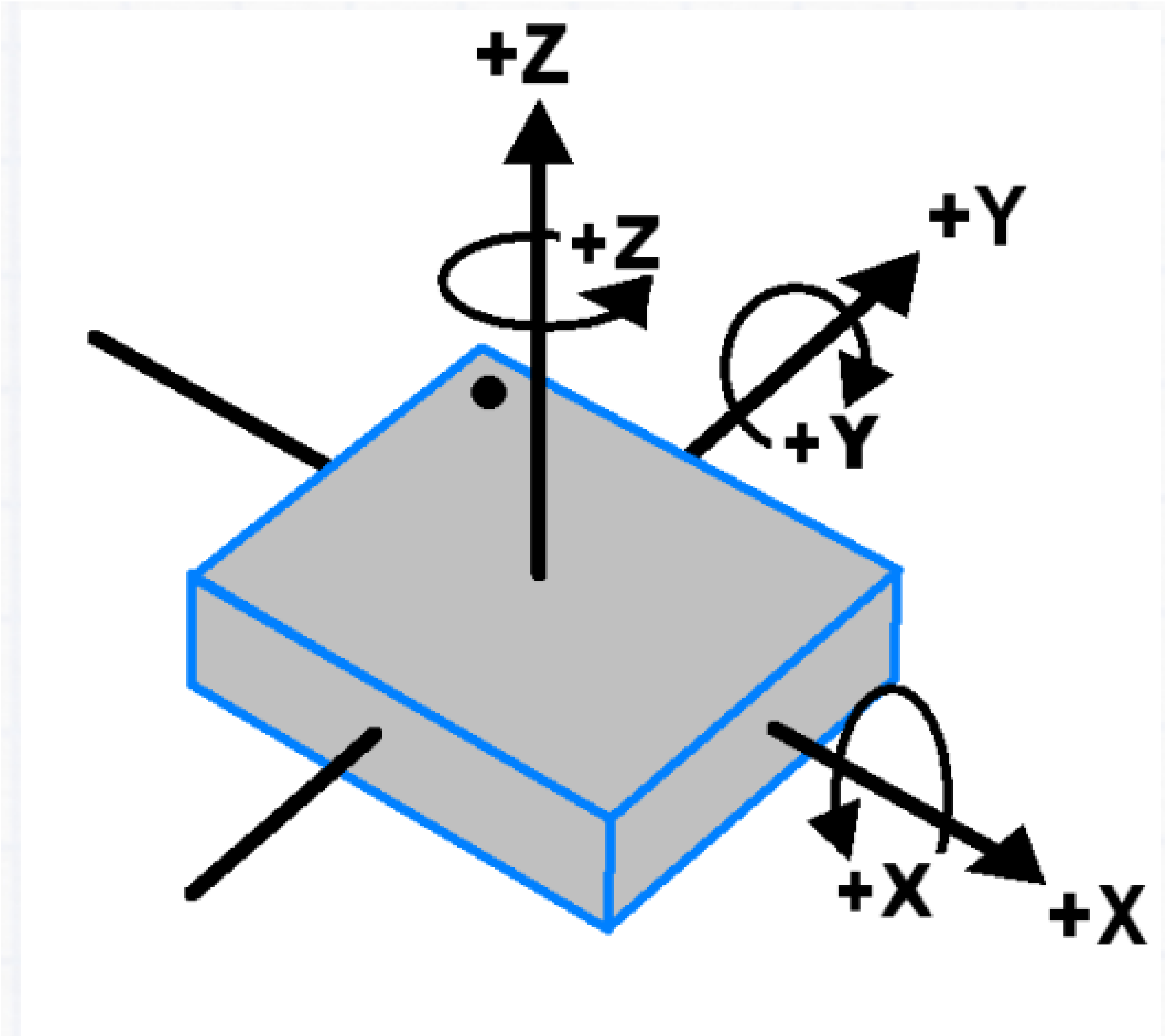


Figure 4.3: Orientation of MPU6050 [15]

The next important component is the L298 Dual Motor driver board which is a high power motor driver for driving DC Motors and Stepper Motors. It uses the L298 motor driver IC and has the onboard 5V regulator which it can supply to an external circuit. It can control up to 4 DC motors, or 2 DC motors with directional and speed control. The micromouse kit acts as the chassis for the prototype, on which the motor driver will be attached. Another set of major components is the IR/US sensors. Infrared sensors are used for object detection and Ultrasonic for distance calculation. Breadboard is the temporary circuit interface used in this study instead of a printed circuit board. Jumper wires and

USBs are used for making the connections. Additionally the design aspect of the scissor lift model provides the bot a second degree of freedom to lift and lower the load placed on top of it.

The core component however remains the microcontroller ESP32. It is a multipurpose microcontroller with various functionalities like Wi-Fi and Bluetooth. The bluetooth functionality alone can be exploited to implement multiple features in this prototype. The bluetooth beacons enable live tracking via Indoor Positioning Systems. The Bluetooth Low Energy (BLE) module provides higher range at lower power consumption. The Received Signal Strength Indicator (RSSI) determines the distance between the bot and the operator.

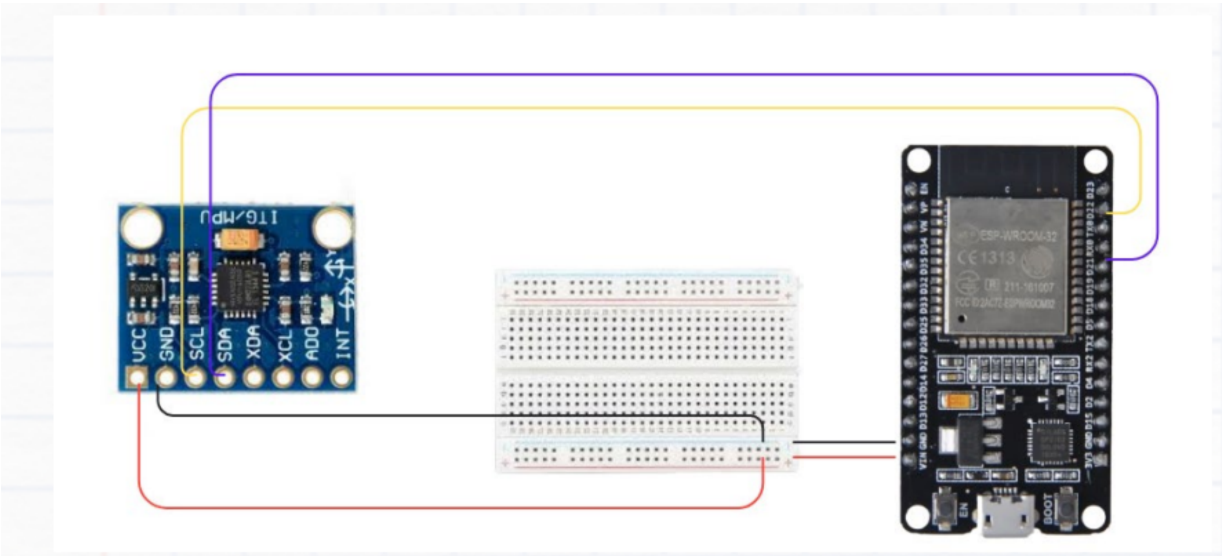


Figure 4.5: Circuit Diagram for the wrist side [15]

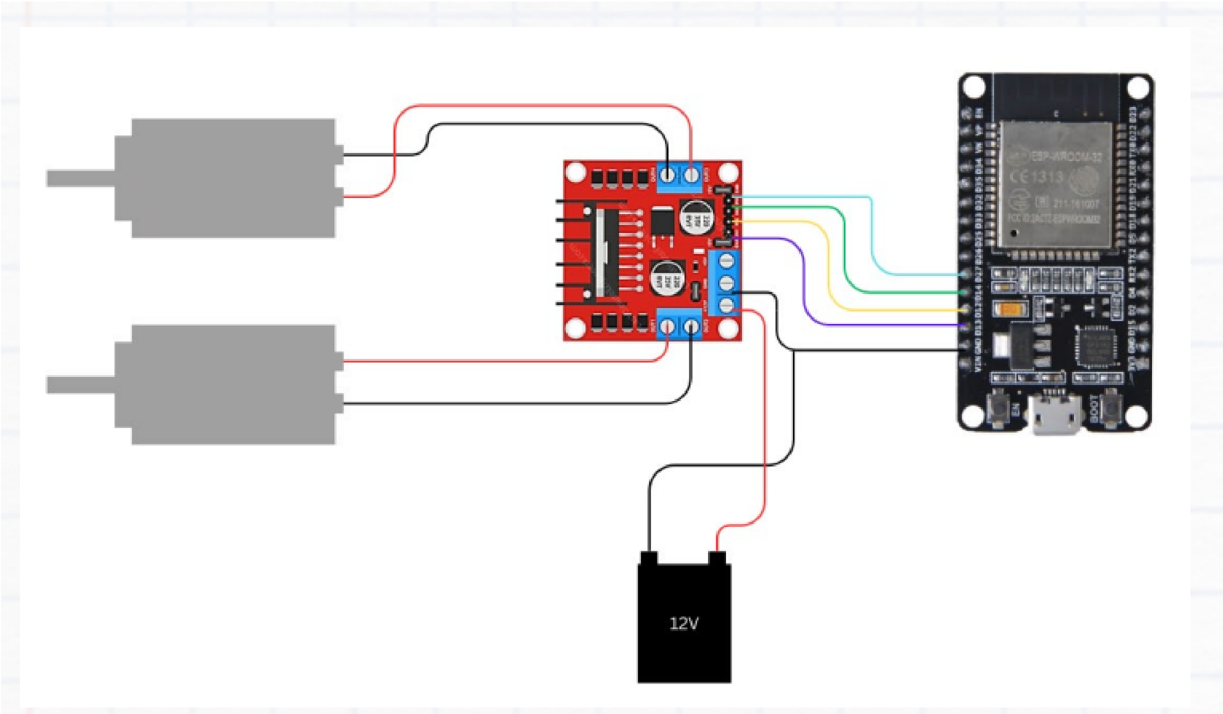


Figure 4.6: Circuit Diagram for the chassis side [15]

4.2 Principle Of Navigation

The core principle is that the gyro sensor detects the motion of the hand and sends that data to the ESP32 which processes and feeds it to the L298 motor driver board. The deviation that the gesture makes against the standard axes of MPU6050 is detected and forwarded to the microcontroller. The sensor is calibrated prior to deployment so as to classify the gesture and convert the deviation degree value to real time 3D coordinates for the servo motor. This converted data is fed to the motor which actuates motion in the gestured direction.

Chapter 5

PROTOTYPING

The hardware implementation of this gesture controlled bot is centered on a modular architecture that integrates mechanical power with intelligent electronic control. At its foundation, a 2-wheel RC car chassis provides the necessary mobility, driven by DC motors that are managed via an L298 dual H-bridge motor driver. This driver serves as the critical bridge between the high-current demands of the propulsion system and the low-voltage logic of the ESP32 microcontroller, which acts as the "brain" of the operation.

To provide vertical functionality, a motorized scissor lift mechanism is integrated into the center of the chassis, allowing the bot to adjust the height of its payload such as a cardboard box or specialized pharmaceutical containers dynamically. The electronic suite is further enhanced by an I2C-connected gesture sensor and a display for real-time feedback, all powered by a high-capacity Li-ion battery regulated through a buck converter. This hardware synergy ensures that the physical movements of the wheels and lift are perfectly synchronized with the digital logic of the multimodal interaction system

A mini prototype is implemented to show the gesture controlled motion feature of the bot and it has been completed in 2 stages owing to software and hardware phases. The Software development further has 2 subsets which are the codes for chassis and gyro sides in C++ language. The chassis side code processes the feed instructions to prompt the wheel motions. The prototype has a 2 wheel RC car chassis. The principles of motions are prompted to give way for 4 types of movements in the forward, backward, left and right directions. The principle of forward and backward motions is that both wheels turn in 1 direction that is forward or backward for the respective motion. Left and right turns are accomplished by 1 wheel turning forward and the other one turning backward. The bot then turns in the direction of the backward rotating wheel. The gyro side code serves the purpose of gesture detection and feeds that data to the ESP32 on the bot.

5.1 Software Development

The Receiver part includes the Wifi library. The main functions in this part of the code are: setup, loop, motor control, move forward, move backward, turn left, turn right and stop motor.

```
void setup() {
  Serial.begin(115200);
  pinMode(motor1Pin1, OUTPUT); pinMode(motor1Pin2, OUTPUT);
  pinMode(motor2Pin1, OUTPUT); pinMode(motor2Pin2, OUTPUT);

  WiFi.begin(ssid, password);
  while (WiFi.status() != WL_CONNECTED) {
    delay(1000);
    Serial.println("Robot connecting to WiFi...");
  }
  Serial.println("Robot Connected!");
  Serial.print(">>> ROBOT IP ADDRESS: ");
  Serial.println(WiFi.localIP()); // COPY THIS TO CONTROLLER CODE
  server.begin();
}

void loop() {
  WiFiClient client = server.available();
  if (client) {
    while (client.connected()) {
      if (client.available()) {
        String commandStr = client.readStringUntil('\n');
        int command = commandStr.toInt();
        Serial.print("Received Command: "); Serial.println(command);
        controlMotors(command);
      }
    }
    client.stop();
  }
}
```

Figure 5.1: Receiver Code 1

```
void controlMotors(int command) {
  switch (command) {
    case 0: Serial.println("Action: STOP"); stopMotors(); break;
    case 1: Serial.println("Action: FORWARD"); moveForward(); break;
    case 2: Serial.println("Action: BACKWARD"); moveBackward(); break;
    case 3: Serial.println("Action: RIGHT"); turnRight(); break;
    case 4: Serial.println("Action: LEFT"); turnLeft(); break;
  }
}

void moveForward() {
  digitalWrite(motor1Pin1, HIGH); digitalWrite(motor1Pin2, LOW);
  digitalWrite(motor2Pin1, HIGH); digitalWrite(motor2Pin2, LOW);
}
void moveBackward() {
  digitalWrite(motor1Pin1, LOW); digitalWrite(motor1Pin2, HIGH);
  digitalWrite(motor2Pin1, LOW); digitalWrite(motor2Pin2, HIGH);
}
void turnRight() {
  digitalWrite(motor1Pin1, HIGH); digitalWrite(motor1Pin2, LOW);
  digitalWrite(motor2Pin1, LOW); digitalWrite(motor2Pin2, HIGH);
}
void turnLeft() {
  digitalWrite(motor1Pin1, LOW); digitalWrite(motor1Pin2, HIGH);
  digitalWrite(motor2Pin1, HIGH); digitalWrite(motor2Pin2, LOW);
}
void stopMotors() {
  digitalWrite(motor1Pin1, LOW); digitalWrite(motor1Pin2, LOW);
  digitalWrite(motor2Pin1, LOW); digitalWrite(motor2Pin2, LOW);
}
```

Figure 5.2: Receiver Code 2

This code functions as the "digital nervous system" of the bot, written for the ESP32 microcontroller to bridge the gap between virtual wireless commands and physical mechanical movement. The script is a TCP/IP Server implementation that listens for numerical instructions over a Wi-Fi network and translates them into electrical signals for the L298 dual H-bridge motor driver. The three primary functional areas in this code are: network communication, data parsing, and hardware actuation.

The WiFi.h library is essential for unlocking the ESP32's on-chip radio. Unlike a standard Arduino, the ESP32 has built-in Wi-Fi hardware that can operate in several modes; here, it is configured to connect to an existing Access Point of the mobile hotspot. The global constants for motor1 and motor2 pins (13, 12, 14, and 27) define the four GPIO (General Purpose Input/Output) pins that will send high or low voltage signals to the L298 driver.

Inside the setup() function, the code performs critical initialization. Setting the serial baud rate to 115200 is standard for ESP32 debugging, ensuring that the developer can monitor the connection status in real-time. The pinMode commands designate the motor pins as outputs, effectively telling the microcontroller to push current through those pins. The connection logic uses a "blocking" while loop to ensure that the robot remains idle until a stable handshake with the router is achieved. Once connected the Local IP Address is printed. This is the unique address on the network that the controller must "call" to send movement commands. Finally, server.begin() opens Port 8080, which acts as a dedicated doorway for incoming data packets.

The loop() function is where the bot spends its active life, constantly checking for a "client". When a connection is detected, the code enters a nested loop that remains active as long as the phone stays connected. The command handling logic is that it uses readStringUntil to capture data. Because Wi-Fi sends data as a stream of characters, the robot might receive the character '1' as a string; toInt() is then used to convert that character into a mathematical integer that the switch statement can understand.

Table 5.1: Actuation Logic for Motor Control

0.9 **Direction**	**Motor 1 (Left)**	**Motor 2 (Right)**	**Logic Explanation**
Forward	HIGH, LOW	HIGH, LOW	Both motors rotate in the same clockwise direction.
Backward	LOW, HIGH	LOW, HIGH	Polarity is reversed; motors rotate counter-clockwise.
Turn Right	HIGH, LOW	LOW, HIGH	The left motor goes forward while the right goes backward, causing a spin-on-axis.
Turn Left	LOW, HIGH	HIGH, LOW	The right motor goes forward while the left goes backward.
Stop	LOW, LOW	LOW, LOW	All pins are grounded; no current flows to the motors.

This Switch-Case structure in controlMotors is the logical clearinghouse of the program. Rather than using multiple "if-else" statements, which can be computationally slower, the switch-case directly jumps to the relevant function (0 to 4). This allows for low-latency response times, which is vital when navigating a bot through narrow industrial aisles or pharmaceutical lab benches.

The final section of the code consists of movement functions that interact with the L298 H-Bridge. An H-bridge works by flipping the polarity of the voltage reaching the motors. By setting motor1Pin1 to HIGH and motor1Pin2 to LOW, current flows in one direction, spinning the wheel forward. Reversing these values flips the motor's rotation.

The code utilizes Differential Steering for navigation. To move forward, both motors rotate in the same direction. To turn, the bot uses "tank steering": for a turnRight command, the left motor spins forward while the right motor spins backward (or remains stationary, depending on the desired turn radius). This allows the 2-wheeled chassis to spin in place, offering the high maneuverability required for the prototype. While this code provides a robust foundation for remote control, it currently operates on Binary Logic (on or off). This means the motors will always run at 100 percent speed when a command is received. In a pharmaceutical scenario where a bot carries sensitive liquids, this sudden acceleration might be too jarring. To improve this, one could implement PWM (Pulse Width Modulation) using the ledcWrite function on the ESP32 to gradually ramp up speed. Furthermore, the code relies on a persistent TCP connection. If the phone moves out of Wi-Fi range or the signal drops, the robot will continue its last received command unless a timeout logic is added. Despite these potential enhancements, the current code is an example of efficient, event-driven programming that perfectly leverages the ESP32's power to create a responsive, multimodal robotic platform.

The Gyro side code includes 4 libraries of MPU6050, sensor, wire and wifi. It has only 2 functions namely setup and loop to detect and send feed to the L298 dual motor driver board on the chassis.

```
Adafruit_MPU6050 mpu;
WiFiClient client;
int lastCommand = -1;

void setup() {
  Serial.begin(115200);
  Wire.begin(21, 22);

  if (!mpu.begin()) {
    Serial.println("MPU6050 not found!");
    while (1) yield();
  }

  WiFi.begin(ssid, password);
  Serial.print("Controller connecting to WiFi");
  while (WiFi.status() != WL_CONNECTED) {
    delay(500);
    Serial.print(".");
  }
  Serial.println("\nController Connected!");
}

void loop() {
  sensors_event_t a, g, temp;
  mpu.getEvent(&a, &g, &temp);

  float pitch = atan2(a.acceleration.x, sqrt(a.acceleration.y * a.acceleration.y + a.acceleration.z * a.acceleration.z)) * 180.0 / M_PI;
  float roll = atan2(a.acceleration.y, a.acceleration.z) * 180.0 / M_PI;

```

Figure 5.3: Gyro Code 1

```

  int command = 0;
  if (pitch > 20)       command = 1; // Forward
  else if (pitch < -20) command = 2; // Backward
  else if (roll > 20)   command = 4; // Left
  else if (roll < -20)  command = 3; // Right
  else                  command = 0; // Stop

  if (command != lastCommand) {
    Serial.printf("Sending Command: %d...", command);
    if (client.connect(robotIP, robotPort)) {
      client.println(command);
      client.stop();
      lastCommand = command;
      Serial.println(" SENT SUCCESS");
    } else {
      Serial.println(" SEND FAILED");
    }
  }
  delay(100);
}
```

Figure 5.4: Gyro Code 2

This code represents the "Controller" side of the multimodal interaction system. This script turns a second ESP32 into an Inertial Measurement Unit (IMU) Controller. It uses an MPU6050 sensor to detect the physical tilt of the user's hand and translates that orientation into movement packets sent over Wi-Fi. It is essentially the wearable that tells the robot what to do based on gravity.

The script includes the Adafruit MPU6050 and Wire libraries. The MPU6050 is a 6 axis Motion Tracking device that contains a 3 axis gyroscope and a 3 axis accelerometer. To communicate with this sensor, the ESP32 uses the I2C (Inter-Integrated Circuit) protocol.

In the setup() function, Wire.begin(21, 22) explicitly defines the SDA (Data) and SCL (Clock) pins.

The code first verifies that the sensor is actually connected. If the mpu.begin() check fails, the ESP32 enters a "halt" state (while(1) yield()), preventing the robot from receiving random, erroneous signals. This is a critical safety feature in industrial or pharmaceutical environments where a malfunctioning sensor could cause a robot to drive into expensive equipment.

The most sophisticated part of this code lies in how it interprets raw gravity data. Inside the loop(), the code calls mpu.getEvent() to capture the current acceleration on the X, Y, and Z axes. Because the sensor is always subject to Earth's gravity ($9.8m/s^2$), trigonometry is used to determine the angle at which the controller is being held.

The code calculates Pitch (tilting forward or backward) and Roll (tilting side to side) using the following trigonometric formulas:

Pitch Calculation: pitch = $atan2(ax, sqrt(ay^2 + az^2)) * (180/pi)$

Roll Calculation: roll = $atan2(ay, az) * (180/pi)$

The atan2 function is used instead of standard atan because it can handle all four quadrants of a circle, providing a full 360° of orientation data. The result is converted from radians to degrees to make the thresholding logic easier to understand and adjust. Once the angles are calculated, the code enters a decision-making phase. It uses a Thresholding Logic of 20°. This creates a "Dead Zone" between -20° and 20° where no command is sent. This is essential for user experience; without a dead zone, the slightest tremor in a user's hand would cause the robot to jitter or move unexpectedly.

Pitch greater than 20: The user tilted the controller forward → Command 1 (Forward)

Pitch lesser than -20: The user tilted backward → Command 2 (Backward)

Roll greater than 20: The user tilted left → Command 4 (Left)

Roll lesser than -20: The user tilted right → Command 3 (Right)

Neutral: Everything else → Command 0 (Stop)

This logic provides a natural, intuitive interface. In a pharmaceutical setting, a worker wearing this as a wristband could control the bot while carrying other items, simply by angling their arm. Unlike the robot, this ESP32 acts as a Client. It does not keep a constant connection open. Instead, it uses a State-Change Detection logic: if (command != lastCommand). This is a vital optimization. If the controller is tilted forward, it doesn't need to spam the robot with the "Forward" command 50 times a second; it only sends the packet once when the tilt is first detected and once more when the state changes.When a change is detected, client.connect(robotIP, robotPort) attempts to open a handshake with the robot's IP address. If the connection is successful, it sends the integer command and immediately calls client.stop(). This "connect-send-disconnect" cycle is efficient for simple control tasks, as it frees up the robot's processing power between commands.

Combining this controller code with the previous code, a Gesture-over-Wi-Fi system has been created. The ESP32 controller transforms physical gravity into a numerical value, and the ESP32 robot transforms that numerical value into voltage for the L298 driver. This loop runs every 100 milliseconds (delay(100)), providing a responsive, low-latency experience that feels like an extension of the user's own body.

5.2 Hardware Integration

The complete hardware integration of the prototype involved a complex synthesis of mechanical engineering, power electronics, and embedded systems programming, centered on the high-performance ESP32 microcontroller. Integration begins with the physical assembly of the 2-wheel RC chassis, which serves as the structural foundation for the mobile platform. The locomotion system utilizes two DC gear motors, which are electrically interfaced with the L298 dual H-bridge motor driver. This driver is the most critical component in the propulsion subsystem because it bridges the gap between the low-power logic of the ESP32 and the high-current demands of the motors. The ESP32 provides the control signals to the L298's input pins, while the L298 draws higher voltage from a Li-ion battery pack to drive the motor coils.

The sensing and feedback layer constitutes the "nervous system" of the hardware integration. This project utilizes the I2C (Inter-Integrated Circuit) protocol to manage multiple peripheral devices with only two wires: SDA (Data) and SCL (Clock). On the robot side, a display is integrated to provide the user with real-time status updates, such as the current IP address, signal strength (RSSI), and operational mode (e.g., "Gesture Control" vs. "Smartphone Control"). Simultaneously, the secondary ESP32—acting as the remote controller—integrates an MPU6050 6-axis IMU (Inertial Measurement Unit). This sensor must be precisely calibrated to ensure that the gravitational acceleration values used in the pitch and roll formulas are accurate. Physical mounting of the MPU6050 is vital; it must be oriented consistently with the intended "forward" direction of the controller to prevent inverted steering commands.

Power management is perhaps the most underrated aspect of hardware integration. A mobile bot with a scissor lift and Wi-Fi capabilities has a high "burst" current requirement. To address this, the system typically uses a 7.4V or 11.1V Lithium-Polymer (LiPo) battery. Since the ESP32 operates at 3.3V and is sensitive to voltage spikes, a buck converter (DC-DC step-down regulator) is used to provide a stable 5V input to the ESP32's Vin pin, while the L298 takes the raw battery voltage directly. Integration also involves adding decoupling capacitors near the motor driver to filter out the high-frequency noise generated by the motor brushes, which could otherwise interfere with the Wi-Fi signal or cause the ESP32 to reset unexpectedly.

Finally, the hardware integration is completed by the communication bridge. By utilizing the ESP32's dual-core architecture, the hardware can simultaneously maintain a Wi-Fi TCP/IP server on one core while handling the PWM (Pulse Width Modulation) for motor speed and the I2C sensor polling on the other. This parallel processing ensures that the mechanical movements whether it is the wheels spinning or the scissor lift rising—are executed with minimal latency. The result is a unified hardware ecosystem where the software logic and physical components work in harmony, allowing the user to control the robot through intuitive hand tilts or smartphone interfaces, effectively turning a collection of wires and motors into a responsive, intelligent machine.

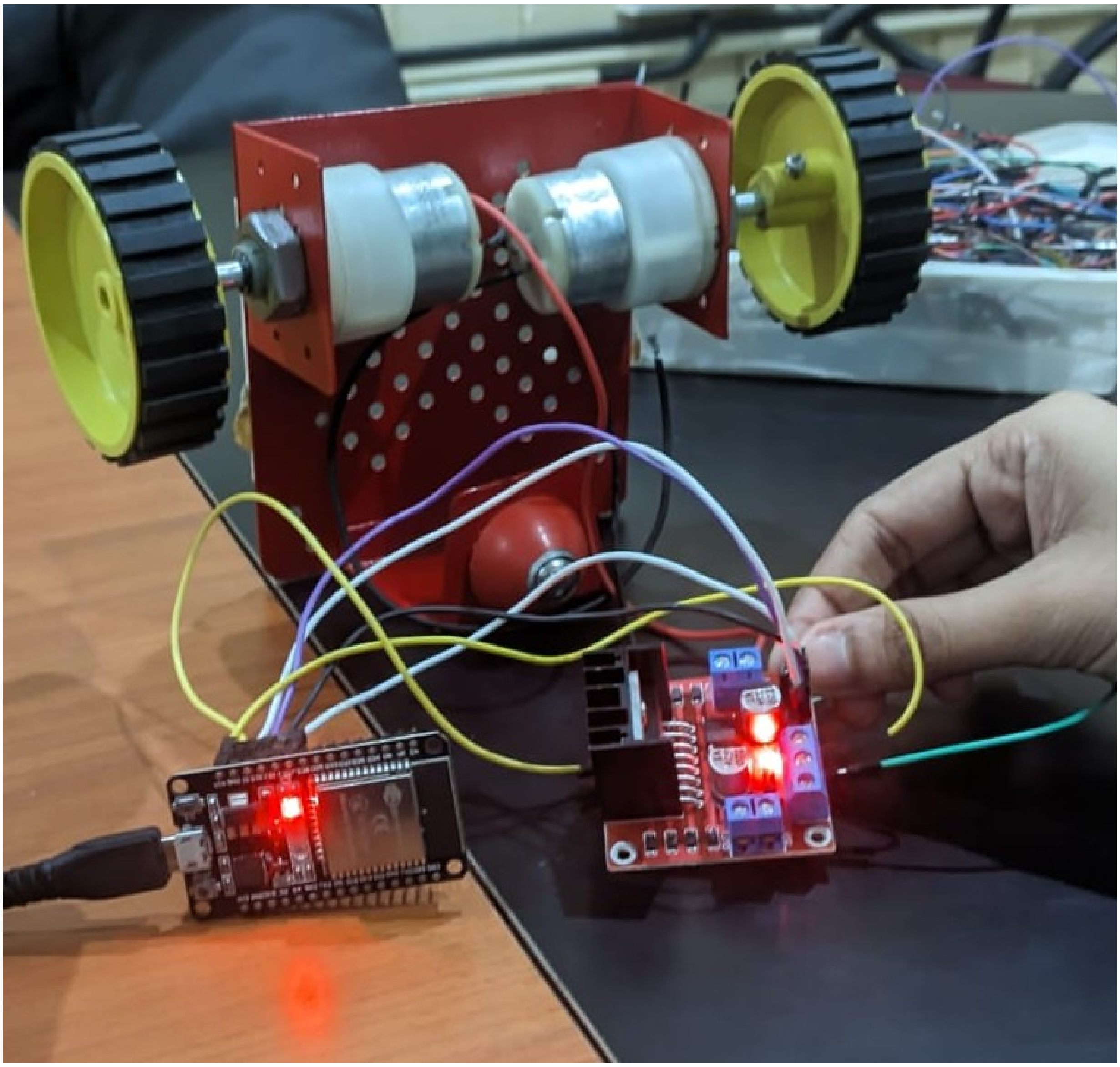


Figure 5.5: Hardware Assembly: Chassis Side

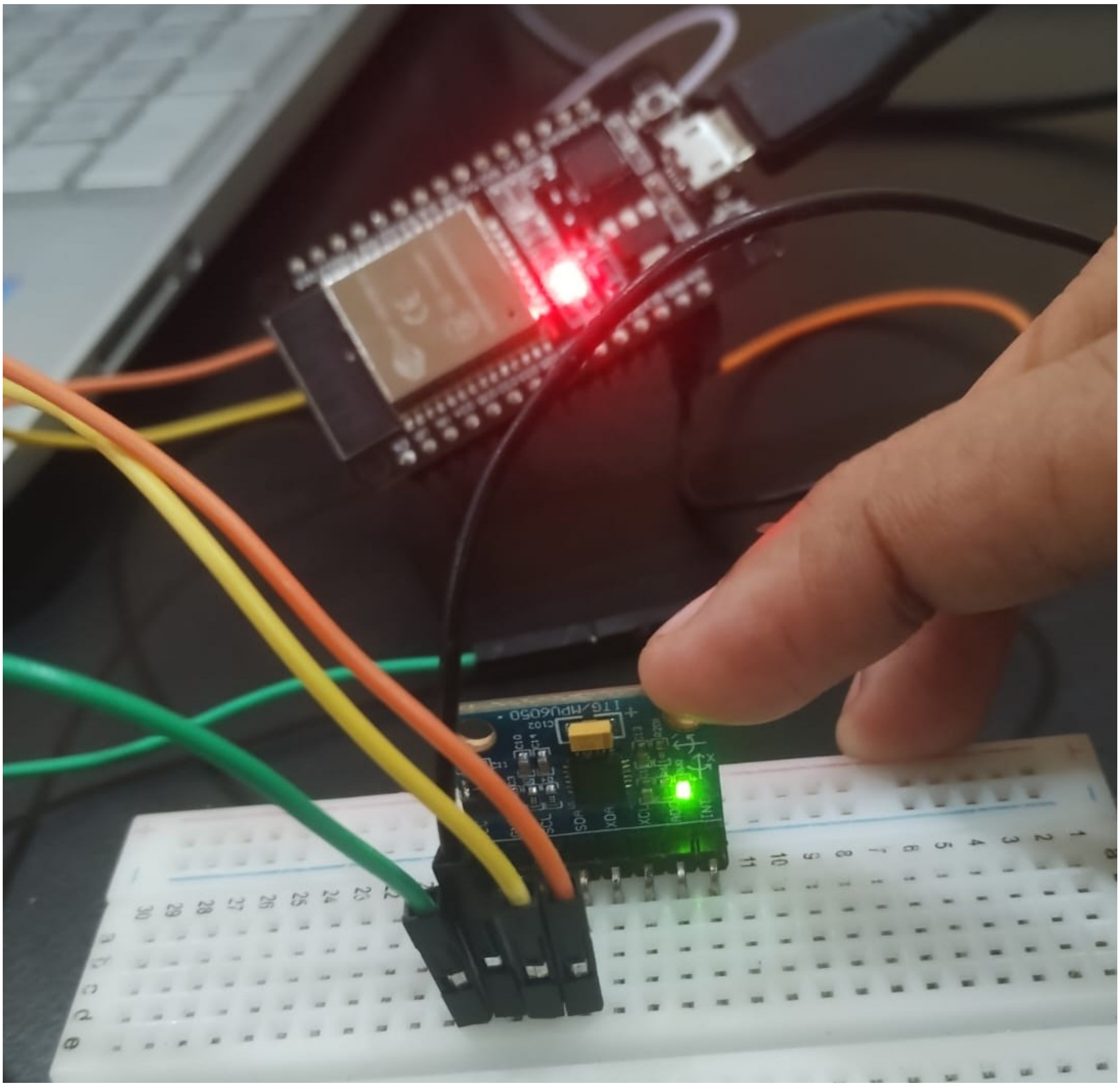


Figure 5.6: Hardware Assembly: Gyro Side

But the actual hardware model implemented was a constrained one that encompasses only the gesture based navigation mechanism. It has 2 sides of circuitry on chassis and gyro sensor parts. The integrated model was able to move according to the gesture inputs. Due to resource crunch and system incompatibility the full fledged model was not implemented on this model. But the blueprint is capable of producing a well working scalable model provided the necessary resources and a strong server.

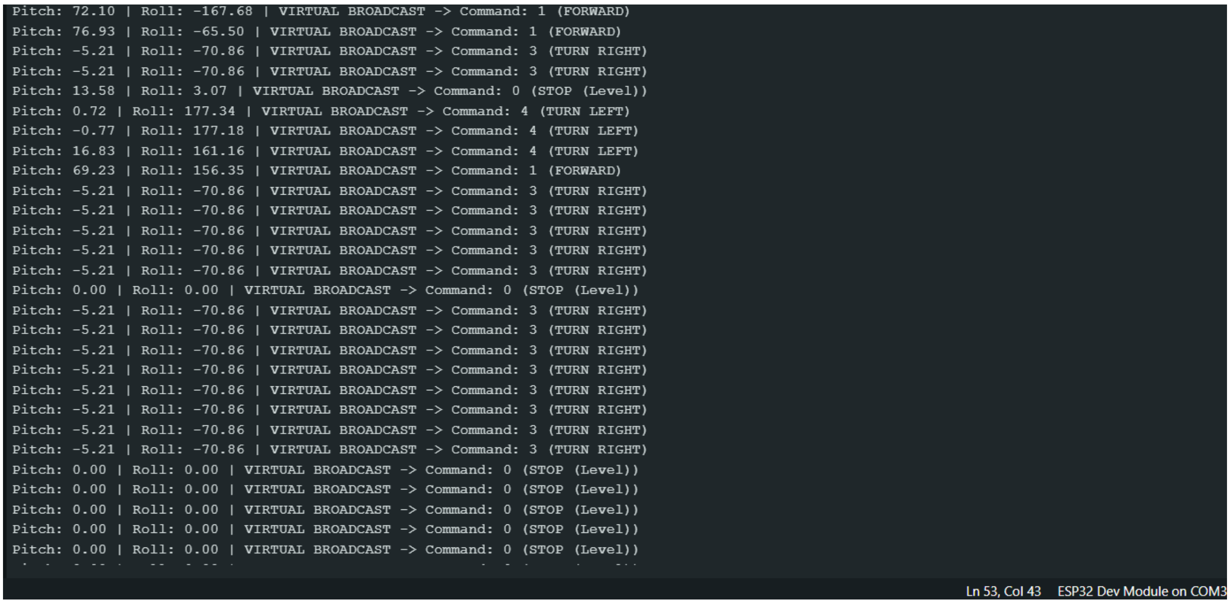
5.3 Outputs

The results from the software development stage showed the working of the gyro sensor and its precision in recognizing the gestures and giving feed directions. The gyro sensor integration successfully demonstrated precise tilt-to-command translation. Using the MPU6050, the prototype accurately calculated pitch and roll, enabling a highly responsive, hands-free interface. This validated the feasibility of inertial-based multimodal control, proving that physical orientation can reliably drive robotic movement with high accuracy.

```
rst:0x1 (POWERON_RESET),boot:0x13 (SPI_FAST_FLASH_BOOT)
configsip: 0, SPIWP:0xee
clk_drv:0x00,q_drv:0x00,d_drv:0x00,cs0_drv:0x00,hd_drv:0x00,wp_drv:0x00
mode:DIO, clock div:1
load:0x3fff0030,len:4980
load:0x40078000,len:16612
load:0x40080400,len:3480
entry 0x400805b4
--- MPU6050 & LOGIC TEST ---
Sensor Online.
Pitch: 54.99 | Roll: -163.18 | VIRTUAL BROADCAST -> Command: 1 (FORWARD)
Pitch: 51.16 | Roll: -144.48 | VIRTUAL BROADCAST -> Command: 1 (FORWARD)
Pitch: -57.63 | Roll: -50.38 | VIRTUAL BROADCAST -> Command: 2 (BACKWARD)
Pitch: -5.21 | Roll: -70.86 | VIRTUAL BROADCAST -> Command: 3 (TURN RIGHT)
Pitch: -62.32 | Roll: -3.05 | VIRTUAL BROADCAST -> Command: 2 (BACKWARD)
Pitch: -63.64 | Roll: 16.73 | VIRTUAL BROADCAST -> Command: 2 (BACKWARD)
Pitch: -64.79 | Roll: 20.63 | VIRTUAL BROADCAST -> Command: 2 (BACKWARD)
Pitch: -68.66 | Roll: 15.89 | VIRTUAL BROADCAST -> Command: 2 (BACKWARD)
Pitch: 71.26 | Roll: 178.74 | VIRTUAL BROADCAST -> Command: 1 (FORWARD)
Pitch: 69.37 | Roll: 173.53 | VIRTUAL BROADCAST -> Command: 1 (FORWARD)
Pitch: 69.63 | Roll: 172.16 | VIRTUAL BROADCAST -> Command: 1 (FORWARD)
Pitch: 68.77 | Roll: -177.70 | VIRTUAL BROADCAST -> Command: 1 (FORWARD)
Pitch: -64.19 | Roll: 2.41 | VIRTUAL BROADCAST -> Command: 2 (BACKWARD)
Pitch: -65.28 | Roll: 2.99 | VIRTUAL BROADCAST -> Command: 2 (BACKWARD)
Pitch: -66.52 | Roll: 3.75 | VIRTUAL BROADCAST -> Command: 2 (BACKWARD)
Pitch: 73.70 | Roll: 170.80 | VIRTUAL BROADCAST -> Command: 1 (FORWARD)
Pitch: 72.10 | Roll: -167.68 | VIRTUAL BROADCAST -> Command: 1 (FORWARD)
Pitch: 76.93 | Roll: -65.50 | VIRTUAL BROADCAST -> Command: 1 (FORWARD)

Ln 53, Col 43     ESP32 Dev Module on COM3
```

Figure 5.7: Output 1



```
Pitch: 72.10 | Roll: -167.68 | VIRTUAL BROADCAST -> Command: 1 (FORWARD)
Pitch: 76.93 | Roll: -65.50 | VIRTUAL BROADCAST -> Command: 1 (FORWARD)
Pitch: -5.21 | Roll: -70.86 | VIRTUAL BROADCAST -> Command: 3 (TURN RIGHT)
Pitch: -5.21 | Roll: -70.86 | VIRTUAL BROADCAST -> Command: 3 (TURN RIGHT)
Pitch: 13.58 | Roll: 3.07 | VIRTUAL BROADCAST -> Command: 0 (STOP (Level))
Pitch: 0.72 | Roll: 177.34 | VIRTUAL BROADCAST -> Command: 4 (TURN LEFT)
Pitch: -0.77 | Roll: 177.18 | VIRTUAL BROADCAST -> Command: 4 (TURN LEFT)
Pitch: 16.83 | Roll: 161.16 | VIRTUAL BROADCAST -> Command: 4 (TURN LEFT)
Pitch: 69.23 | Roll: 156.35 | VIRTUAL BROADCAST -> Command: 1 (FORWARD)
Pitch: -5.21 | Roll: -70.86 | VIRTUAL BROADCAST -> Command: 3 (TURN RIGHT)
Pitch: -5.21 | Roll: -70.86 | VIRTUAL BROADCAST -> Command: 3 (TURN RIGHT)
Pitch: -5.21 | Roll: -70.86 | VIRTUAL BROADCAST -> Command: 3 (TURN RIGHT)
Pitch: -5.21 | Roll: -70.86 | VIRTUAL BROADCAST -> Command: 3 (TURN RIGHT)
Pitch: -5.21 | Roll: -70.86 | VIRTUAL BROADCAST -> Command: 3 (TURN RIGHT)
Pitch: 0.00 | Roll: 0.00 | VIRTUAL BROADCAST -> Command: 0 (STOP (Level))
Pitch: -5.21 | Roll: -70.86 | VIRTUAL BROADCAST -> Command: 3 (TURN RIGHT)
Pitch: -5.21 | Roll: -70.86 | VIRTUAL BROADCAST -> Command: 3 (TURN RIGHT)
Pitch: -5.21 | Roll: -70.86 | VIRTUAL BROADCAST -> Command: 3 (TURN RIGHT)
Pitch: -5.21 | Roll: -70.86 | VIRTUAL BROADCAST -> Command: 3 (TURN RIGHT)
Pitch: -5.21 | Roll: -70.86 | VIRTUAL BROADCAST -> Command: 3 (TURN RIGHT)
Pitch: -5.21 | Roll: -70.86 | VIRTUAL BROADCAST -> Command: 3 (TURN RIGHT)
Pitch: -5.21 | Roll: -70.86 | VIRTUAL BROADCAST -> Command: 3 (TURN RIGHT)
Pitch: -5.21 | Roll: -70.86 | VIRTUAL BROADCAST -> Command: 3 (TURN RIGHT)
Pitch: 0.00 | Roll: 0.00 | VIRTUAL BROADCAST -> Command: 0 (STOP (Level))
Pitch: 0.00 | Roll: 0.00 | VIRTUAL BROADCAST -> Command: 0 (STOP (Level))
Pitch: 0.00 | Roll: 0.00 | VIRTUAL BROADCAST -> Command: 0 (STOP (Level))
Pitch: 0.00 | Roll: 0.00 | VIRTUAL BROADCAST -> Command: 0 (STOP (Level))
Pitch: 0.00 | Roll: 0.00 | VIRTUAL BROADCAST -> Command: 0 (STOP (Level))
```

Ln 53, Col 43 ESP32 Dev Module on COM3

Figure 5.8: Output 2

The prototype successfully validated the foundational locomotion and wireless communication architecture required for a remote operated mobile platform. Although the scissor lift and multimodal gesture interactions were not integrated into this specific iteration, the outputs confirm that the ESP32 and L298 hardware stack can reliably execute real-time movement commands over a Wi-Fi network. This baseline establishes a stable, modular framework, proving that the propulsion and networking logic are robust. The model runs were analysed and the speed was found to be proportional to the voltage applied. In an industrial scenario a battery of 80V can power up to 2m/s speed. The data obtained from this deployment will be used for the simulation in the upcoming chapter. The project has thus transitioned from a theoretical concept to a functional proof of concept, providing the essential groundwork for the subsequent integration of vertical mechanics and advanced sensory modalities.

Chapter 6

MULTI MODAL INTERACTION

Multimodal interaction using ESP32 beacons refers to a system where a user interacts with an environment through multiple communication channels (modes) such as proximity, touch, gesture, or sound—facilitated by the ESP32's Bluetooth Low Energy (BLE) and Wi-Fi capabilities Instead of relying on a single input, a multimodal system perceives the user's intent through various sensors and signals provided by the beacon. While dedicated beacons (like iBeacons) exist, the ESP32 is preferred for custom multimodal setups because of Programmability in which a program can be written using custom logic in Arduino, MicroPython, or ESP-IDF. It has plenty of peripheral support like GPIOs for adding microphones, displays, or biometric sensors. The Dual-Core Power enables One core to handle the BLE radio while the other processes sensor data or user gestures.

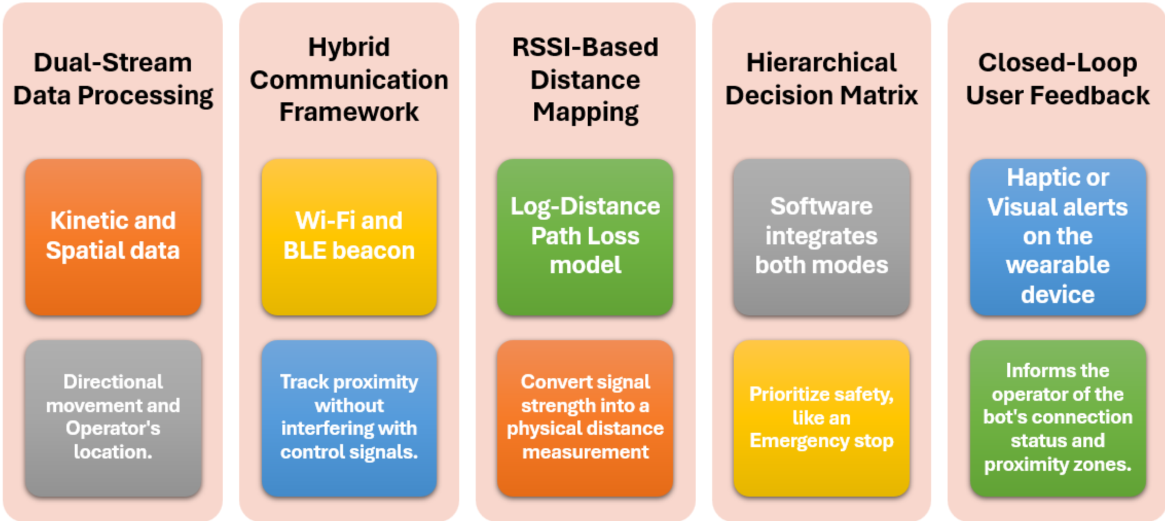


Figure 6.1: Multimodal Interaction Implementation Phases

Implementing multimodal interaction transforms the bot from a simple remote controlled device into an operator-aware intelligent system. The phases of implementation are as follows. Dual-Stream Data Processing is the initial phase where the system concurrently handles kinetic data (accelerometer/gyroscope values from the MPU6050) for directional movement and spatial data (Bluetooth RSSI) to establish the operator's location. Next one is the Hybrid Communication Framework which utilizes a low-latency Wi-Fi connection (or ESP-NOW) for real-time gesture execution while simultaneously employing Bluetooth Low Energy (BLE) beaconing to track proximity without interfering with control signals. This is followed by RSSI-Based Distance Mapping where the ESP32 uses the Log-Distance Path Loss model to convert signal strength into a physical distance measurement, allowing the bot to maintain a "digital tether" to the operator. The Hierarchical Decision Matrix is where the software integrates both modes to prioritize safety for instances like the requirement of an automatic trigger for Emergency Stop if the operator moves beyond 5 meters or reduce speed if they are within 1 meter. The final phase is the Closed-Loop User Feedback where Multimodal interaction is completed via haptic or visual alerts (like a vibrating motor or RGB LEDs) on the wearable device to inform the operator of the bot's connection status and proximity zones.

The primary algorithm used to calculate the distance between the operator and the bot via Bluetooth is the Log-Distance Path Loss Model. Because radio signal strength (RSSI) does not decrease linearly with distance, but rather logarithmically, this mathematical model is essential for converting decibel-milliwatts (dBm) into meters. The Mathematical Formula showing the relationship between distance and signal strength is expressed as: $d = 10^((A - RSSI)/(10 * n))$. Where, d is the calculated distance in meters, RSSI is the Received Signal Strength Indicator currently measured by the ESP32 (usually a value between -30 and -90), A is the reference RSSI value measured at a known distance of exactly 1 meter from the bot and n is the Path Loss Exponent, which represents the environment's interference level (typically 2.0 for open space and 2.7 to 4.3 for indoor warehouse environments with obstacles).

The ESP32 is a versatile microcontroller because it can act as both a broadcaster (sending out its ID) and an observer (listening for other devices). Proximity Presence (The Primary Mode) in which the most basic mode is location-awareness. Using BLE, the ESP32 broadcasts a "heartbeat" signal. A receiver (like a smartphone or another ESP32) measures the Received Signal Strength Indicator (RSSI) to estimate distance.A light turns on automatically as you walk into a room (passive interaction). Physical and Haptic Input features built-in capacitive touch sensors. You can turn a beacon into an interactive surface. By Tapping the beacon housing to "check in" or trigger an emergency alert. Environmental Context involves Beacons which can be equipped with sensors (temperature, humidity, or PIR motion sensors). The system adjusts the HVAC based on whether the beacon detects human motion and a specific temperature threshold. Visual and Auditory Feedback involves Multimodal interaction which is a two-way street. The beacon can communicate back to the user via onboard LEDs (WS2812B/NeoPixels) or small buzzers. A beacon flashes green to guide a warehouse worker toward the correct shelf.

Technical Components of the System to implement multimodal interaction use the following framework:

The steps to implement multimodal interactions are as follows:

1. System Architecture: The setup requires a minimum of three fixed anchors at known

Table 6.1: Technical Components of the System to implement multimodal interaction

Component	Role
BLE Advertising	The ESP32 sends packets containing a UUID, Major, and Minor ID to identify itself.
RSSI Filtering	Mathematical algorithms (like Kalman filters) are used to smooth out signal "noise" to accurately determine if a user is "Immediate," "Near," or "Far."
MQTT/Wi-Fi	The ESP32 uses Wi-Fi to send sensor data to a central server (like Home Assistant) to trigger complex actions.
Deep Sleep	To remain a "beacon," the ESP32 uses low-power modes, waking up only to broadcast or scan.

coordinates to calculate a 2 dimensional position. It requires Fixed Anchors (3+ ESP32s) Placed in the corners of the room which act as iBeacons, constantly broadcasting their unique ID. the Mobile Tag (Bot's ESP32) Scans for the anchors, measures their RSSI, and calculates its distance from each. The bot sends its calculated position to a UI (Smartphone or Web Dashboard) via MQTT or WebSockets which constitutes the communication layer

2. Mathematical Blueprint (Trilateration): To turn "signal strength" into "map coordinates," these three mathematical steps are followed. RSSI to Distance Convert the signal strength into meters using the path loss formula. Signal Filtering by Raw RSSI fluctuates due to walls and interference. Implement a Kalman Filter on the ESP32 to smooth the distance data before it hits the UI. This prevents "jittery" movement on your map.Calculating the coordinates includes three distances from three known points, by solving the intersection of three circles.

3. Real-Time UI and Control Integration: To visualize and control the bot, there's the need of a bidirectional bridge.

Control Logic:

1. UI Interaction: You tap a point on the map

Table 6.2: Function of UI layers

Layer	Technology	Function
Messaging	MQTT (HiveMQ/Mosquitto)	Handles high-speed data packets for position and commands.
Backend	Node-RED or Python	A simple script to receive coordinates and host the UI.
Frontend	HTML5 Canvas / Flutter	Renders a “Floor Plan” where the bot appears as a moving dot.

2. Command Sent: The app publishes the target coordinates to an MQTT topic

3. Bot Execution: The bot compares its current position with the target, uses its Gyro to calculate the heading, and drives the L298 motors until it reaches the goal.

4. Implementation Steps:

    1. Flash Anchors: Program 3 ESP32s and Assign each a unique ”Minor” ID.

    2. Calibrate: Place the bot exactly 1m from each anchor and record the RSSI. Hardcode this as the ”Reference RSSI.”

    3. Setup MQTT: Use a free broker like EMQX or HiveMQ Cloud.

    4. Build Dashboard: Use Node-RED Dashboard for a ”no-code” UI that can display the coordinates on a static image of your floor plan.

For cm-level accuracy (much better than standard Bluetooth), consider upgrading the ESP32s with UWB (Ultra-Wideband) modules like the DWM3000. They use time-of-flight instead of signal strength, making them immune to signal reflections.

This chapter outlines the Multimodal Interaction System for the robot, integrating the ESP32 as the central hub for processing commands from multiple inputs (Smartphone, Gestures, and RSSI tracking) and translating them into physical actions. The Interaction

Blueprint Architecture in which The system is divided into three main layers: the Input Modality, the Processing Layer, and the Actuation Layer. The Layer is broken down into Input Modalities (The Senses), The Processing Layer (The Brain) and The Actuation Layer (The Body). Input Modalities includes Smartphone (Secondary Control) that Sends HTTP or BLE packets for movement (Joystick) and Lift Control (Slider), Gesture Sensor (Primary/Local Control) that like the PAJ7620 or a camera detects hand motions,RSSI Tracking where The ESP32 broadcasts a signal from which the smartphone measures the strength to estimate distance. The Processing Layer has the ESP32 Microcontroller which Runs a dual-core loop, Core that Manages Wi-Fi/Bluetooth stacks and listens for incoming packets from the phone and Core which Processes local gesture data and executes the PWM (Pulse Width Modulation) logic for the motors. The Actuation Layer has an L298 Motor Driver to Control the high-current DC motors for the 2-wheel chassis, Lift Actuator which is a secondary DC motor or Servo connected to the scissor mechanism and Visual Feedback from the 1.3" OLED/LCD display showing connection status and "Distance to Phone" based on RSSI.

Table 6.3: Data Flow Logical Blueprint

Interaction	**Source**	**Protocol**	**Resulting Action**
Movement	Smartphone UI	WebSockets/BLE	L298 drives wheels (Forward/Back/Turn)
Lifting	Gesture / Phone	I2C / HTTP	Scissor lift motor extends/retracts
Tracking	ESP32 Signal	WiFi/BLE RSSI	Phone UI updates "Hot/Cold" distance bar
Status	ESP32 Sensors	Serial/I2C	LCD displays "Mode: Gesture" or "Battery Low"

The Implementation Logic (The "State Machine") is to prevent the robot from getting confused between phone commands and hand gestures, for which a Priority Logic should be implemented. Manual Override that If a command is received from the Phone, it takes priority over Gesture control for 5 seconds. Safety Stop that If RSSI drops below a certain threshold (meaning the bot is too far away), it enters "Search Mode" or stops automatically. Gesture Mode which is the default mode when the phone is idle, allowing for quick "stop" or "lift" hand signals. Hardware Connectivity Maps for Communication is ESP32 (Internal Wi-Fi/BT), for Motor Controls are GPIOs 25, 26, 27, 33 L298 Inputs, with Sensory part handled by I2C Pins (SDA/SCL) Gesture Sensor and OLED Display and Powered by 7.4V Li-ion Battery Buck Converter 5V (ESP32) and 7.4V (L298).

In an industrial landscape increasingly defined by Industry 4.0, the deployment of a multimodal interacting bot combining an RC chassis, a vertical scissor lift, and dual-layer control (Gesture and Smartphone) addresses the critical need for flexible, low-cost automation. Modern warehouses and manufacturing plants often struggle with "the last meter" problem: moving small components between workstations that are not connected by fixed conveyors. This bot serves as a dynamic bridge, capable of navigating tight aisles while adjusting its height to interface with different workbench levels. The integration of the ESP32 allows for a "connected" factory floor where the bot isn't just a tool, but a data point capable of being tracked via RSSI signals to optimize logistical workflows and prevent collisions.

The advantages of this specific multimodal design lie in its versatility and intuitive user experience. By offering gesture control, the bot allows floor workers to interact with it "hands-free." In a scenario where a technician's hands are occupied with tools or heavy parts, a simple wave or palm gesture can command the bot to raise its scissor lift or move to a designated bay. This reduces the cognitive load on workers and speeds up the handover process. Simultaneously, the smartphone interface provides a "birds-eye" administrative control, allowing a floor manager to override local actions, perform remote diagnostics, or redirect the bot to a different zone without being physically present. This redundancy ensures that the bot remains functional even if one control modality fails due to environmental interference.

However, the challenges of implementing such a system in a rugged industrial environment are substantial. Industrial settings are notorious for "Electromagnetic Interference" (EMI) caused by heavy machinery and high-voltage lines, which can severely degrade the ESP32's Wi-Fi and Bluetooth signals. This makes RSSI-based tracking notoriously unstable, requiring complex filtering algorithms to prevent the bot from "teleporting" in the digital twin system. Furthermore, the mechanical simplicity of an RC-style chassis may struggle with floor debris or uneven surfaces common in factories. The L298 motor driver, while excellent for prototyping, is relatively inefficient and generates significant heat under constant load, which could lead to thermal throttling or hardware failure in high-duty-cycle industrial shifts.

The disadvantages primarily revolve around safety and scalability. Unlike high-end Industrial Mobile Robots (IMRs) that use LiDAR for SLAM (Simultaneous Localization and Mapping), a bot relying on gesture and smartphone control lacks the autonomous "vision" to detect obstacles in its blind spots. In a fast-paced environment with forklifts and moving personnel, this poses a significant liability. Additionally, the open-frame nature of the scissor lift and exposed electronics makes the bot vulnerable to dust, oil mist, and moisture. To move from a prototype to a production-ready tool, the system would require a costly transition to industrial-grade enclosures and more robust communication protocols like LoRa or wired industrial Ethernet, potentially stripping away the low-cost appeal of the original ESP32 design.

The development of this multimodal interacting bot represents a significant shift from rigid, single-input automation toward a more intuitive and adaptive robotic ecosystem. By synthesizing gesture-based commands, smartphone remote interfacing, and RSSI-based signal tracking, we bridge the gap between complex industrial machinery and natural human behavior. This project demonstrates that sophisticated, multi-layered control systems no longer require prohibitively expensive hardware; rather, through the strategic use of versatile microcontrollers like the ESP32 and modular mechanical components like the scissor lift, we can create highly functional prototypes capable of solving real-world logistical challenges.

While the transition from a laboratory prototype to a ruggedized industrial or pharmaceutical tool presents hurdles in signal reliability, sterilization, and safety certification, the core architecture provides a robust foundation for future iterations. The integration of these diverse interaction modalities will be essential in creating "cobots" that do not just follow a script, but actively respond to the presence and needs of their human partners. Ultimately, this chapter highlights that the success of next-generation robotics lies not just in the strength of the motors or the speed of the processor, but in the seamlessness of the communication between the user and the machine, ensuring that technology remains an empowering extension of human intent.

In the high-stakes environment of a pharmaceutical industry, the implementation of a multimodal interacting boat equipped with a scissor lift, RC-style mobility, and dual-control interfaces presents a transformative opportunity for sterile and efficient logistics.

The primary need for such a system stems from the industry's stringent requirements for contamination control and precision. In cleanrooms (Grade A or B), human presence is the largest source of particulates and microbial shedding. By deploying a bot that can be controlled via touchless gestures or remotely via a smartphone, the need for human proximity to sensitive chemical batches or sterile vials is significantly reduced. Furthermore, the scissor lift addresses the need for vertical integration, allowing the bot to interact with autoclaves, conveyor belts, and storage racks of varying heights without requiring expensive fixed automation infrastructure.However, the challenges of implementation are significant, particularly regarding regulatory compliance and environmental constraints.

Pharmaceutical facilities are "RF-unfriendly" environments; thick lead-shielded walls and stainless steel equipment can cause signal multipath interference, making RSSI-based tracking or Wi-Fi control prone to "dead zones" or latency. Additionally, the bot itself must meet "wash-down" standards. The exposed wires of an L298 driver and the open crevices of a scissor lift mechanism, as seen in the prototype, are magnets for dust and difficult to sterilize. Achieving IP65 or higher ingress protection for such a complex mechanical device is a costly engineering hurdle. There is also the challenge of sensor reliability; gesture sensors may struggle with the glare from reflective stainless steel surfaces or the interference from specialized cleanroom lighting.

The disadvantages involve the steep learning curve for staff and potential safety risks. While multimodal interaction is intuitive, "gesture drift" or accidental triggers could lead to the scissor lift engaging at the wrong time, potentially tipping over expensive or volatile chemical samples. Unlike fully autonomous AGVs (Automated Guided Vehicles), this bot relies on human-in-the-loop interaction, which introduces the possibility of human error during remote smartphone operation. Moreover, the payload capacity of a standard RC chassis is limited; if the pharmaceutical application requires moving heavy lead-lined containers for radiopharmaceuticals, the motor torque and battery life of an ESP32-powered system may prove insufficient for full-shift duty cycles. The advantages of this specific design are multifaceted. Firstly, the multimodal control (Smartphone + Gestures) provides a fail-safe operational layer. A lab technician can use hand gestures to "call" the bot while their hands are busy, or use a smartphone interface for high-precision navigation from outside the cleanroom. This flexibility increases operational throughput. Secondly, the RC chassis with a scissor lift offers "flexible automation," meaning the bot can be repurposed for different lab layouts much faster than traditional belt systems. The use of ESP32-based RSSI tracking also allows the facility to maintain a real-time digital twin of the bot's location, ensuring that hazardous materials are always accounted for within the floor plan. However in a micro, small and medium enterprises facility scenario this bot can serve as a real productivity and profit enhancer considering the pros that outweigh the disadvantages.

Chapter 7

SIMULATION

Simulation has been done in pre deployment and post deployment phases. Initial simulation was done on assumptive data while the second phase was done using the actual data obtained after bot deployment.

The pre deployment simulation was based on assumptive data to compare the overall efficiency, time Consumed and utility of material handling done manually and using a semi autonomous bot. This simulation has been implemented using Simio, which is a discrete event simulation software. It has pre built, custom "intelligent objects" to represent the different system components. The 3D animated models help in analyzing and optimizing complex systems with an additional advantage of visual representations. This chapter covers the detailed approach, results and inferences of both phase simulations.

The key parameters of this comparison analysis are Number In System, Time In System and Throughput, which are statistical output metrics that are generated from model runs. The experiments were run on a confidence level of 95 percentage. This 95 percent confidence level means that if the same sampling procedure is repeated many times, 95 percent of the confidence intervals calculated would contain the true population parameter. For a single study, it means that there can be 95 percentage confidence that the interval calculated contains the population mean. The average Number In System is a key performance metric that tracks the total number of entities within a defined system at any given time. It measures the total number of entities that are being processed, waiting in queues, or traveling through the system. The average Time In System is another important performance metric that measures how long, on average, it takes for an entity to move through the entire system. It measures the total time elapsed for an item, customer, or entity from its arrival until its exit from the system. This includes time spent waiting, traveling, and being processed. The throughput is also a performance metric representing the rate at which a system processes entities and it can be improved through simulation and optimization by analyzing various scenarios. In this analysis throughput is evaluated by the difference between the total number of entities created and destroyed in each experiment. These data provide a central tendency for the metric over the course of the simulation.

The post deployment simulation is based on a production line simulated using actual data obtained from "Performance optimization of a pharmaceutical production line by integrated simulation and data envelopment analysis" (Khalili et al., 2020, published in the International Journal of Simulation and Process Modelling). The Core Objective of this paper is to develop a hybrid decision-support framework that combines Discrete Event Simulation (DES) and Data Envelopment Analysis (DEA) to identify and resolve bottlenecks and optimize resource allocation in a complex pharmaceutical manufacturing environment.

The pharmaceutical industry faces high pressure to maintain quality while increasing throughput and reducing costs. Pharmaceutical production lines are characterized by High levels of stochasticity, Complex multi-stage processes, Strict regulatory constraints and The challenge of "multi-objective" optimization. The authors propose a four-stage methodology that bridges the gap between raw data and strategic decision-making including Data Collection and System Analysis, Discrete Event Simulation Modeling with validation and scenario generation, Data Envelopment Analysis using Decision Making Units and inputs/outputs and Optimization and Ranking. The Key Findings include Bottleneck Identification, Performance Gains and Efficiency Scores. Unlike static mathematical models, the DES component captures the randomness of the pharmaceutical floor. The integration of DEA solves the common simulation problem of "data overload." It provides a clear, ranked hierarchy of which scenario is best based on multiple KPIs. The framework is modular and can be applied to various types of manufacturing beyond pharmaceuticals. The accuracy of the DEA-Simulation output is entirely dependent on the quality of the input data. Inaccurate data leads to "Garbage In, Garbage Out." Developing and running dozens of simulation iterations to feed into a DEA model requires significant time and specialized software expertise.

7.1 Pre Deployment Simulation

This simulation involves a comparative analysis using 2 experiments on the same scenario. The scenario assumed is of a facility where inventory arrives in boxes weighing 1Kg each. Each box is an entity and there are a total of 10 boxes in this scenario. These boxes are to be moved from its point of arrival to a shelf at a distance of 50m. In the first experiment the boxes are moved manually one by one. In the second experiment the bot moves the boxes. The bot is assumed to be faster than the man and capable of carrying more weight than the man. Here it is taken as the bot can carry 5 boxes in a single go. The placement of boxes at starting and ending nodes are done manually in both cases. This is assumed to take 3 seconds at each node for both cases. The speed of the worker and bot area is assumed to be 1m/s and 1.5m/s respectively. This model is run for 2 hours and the results are collected. The results of the simulation will give precise informatics to map and scale the bot's parametrics so as to derive maximum operational efficiency.

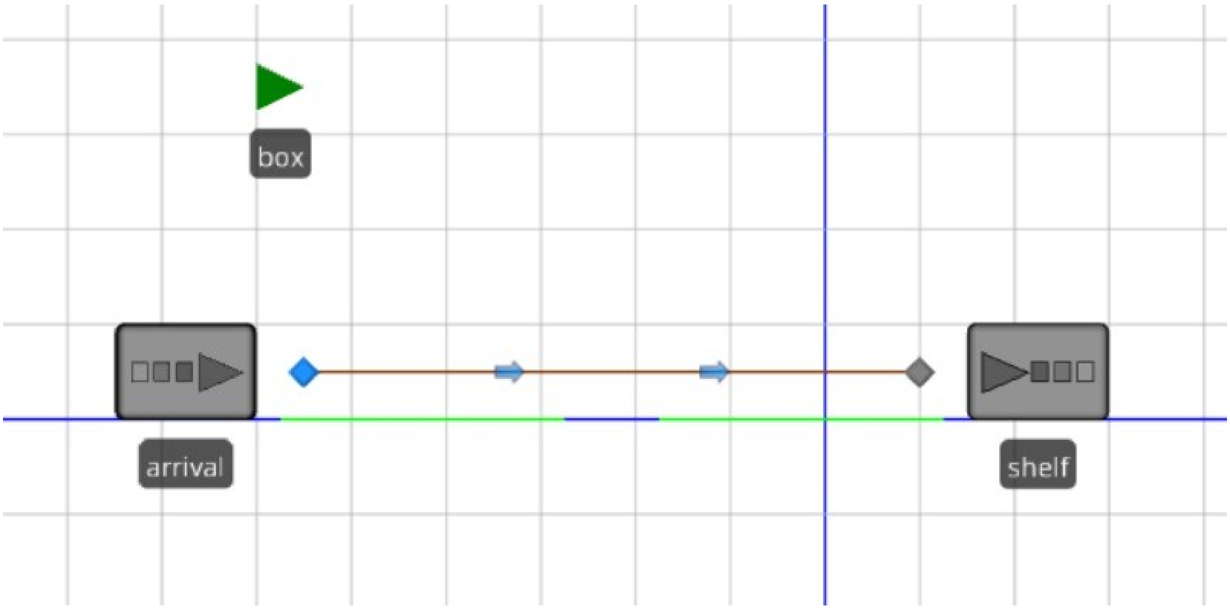


Figure 7.1: A Simple Production Line Layout

The default entity is the box and its parameters and properties are defined according to each experiment. Initial number in system is taken as 1 for and maximum number of entities is at 10. A time path is used to connect the start and end nodes, the property of which is changed according to the assumed speeds in each case. The placement of the box from the truck and into the shelf are taken as delays and are unchanged in both experiments.

7.1.1 Results And Inferences Of Pre Deployment Simulation

The simulation results are obtained in tabular format in Simio itself. The first experiment was run for the manual material handling scenario. The second experiment was run after modifying the facility for the semi autonomous material handling scenario.

Average

Object Type	Object Name	Data Source	Category	Data Item	Statistic	Average Total
ModelEntity	DefaultEntity	[Population]	Content	NumberInSystem	Average	4.5368
					Maximum	13.0000
			FlowTime	TimeInSystem	Average (Hou...	0.0148
					Maximum (Ho...	0.0170
					Minimum (Hou...	0.0147
					Observations	5,724.0000
			Throughput	NumberCreated	Total	5,731.0000
				NumberDestroyed	Total	5,724.0000
Sink	Sink1	[DestroyedEntities]	FlowTime	TimeInSystem	Average (Hou...	0.0148
					Maximum (Ho...	0.0170
					Minimum (Hou...	0.0147
					Observations	5,724.0000
		InputBuffer	Content	NumberInStation	Average	0.1987
					Maximum	2.0000
			EntryQueue	NumberWaiting	Average	0.0242
					Maximum	3.0000
				TimeWaiting	Average (Hou...	0.0001
					Maximum (Ho...	0.0022
					Minimum (Hou...	0.0000
			HoldingTime	TimeInStation	Average (Hou...	0.0008
					Maximum (Ho...	0.0008
					Minimum (Hou...	0.0008
			Throughput	NumberEntered	Total	5,724.0000
				NumberExited	Total	5,724.0000
Source	Source1	OutputBuffer	Throughput	NumberEntered	Total	5,730.0000
				NumberExited	Total	5,730.0000
		Processing	Throughput	NumberEntered	Total	5,730.0000
				NumberExited	Total	5,730.0000

Figure 7.2: Results of Experiment 1.1

TimePath	TimePath1	[Travelers]	Content	NumberAccumulated	Average	0.2236
					Maximum	4.0000
				NumberOnLink	Average	3.5368
					Maximum	12.0000
			FlowTime	TimeOnLink	Average (Hou...	0.0148
					Maximum (Ho...	0.0170
					Minimum (Hou...	0.0147
			Throughput	NumberEntered	Total	5,730.0000
				NumberExited	Total	5,724.0000

Figure 7.3: Results of Experiment 1.2

Object Type	Object Name	Data Source	Category	Data Item	Statistic	Average Total
ModelEntity	DefaultEntity	[Population]	Content	NumberInSystem	Average	7.4323
					Maximum	15.0000
			FlowTime	TimeInSystem	Average (Hou...	0.0102
					Maximum (Ho...	0.0123
					Minimum (Hou...	0.0101
					Observations	5,726.0000
			Throughput	NumberCreated	Total	5,735.0000
				NumberDestroyed	Total	5,726.0000
Sink	Sink1	[DestroyedEntities]	FlowTime	TimeInSystem	Average (Hou...	0.0102
					Maximum (Ho...	0.0123
					Minimum (Hou...	0.0101
					Observations	5,726.0000
		InputBuffer	Content	NumberInStation	Average	0.1988
					Maximum	2.0000
			EntryQueue	NumberWaiting	Average	0.0242
					Maximum	3.0000
				TimeWaiting	Average (Hou...	0.0001
					Maximum (Ho...	0.0022
					Minimum (Hou...	0.0000
			HoldingTime	TimeInStation	Average (Hou...	0.0008
					Maximum (Ho...	0.0008
					Minimum (Hou...	0.0008
			Throughput	NumberEntered	Total	5,726.0000
				NumberExited	Total	5,726.0000
Source	Source1	OutputBuffer	Throughput	NumberEntered	Total	5,730.0000
				NumberExited	Total	5,730.0000
		Processing	Throughput	NumberEntered	Total	5,730.0000
				NumberExited	Total	5,730.0000

Figure 7.4: Results of Experiment 2.1

				NumberExited	Total	5,730.0000
TimePath	TimePath1	[Travelers]	Content	NumberAccumulated	Average	0.2230
					Maximum	4.0000
				NumberOnLink	Average	2.4323
					Maximum	10.0000
			FlowTime	TimeOnLink	Average (Hou...	0.0102
					Maximum (Ho...	0.0123
					Minimum (Hou...	0.0101
			Throughput	NumberEntered	Total	5,730.0000
				NumberExited	Total	5,726.0000

Figure 7.5: Results of Experiment 2.2

The values obtained for manual material handling scenarios are as follows. The average values are considered for optimum decision making. The value of the number in system is 4.5368 and time in system is 0.0148 hours. The number of entities created and destroyed in this run are 5731 and 5724 respectively. Thus the throughput being the difference of these values is 7 entities. The values obtained for the semi automated material handling scenarios are as follows. The value of the number in system is 7.4323 and the time in system is 0.0102 hours. The number of entities created and destroyed in this run are 5735 and 5726 respectively. Thus the throughput being the difference of these values is 9 entities.

The overall changes in number in system, time in system and throughput are computed as percentages to compare the efficiency brought about by the bot in manual material handling scenario. An increase in number in system was observed by 66 percent by using the bot. The time in system reduced by 30 percent in the semi autonomous system. The throughput also saw a rise by 2 entities. Thus the simulation results have proved that a minute rise in speed and capacity of the bot has brought about drastic changes in the existing system. The rise in the number in system shows higher processing capacity of the bot as it is able to move more materials than that which can be handled manually. The time in system variations shows the overall time saving and faster handling of materials by the bot. The throughput variation shows how the bot processes more entities than done manually. These inferences assist in concluding that the bot's capability improves the existing system by raising the productivity by faster and efficient processing of material handling.

7.2 Post Deployment Simulation

This facility taken for this simulation is of an actual Pharmaceutical facility. The job line has 9 elements in addition to the entity. A batch is taken as a single entity. The model is run twice on manual material handling specifications and that of the semi autonomous bot. The different elements are the source, sink and 7 servers representing the different production processes. The data used for creating the Simio model is given below.

Table 7.1: Input Data From a Pharmaceutical Production Line

Seq.	Activity / Process Stop	Processing Time	Distance (m)	Movement Method
1	Dispensing & Weighing	30, 45, 60 mins	20	IBC / Pallet
2	Initial Blending, Wet/Dry Granulation, Fluid Bed Drying	30, 45, 65 mins	15	Gravity/Vacuum
3	Sizing & Milling	15, 20, 30 mins	15	IBC
4	Final Lubrication/Blending	10, 15, 25 mins	30	Forklift/AGV
5	Tablet Compression	120, 240, 360 min	20	IBC / Drums
6	Tablet Coating	90, 150, 210 mins	50	Conveyor/Cart
7	Blister/Bottle Packaging	60, 120, 180 mins	15	Conveyor
8	Finished Goods / QA	15, 30, 45 mins	0	Pallet Jack

The specification of manual material handling was obtained from analysing historical data and taking readings of moving a load over 6 meters. The average speed was obtained to be 0.9m/s. The speed of the bot was calculated from the time taken for 20 revolutions of the wheel. The results obtained were that the speed of the bot is directly proportional to the voltage applied, hence for an 80V the bot runs at a speed of 2m/s. A time path is used to connect the start and end nodes, the property of which is changed according to the assumed speeds in each case.

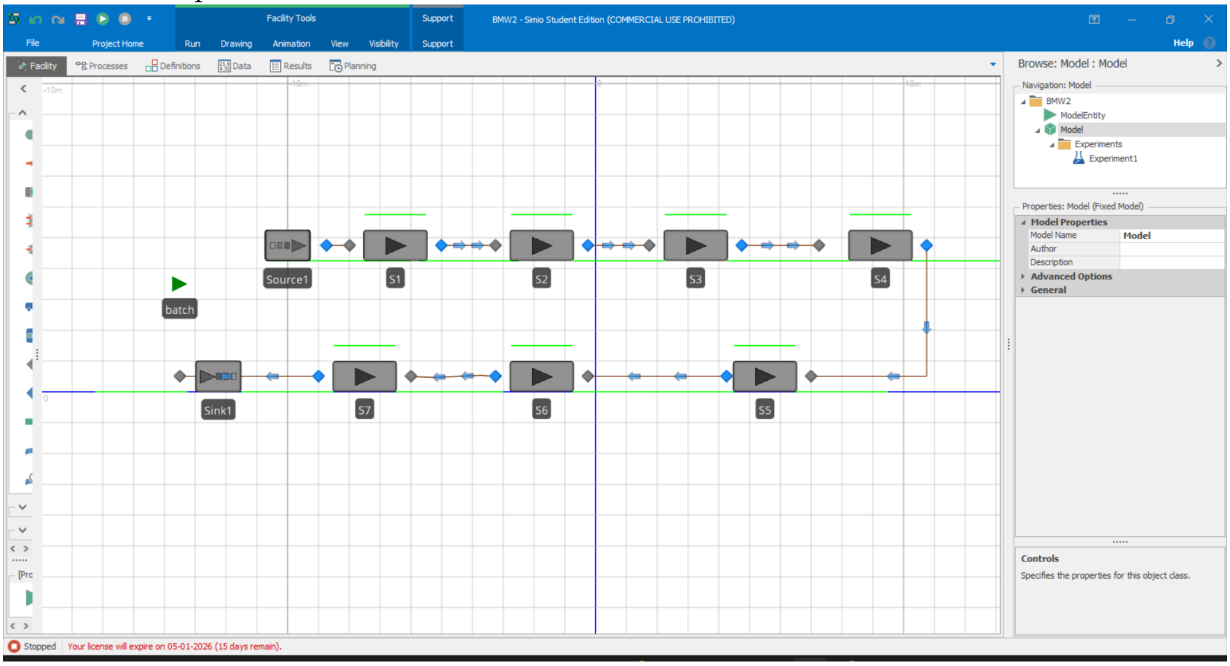
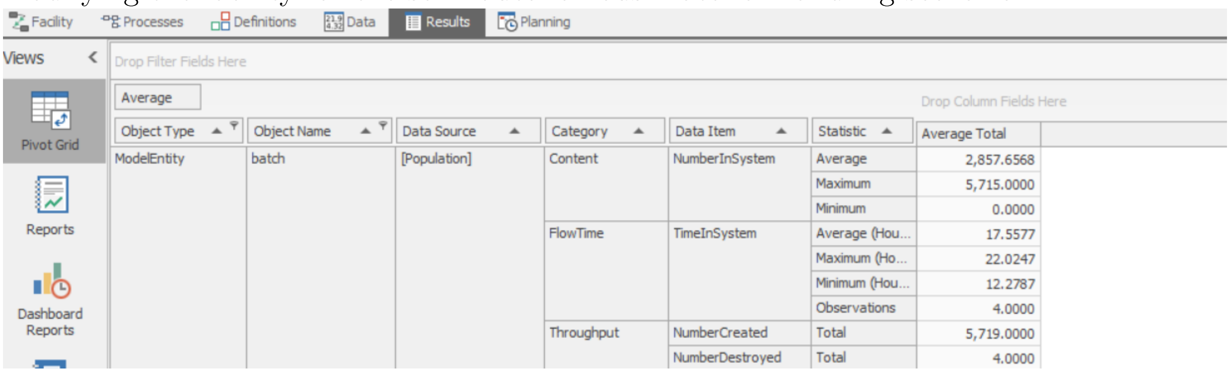


Figure 7.6: A Pharmaceutical Production Line Layout

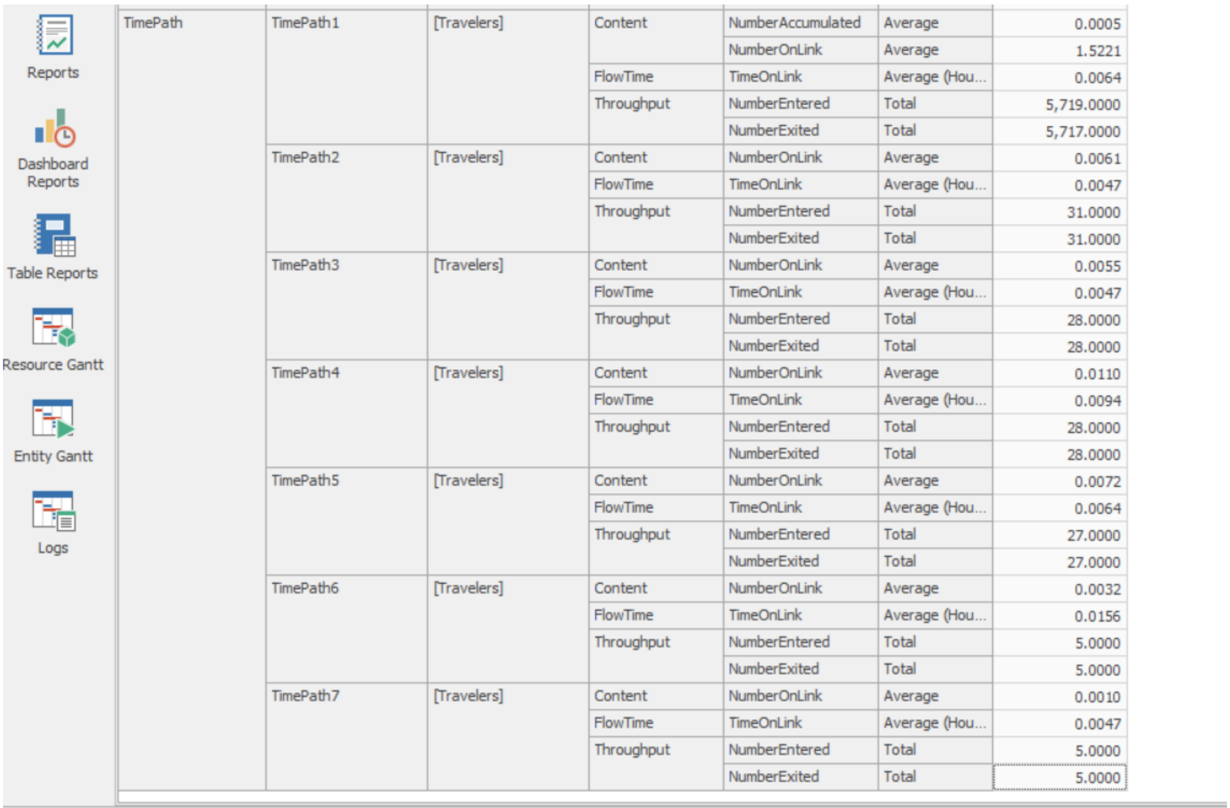
7.2.1 Results And Inferences Of Post Deployment Simulation

The simulation results are obtained in tabular format in Simio itself. The first experiment was run for the manual material handling scenario. The second experiment was run after modifying the facility for the semi autonomous material handling scenario.



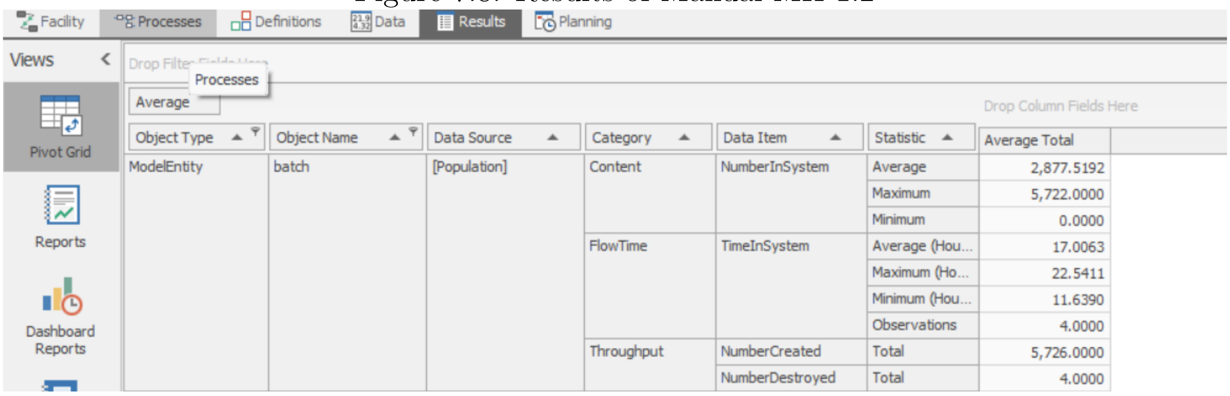
Object Type	Object Name	Data Source	Category	Data Item	Statistic	Average Total
ModelEntity	batch	[Population]	Content	NumberInSystem	Average	2,857.6568
					Maximum	5,715.0000
					Minimum	0.0000
			FlowTime	TimeInSystem	Average (Hou...	17.5577
					Maximum (Ho...	22.0247
					Minimum (Hou...	12.2787
					Observations	4.0000
			Throughput	NumberCreated	Total	5,719.0000
				NumberDestroyed	Total	4.0000

Figure 7.7: Results of Manual MH 1.1



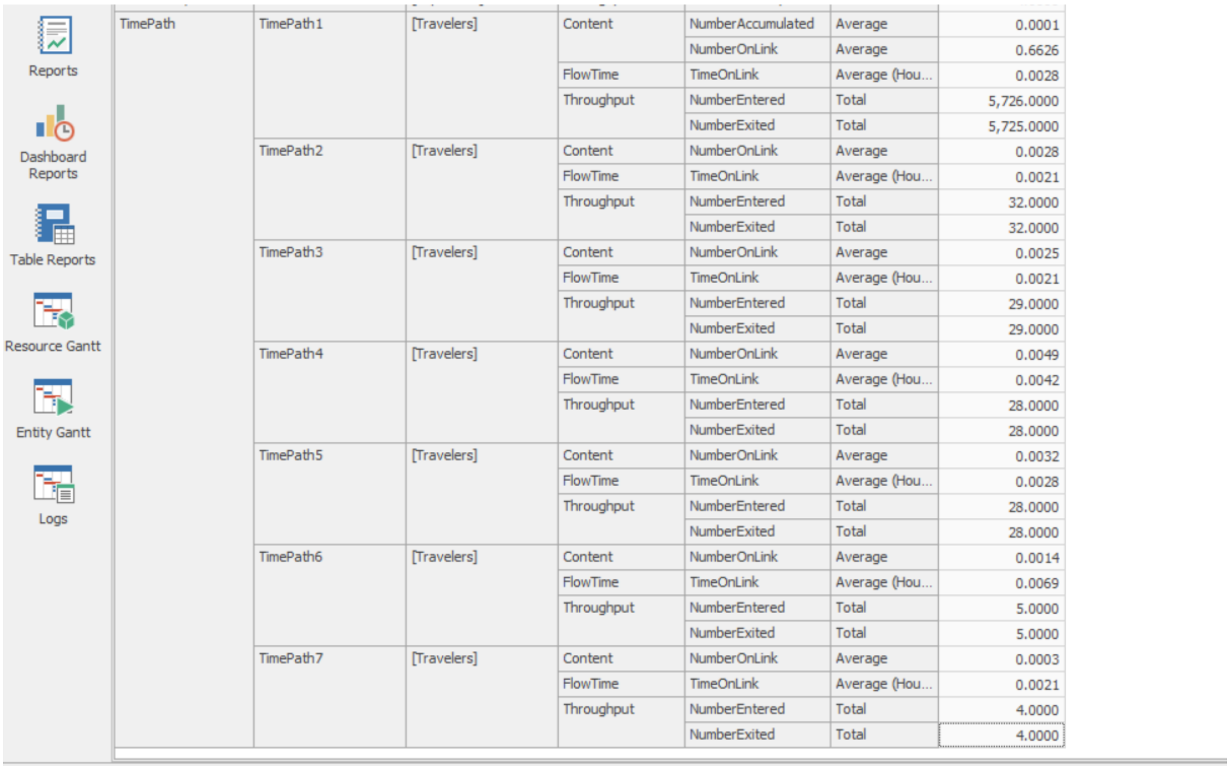
TimePath	TimePath1	[Travelers]	Content	NumberAccumulated	Average	0.0005
				NumberOnLink	Average	1.5221
			FlowTime	TimeOnLink	Average (Hou...	0.0064
			Throughput	NumberEntered	Total	5,719.0000
				NumberExited	Total	5,717.0000
	TimePath2	[Travelers]	Content	NumberOnLink	Average	0.0061
			FlowTime	TimeOnLink	Average (Hou...	0.0047
			Throughput	NumberEntered	Total	31.0000
				NumberExited	Total	31.0000
	TimePath3	[Travelers]	Content	NumberOnLink	Average	0.0055
			FlowTime	TimeOnLink	Average (Hou...	0.0047
			Throughput	NumberEntered	Total	28.0000
				NumberExited	Total	28.0000
	TimePath4	[Travelers]	Content	NumberOnLink	Average	0.0110
			FlowTime	TimeOnLink	Average (Hou...	0.0094
			Throughput	NumberEntered	Total	28.0000
				NumberExited	Total	28.0000
	TimePath5	[Travelers]	Content	NumberOnLink	Average	0.0072
			FlowTime	TimeOnLink	Average (Hou...	0.0064
			Throughput	NumberEntered	Total	27.0000
				NumberExited	Total	27.0000
	TimePath6	[Travelers]	Content	NumberOnLink	Average	0.0032
			FlowTime	TimeOnLink	Average (Hou...	0.0156
			Throughput	NumberEntered	Total	5.0000
				NumberExited	Total	5.0000
	TimePath7	[Travelers]	Content	NumberOnLink	Average	0.0010
			FlowTime	TimeOnLink	Average (Hou...	0.0047
			Throughput	NumberEntered	Total	5.0000
				NumberExited	Total	5.0000

Figure 7.8: Results of Manual MH 1.2



Object Type	Object Name	Data Source	Category	Data Item	Statistic	Average Total
ModelEntity	batch	[Population]	Content	NumberInSystem	Average	2,877.5192
					Maximum	5,722.0000
					Minimum	0.0000
			FlowTime	TimeInSystem	Average (Hou...	17.0063
					Maximum (Ho...	22.5411
					Minimum (Hou...	11.6390
					Observations	4.0000
			Throughput	NumberCreated	Total	5,726.0000
				NumberDestroyed	Total	4.0000

Figure 7.9: Results of Semi Autonomous MH 2.1



TimePath	TimePath1	[Travelers]	Content	NumberAccumulated	Average	0.0001
				NumberOnLink	Average	0.6626
			FlowTime	TimeOnLink	Average (Hou...	0.0028
			Throughput	NumberEntered	Total	5,726.0000
				NumberExited	Total	5,725.0000
	TimePath2	[Travelers]	Content	NumberOnLink	Average	0.0028
			FlowTime	TimeOnLink	Average (Hou...	0.0021
			Throughput	NumberEntered	Total	32.0000
				NumberExited	Total	32.0000
	TimePath3	[Travelers]	Content	NumberOnLink	Average	0.0025
			FlowTime	TimeOnLink	Average (Hou...	0.0021
			Throughput	NumberEntered	Total	29.0000
				NumberExited	Total	29.0000
	TimePath4	[Travelers]	Content	NumberOnLink	Average	0.0049
			FlowTime	TimeOnLink	Average (Hou...	0.0042
			Throughput	NumberEntered	Total	28.0000
				NumberExited	Total	28.0000
	TimePath5	[Travelers]	Content	NumberOnLink	Average	0.0032
			FlowTime	TimeOnLink	Average (Hou...	0.0028
			Throughput	NumberEntered	Total	28.0000
				NumberExited	Total	28.0000
	TimePath6	[Travelers]	Content	NumberOnLink	Average	0.0014
			FlowTime	TimeOnLink	Average (Hou...	0.0069
			Throughput	NumberEntered	Total	5.0000
				NumberExited	Total	5.0000
	TimePath7	[Travelers]	Content	NumberOnLink	Average	0.0003
			FlowTime	TimeOnLink	Average (Hou...	0.0021
			Throughput	NumberEntered	Total	4.0000
				NumberExited	Total	4.0000

Figure 7.10: Results of Semi Autonomous MH 2.2

The values obtained for manual material handling scenarios are as follows. The average values are considered for optimum decision making. The value of the number in system is 2857.6568 and time in system is 17.5577 hours. The number of entities created in this run are 5719. The values obtained for the semi automated material handling scenarios are as follows. The value of the number in system is 2877.5192 and the time in system is 17.0063 hours. The number of entities created in this run are 5726. The number of entities destroyed in both runs are 4. The throughput obtained in both runs are 5715 and 5722 for manual material handling and bot based material handling respectively.

The overall changes in number in system, time in system and throughput are computed as percentages to compare the efficiency brought about by the bot in manual material handling scenario. An increase in number in system was observed by 69 percent by using the bot. The time in system reduced by 33 minutes per batch in the semi autonomous system. The throughput also saw a rises by 7 entities. Thus the simulation results have proved that the rise in speed and capacity of the bot has brought about drastic changes in the existing system. The rise in the number in system shows higher processing capacity of the bot as it is able to move more materials than that which can be handled manually. The time in system variations shows the overall time saving and faster handling of materials by the bot. The throughput variation shows how the bot processes more entities than done manually. These inferences assist in concluding that the bot's capability improves the existing system by raising the productivity by faster and efficient processing of material handling. The bot thus has been proved capable of handling double the capacity at half the total time invested.

Chapter 8

DESIGN

Autodesk Fusion (formerly Fusion 360) is a commercial computer-aided design (CAD), computer-aided manufacturing (CAM), computer-aided engineering (CAE) and printed circuit board (PCB) design software application, developed by Autodesk. It is available for Windows, macOS and web browsers, with simplified view-only applications available for Android and iOS. Fusion is licensed as a paid subscription, with a free limited home-based, non-commercial personal edition available. Fusion has built-in capabilities for 3D modeling, collaboration, simulation and documentation. It can manage manufacturing processes such as machining, milling, turning and additive manufacturing. It also has electronic design automation (EDA) features, such as schematic design, PCB design and component management. It can be also used for rendering, animation, generative design and a number of advanced simulation tasks (FEA). Fusion uses a number of cloud-based AI functions, e.g. for drawing views, automated modelling, generative design, sketch constraints. To design a gesture-controlled robot for material handling, we decided to adopt the CAD software Fusion 360. The reason why we chose Fusion 360 as our main CAD software is multifaceted- 1. Fusion 360's main advantages are its unified, cloud-based platform that combines CAD, CAM, CAE and PCB design. 2. It also features a user-friendly interface and includes powerful tools like parametric modeling, simulation, and generative design. 3. The software is free for students, educators, and startups (non-commercial use). We faced a fundamental challenge as our current curriculum focused on mainly on Simulation software's like Vensim and Simio, thus we had to learn the fundamentals of CAD design before designing our Gesture Controlled Robot. To learn Fusion 360, we browsed through CAD literature available in our College Library, Watched YouTube Tutorials on Fusion 360. To get a comprehensive understanding of the tools and properties of the software, we referred multiple YouTube Tutorials that explained the design of a Lego Brick, Beer Bottle, Paperclip, Ice cube Tray, Door Grip, Hex Nut, Light Bulb, Phone Case and a Robotic Arm Gripper. We spent a significant part of our design phase learning to model simple daily life objects in Fusion 360, slowly leading to a better understanding of the Software and its Features. This approach has led to a much greater appreciation of the Tool that is Fusion 360 that we have learned as part of our Project. We continue to use Fusion 360 to develop our gesture-controlled robot.

8.1 Scissor Lift Design

Figure 8.1: A Hydraulic Scissor Lift System [16]

The Semi-Autonomous Robotic Lift represents a significant advancement in the realm of material handling technology, specifically as a scissor lift robot. Scissor lifts have long been recognized as robust devices employed across various industries, encompassing tasks in cleaning, maintenance, and construction. These machines, characterized by their ability to efficiently transport heavy materials, excel in distributing force among their scissor legs, minimizing the likelihood of structural deformation when subjected to heavy loads. Traditionally, scissor lifts have been designed with the capacity to carry substantial weights, including human occupants. However, the Semi-Autonomous Robotic Lift departs from this conventional application, focusing on smaller-scale utility within indoor industrial or warehouse environments. Tailored to assist workers in transporting heavy materials, the robot redefines heavy material transports underscoring its dedication to enhancing efficiency and reducing manual labor.

The design of a gesture-controlled service robot tailored for the pharmaceutical "Fill and Finish" environment represents a sophisticated integration of mechanical stability, precise electronic control, and intuitive human-machine interfacing. To move beyond the manual baseline of high-latency material handling, the physical architecture of the bot must be engineered to balance payload capacity with cleanroom agility. The structural foundation begins with the Base Platform, which is specified at a length of 35–45 cm and a width of 25–30 cm. This footprint is a deliberate design choice; it is narrow enough to navigate the compact 165-meter U-flow production lines common in Active Pharmaceutical Ingredient (API) facilities, yet wide enough to maintain a low center of gravity when the 15 kg load is elevated.

The chassis serves as the skeletal core of the bot, requiring materials that offer a high strength-to-weight ratio while adhering to pharmaceutical sterility standards. For the Base Chassis, the choice between an 8 mm Acrylic sheet and a 3 mm Aluminium sheet involves a trade-off between ease of prototyping and industrial durability. While acrylic provides excellent visibility for debugging electronic layouts, Aluminium 6061 is the preferred baseline for a "Fill and Finish" scenario. Aluminium's non-shedding properties and resistance to chemical cleaning agents like Isopropyl Alcohol (IPA) make it superior for contamination control. The Frame and Scissor Lift Structure further refine this material logic. Utilizing Mild Steel (MS) square tubes (10$\times$10 mm) for the load-bearing joints provides the rigidity necessary to prevent "structural sway" during high-velocity transit 2m/s However, to ensure the motors are not over-strained by the weight of the bot itself, Aluminium L-angles are integrated into the non-load-bearing segments. This hybrid approach ensures that the total bot mass remains near the 10 kg baseline, allowing the torque-dense DC geared motors to focus their energy on moving the payload rather than the machine's dead weight.

A defining feature of this design is the 2-stage Scissor Lift, which addresses the ergonomic ”shelf-height” bottleneck. The lift is designed with 20×20 cm flat bars, typically made of Aluminium flat bars (3–5 mm thickness) to minimize inertia. The mechanical advantage of the scissor lift allows for a compact minimum height of 8–10 cm, which can expand to a maximum height of 30–40 cm. This range is calibrated to align with standard pharmaceutical dispensing counters and storage racks, effectively neutralizing the vertical ”Physical Demand” on the operator.

To drive this vertical movement, the design specifies a Lead Screw (threaded rod) coupled with a high-torque DC geared motor. The lead screw 8-12mm provides a ”mechanical lock” feature; unlike a belt drive, a lead screw will not back-drive or drop the load if power is lost, which is critical when transporting high-value APIs. This ensures Smooth Lifting and precise control, allowing the bot to stop exactly at the height dictated by the operator's upward hand gesture.

The transition from manual pushing to robotic propulsion is managed by the 4-wheel drive system. With a wheel diameter of 6–10 cm 7-3.5 cm radius is optimal for torque), the bot utilizes rubber-treaded wheels to maximize traction on polished epoxy cleanroom floors. The drive is powered by $2 \times$ DC geared motors (100–300 RPM). As established in the torque requirements, these motors must overcome the static friction of a 25 kg total mass (Bot + Load). By using L298N motor drivers or more efficient H-bridge controllers, the system can achieve the target velocity of 1.5-2m/s identified in the productivity analysis. This consistent speed is what allows the bot to slash the transit time on Path 6 by over 55 percent, transforming the facility's logistics from a variable human-paced flow to a constant, machine-optimized rhythm.

The ”intelligence” of the bot resides in its Gesture Control System, which facilitates the Human-Robot Collaboration (HRC) framework. The brain of the unit is an ESP32 microcontroller, chosen for its dual-core processing and integrated ESP-NOW capabilities. ESP-NOW is the technical lynchpin that reduces communication latency to ¡20 ms, ensuring the bot responds to hand movements in real-time.

The input is captured via an MPU6050 sensor (IMU) mounted on a wearable glove. This sensor tracks the operator's hand tilt and acceleration, translating XYZ-axis data into specific commands: Forward/Backward: Pitch adjustment of the hand, Lift/Lower: Z-axis acceleration or specific tilt thresholds, Stop: Neutral hand position (Gesture 0). This "multimodal interaction" is further augmented by Ultrasonic sensors for collision avoidance and a 12V Li-ion battery for high-density power delivery. By combining these electronic components with the robust mechanical frame, the design fulfills the Industry 5.0 promise of a collaborative assistant that is ergonomically superior to manual labor and technologically resilient enough for the rigorous demands of pharmaceutical manufacturing.

The scissor lift mechanism adds a second degree of freedom to the prototype by moving up and down in addition to the forward, backward, left and right turns. This enables better placement of goods on racks and shelves located at a height. It can be scaled according to the physical parameters of the specific facility. It also increments the load carrying capacity of the bot by virtue of the laws of mechanics.

Table 8.1: Mechanical Principles of the Lifting System

Feature	**Mechanical Law / Principle**	**Practical Impact**
Start-up Force	Statics / Trigonometry	Requires high torque/force to begin lifting from the flat position.
Height Change	Kinematics ($L \cdot \sin(x)$)	Vertical speed increases as the lift gets higher for the same actuator speed.
Stability	Center of Gravity (CoG)	The load must be centered to prevent moments (torque) that could bend the arms.
Power Needs	Work-Energy Principle	Faster lifting requires more instantaneous power from your ESP32 controlled motor.

Moving vertically, the integration of the scissor lift mechanism adds a layer of mechanical and electrical complexity. This mechanism is typically powered by a lead screw or a linear actuator driven by an additional DC motor. In this hardware stack, the lift motor can be controlled by a second L298 driver or the unused channels of the primary driver, depending on the current rating. The mechanical challenge in this integration is ensuring that the center of gravity remains stable as the lift extends. From an electrical standpoint, the lift requires its own set of limit switches or a feedback loop to prevent over-extension, which could lead to mechanical binding or motor burnout. These limit switches are integrated into the ESP32's digital input pins, using pull-up resistors to ensure the signals remain noise-free in the electromagnetically "noisy" environment created by the high-torque DC motors.

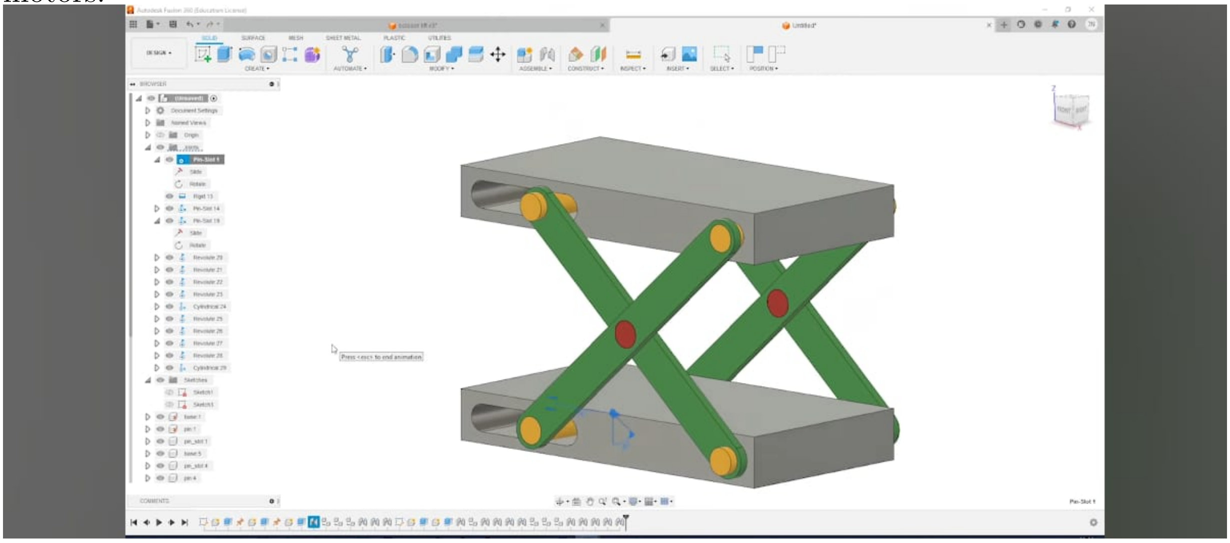


Figure 8.2: Scissor Lift Design in Fusion 360

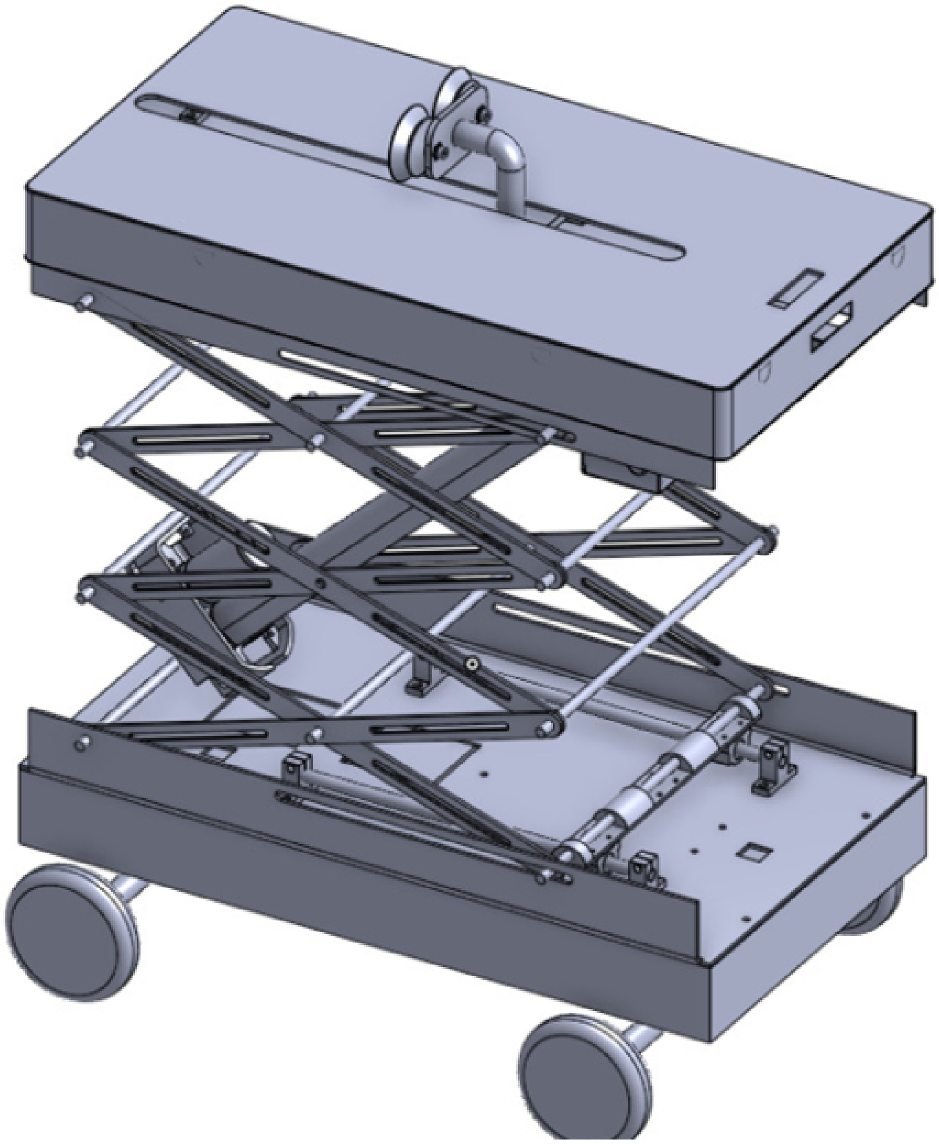


Figure 8.1: Figure 8.3: Prototype design in Solidworks

A scissor lift functions as a pantograph linkage, converting horizontal input force into vertical displacement through trigonometry. Kinematically, the total height is defined by H=n*L*sin(x), where height increases non-linearly with the arm angle. Mechanically, the system faces its greatest challenge at the start; because mechanical advantage is lowest when the arms are horizontal, the required input force—governed by F=W/(2*tan(x)) is at its maximum. As the lift rises, the mechanical advantage improves, reducing motor torque requirements. Structurally, this creates peak shear stress on joint pins during the initial phase of elevation.

Chapter 9

QUALITY CONTROL STUDY

Quality is defined as the degree to which a set of inherent characteristics fulfills requirements. Degree means that quality can be used with adjectives such as poor, good, and excellent. Inherent is defined as existing in something, especially as a permanent characteristic. Characteristics can be quantitative or qualitative. Requirement is a need or expectation that is stated; generally implied by the organization, its customers, and other interested parties; or obligatory.

Here the Quality control analysis is done based on the dimensions of quality which are: Performance, Features, Conformance, Reliability, Durability, Service, Response, Aesthetics, Reputation. Except for Response all other parameters fall in a category called Garvin's Framework. David Garvin's framework, introduced in 1987, is the "Gold Standard" for understanding that quality is not a single, vague idea. It's a multidimensional spectrum that helps businesses and developers identify exactly where a product excels or needs work.

Performance refers to the primary product characteristics, Features are the additional or secondary characteristics, Conformance is meeting specifications and industry standards, Reliability is the consistency of performance over time, Durability is the useful time including repair, Service is the resolution of problems, complaints and the ease of repair, Response is efficacy of system interfaces, Aesthetics are the sensory characteristics, Reputation refers to the past performance and other intangibles.

Table 9.1: Quality Characteristics Analysis 1

Parameter	**Current Status**	**Improvement**
Performance	Speed: 1.5-2m/s 70-90% time reduction Economical	Consistent speed depending on each load
Features	Gesture Controlled Multimodal interaction Scissor lift	Ultrasonic sensors for collision avoidance
Conformance	High adherence to the 5 command gesture vocabulary	Implement a Calibration Routine to account for varying gesture intensities.
Reliability	Safety Lock Backup power source	Watchdog timer to auto-reset on firmware hangs
Durability	High shear stress at scissor lift pins during the lift phase.	Standard pivot pins can be replaced with high-tensile steel bolts to prevent material fatigue.
Serviceability	Modular components allow for quick part replacement.	Transition from a breadboard to a Custom PCB to eliminate "loose wire" failure points.

Table 9.2: Quality Characteristics Analysis 2

Parameter	**Current Status**	**Improvement**
Responsiveness	Loop delay of 50ms ensures the bot responds to gestures in near real-time	ESP-NOW protocol reduces latency to 20ms
Aesthetics	Functional industrial look, open-frame chassis and exposed jumpers.	A 3D-printed enclosure for the wearable unit to improve cable management
Reputation	High innovation value as an Industry 5.0 project focused on human centric ergonomic assistance and reduction of job replacement.	Document user feedback to build a “User-Trusted” project profile

Performance is the primary operational measure of the bot’s utility. The system achieves a significant travel speed of 1.5–2 m/s, which simulations suggest can lead to a 70–90 percent reduction in material handling time. This efficiency is the core value proposition of a warehouse environment. However, high-quality performance must be consistent regardless of the payload. While the bot may hit 2 m/s empty, the real test is maintaining that velocity when the scissor lift is fully loaded. Implementing a dynamic Pulse Width Modulation adjustment in the ESP32 code will ensure that the motor output compensates for weight, providing a smooth, predictable user experience that remains economical in power consumption.

Features are the secondary characteristics that enhance the bot’s basic function. The bot stands out due to its multimodal interaction ability to interpret kinetic gestures while simultaneously using Bluetooth RSSI for spatial tracking. This dual layered control system is rare in low-cost industrial solutions. The addition of the scissor lift mechanism provides vertical versatility, turning a simple transporter into a multi-level handling tool. To reach Gold Standard quality, a new step to involve integrating Ultrasonic sensors can be undertaken. This feature adds a layer of autonomous intelligence, allowing the bot to override manual gesture commands if it detects an obstacle, thereby preventing collisions and protecting both the inventory and the warehouse staff.

Conformance measures how closely the product adheres to established internal and external standards. The system currently shows high adherence to a 5-command gesture vocabulary, ensuring that specific hand tilts result in predictable motor actions. However, human movement is not standardized; "Forward" for one operator might be a 20° tilt, while another might naturally tilt 40°. High-conformance quality requires a Calibration Routine in the firmware. By allowing the system to "zero out" or define a neutral pose at the start of each session, the bot conforms to the unique physical range of the operator. This reduces "false positives" and ensures the control logic is always in sync with user intent.

Reliability is the probability that the bot will function without failure over a set period. In wireless robotics, the greatest reliability risk is signal drop or firmware "hangs." The Safety Lock feature is the first line of defense, ensuring the bot stops immediately if the Wi-Fi connection is severed. To improve this dimension, the "brain" of the bot must be addressed. Integrating a Watchdog Timer in the ESP32 code ensures that if the code enters an infinite loop or crashes, the hardware automatically reboots to a safe state. Coupled with a backup power source for the logic side, this ensures the bot remains operational even during voltage spikes or high-drain maneuvers.

Durability measures the physical lifespan of the bot under industrial stress. The most vulnerable points identified are the scissor lift pins, which endure high shear stress during the phase of elevation. Standard pivot pins often fail due to material fatigue after repeated cycles. Quality improvement here shifts from software to mechanical engineering: replacing standard pins with high-tensile steel bolts. This upgrade ensures the structural integrity of the lift mechanism over thousands of cycles. Additionally, the chassis should be evaluated for vibration resistance. Using lock-nuts and vibration-dampening mounts for the ESP32 will prevent the electronics from shaking loose during high-speed travel across warehouse floors.

Serviceability is the ease and speed with which the bot can be repaired. Using modular components like the ESP32, MPU6050, and L298N is a relevant choice for serviceability, as a single failed module can be swapped in minutes. However, the move from a breadboard to a Custom PCB is non-negotiable for industrial quality. Breadboards are prone to "loose wire" faults, which are difficult to diagnose in a busy work environment. A PCB with JST or screw-terminal connectors simplifies maintenance, allowing even non-technical staff to replace a motor driver or sensor without a soldering iron. This reduces downtime and makes the bot a "servicing-friendly" tool rather than a complex technical burden.

Responsiveness is the "feel" of the interaction, measured by latency. A 50ms loop delay provides a near real-time experience, but Wi-Fi protocols can sometimes introduce lag spikes. To reach peak responsiveness, the ESP-NOW protocol can be implemented. Unlike standard Wi-Fi, which requires a heavy handshake process, ESP-NOW is connectionless and can reduce latency to 20ms. This 30ms difference is highly perceptible to an operator; it's the difference between the bot "feeling" like it is following the hand and the bot feeling like a part of the body. Faster responsiveness directly improves safety, as the "Stop" command is processed almost instantaneously.

Aesthetics in an industrial setting are about "perceived professionalism" and safety. While the current "open-frame" look is great for debugging, exposed jumpers and green-boards need to be modified to get a professional outlook. A 3D-printed enclosure for both the wearable and the chassis is the solution. An enclosure protects the sensors from warehouse dust and prevents snagging on clothing or shelving. Proper cable management by hiding the wires inside the chassis gives the bot a sleek, finished appearance that signals to potential MSME clients that this is a mature, market-ready piece of technology.

Reputation is the Perceived Quality based on the project's mission. As an Industry 5.0 project, this project builds a reputation for human centric innovation. Unlike robots that replace workers, this bot is an assistant, focusing on ergonomic support and reducing musculoskeletal injuries. To solidify this reputation, user feedback documentation is mandatory. Building a "User-Trusted" profile involves recording testimonials from workers who feel less back pain or fatigue when using the bot. This qualitative data, combined with the quantitative Simio stats, builds a project reputation that stands for safety, social responsibility, and technical excellence.

The quality control study of the gesture-controlled bot demonstrates a robust technical foundation while highlighting critical paths for industrial hardening and market readiness. From a performance standpoint, the bot's ability to achieve travel speeds up to 2m/s coupled with a simulated 90 percent reduction in material handling time, positions it as a disruptive tool for MSME environments. This efficiency is significantly amplified by its distinctive feature set, specifically the synergy between kinetic gesture control via the MPU6050 and spatial Bluetooth RSSI tracking which creates a seamless digital tether between the human operator and the machine. The study does reveal a dichotomy between its advanced software logic and its mechanical maturity. While the conformance and responsiveness of the system are highly refined boasting near real-time latency that is set to drop to 20ms with the transition to ESP-NOW the physical dimensions of durability and aesthetics require focused reinforcement.

Specifically, the mechanical analysis of the scissor lift mechanism identified peak shear stress at the pivot joints during the initial lift phase, marking standard pins as a potential fail point that must be upgraded to high-tensile steel to ensure long term structural integrity. Furthermore, while the bot's serviceability is excellent due to its modular ESP32-based architecture, the current reliance on breadboard connections and an open-frame chassis is a liability in a dusty, high-vibration warehouse setting. Transitioning to a custom PCB and a 3D-printed protective enclosure is not merely an aesthetic choice but a requirement for reliability and user safety. The study also emphasizes that reliability must be bolstered through the implementation of a watchdog timer to prevent firmware hangs during continuous operation. From a reputational perspective, the bot's alignment with Industry 5.0 principles prioritizing ergonomic assistance and musculoskeletal health over traditional automation gives it a unique competitive edge in the MSME sector.

By validating these nine parameters, this study confirms that the project has successfully moved beyond a simple proof of concept. The improvements reinforcing the mechanical "weakest links," streamlining the communication protocols for even lower latency, and professionalizing the hardware housing are unavoidable. Ultimately, the bot is not just a functional material handler; it is a scientifically validated, human-centric ergonomic assistant that is poised to significantly lower the barrier to entry for advanced robotics in smaller-scale industrial facilities. This quality audit provides the necessary data to transition from laboratory testing to real-world pilot deployment with a high degree of confidence in the system's safety, efficiency, and operational return on investment.

Chapter 10

CASE STUDY

10.1 Introduction

Modern pharmaceutical manufacturing relies on tightly controlled material handling to move active pharmaceutical ingredients (APIs), components, and finished products between research, bulk formulation, aseptic filling, packaging, warehousing, and outbound logistics stages. In a typical aseptic fill-and-finish facility for biologics such as erythropoietin, materials pass through graded cleanrooms (Grades A–D), personnel and material airlocks, and segregated warehouse and production areas, with strict requirements on temperature, humidity, differential pressure, and particulate and microbial limits.In many small and medium pharmaceutical plants, internal material movement within these layouts still depends heavily on manual labour using trolleys, pallets, and forklifts, especially between weighing, blending, drying, compression, coating, and packaging stations. This manual handling is time-consuming, ergonomically stressful, and contributes to high transit components in overall lead time, particularly where workers repeatedly push or lift 10–15 kg loads over distances of more than 150 m inside controlled environments. The resulting fatigue increases the risk of human error, contamination incidents, and inconsistent throughput, which is critical because the finished product at this stage is in its most valuable state and any deviation can cause large economic losses.

To address these issues, this case study evaluates the deployment of a gesture-controlled, semi-autonomous material handling bot in a pharmaceutical production line modeled on a fill-and-finish facility, comparing it with manual handling in terms of productivity, logistics efficiency, ergonomics, and economic impact. The proposed solution draws on Industry 5.0 and Industrial Internet of Things (IIoT) concepts to keep humans in the loop while off-loading low-value, high-fatigue transport tasks to an intelligent robot that can move safely and efficiently inside a regulated pharmaceutical environment.

In the reference facility, inbound logistics bring in API, excipients, and components to segregated receiving areas, after which materials follow defined product, personnel, and waste flow paths to avoid cross-contamination. Good Manufacturing Practice (GMP) guidelines emphasise unidirectional flows, separation of raw materials, work-in-process, and finished goods, and the use of dedicated airlocks and staging zones, all of which make material handling routes long and complex. Manual handling in such layouts offers flexibility and low initial investment, because human operators can adapt to layout changes, manage exceptions, and perform inspection tasks simultaneously while moving materials. However, it suffers from low transit velocity (around 0.9 m/s observed in the simulated line), high physical demand, and large variation in performance due to fatigue and shift-to-shift differences. Manual push–pull of 10–15 kg loads up to 1.5 m height yields high Rapid Entire Body Assessment (REBA) risk scores around 8–10, indicating a significant probability of musculoskeletal disorders and longterm injury.

The advantages of more automated material handling, such as conveyor lines or fully autonomous guided vehicles (AGVs), include stable speeds, predictable travel times, and reduced ergonomic risk, but conventional industrial AGV systems are often capitalintensive, rigid, and better suited to large multinational corporations than micro, small, and medium enterprises (MSMEs). For MSME pharmaceutical plants operating under cost pressure, there is a gap for economical, human-centric automation that enhances safety and throughput without completely replacing operators or demanding extensive layout modifications.

The proposed intervention is a gesture-controlled material handling bot designed as a semi-autonomous platform with a scissor-lift mechanism for vertical handling, targeting MSME-scale pharmaceutical facilities rather than fully automated MNC plants. The bot uses an ESP32 microcontroller with integrated Bluetooth, a gyro sensor, and an L298 dual motor driver to execute a five-direction command set derived from the operator's hand gestures, enabling intuitive navigation without physical contact with the load. The scissor-lift design provides a second degree of freedom, allowing the bot to handle payloads up to around 15 kg at controlled lift heights suitable for feeding blenders, granulators, or conveyors within cleanroom constraints. For tracking and supervisory control, the design incorporates a multimodal interaction strategy using Received Signal Strength Indicator (RSSI) and a log-distance path-loss model to estimate distance between operator and bot, thereby enabling Bluetooth-based localisation and speed control. Line-following capabilities are planned for path synchronisation with the existing material routes, while ESP-NOW (a low-latency ESP32 protocol) is proposed to reduce communication delays in dense IIoT environments. Safetyoriented features such as collision avoidance, motor torque calibration to maintain constant speed under varying loads, and watchdog timers for load compensation are included to satisfy pharmaceutical safety and quality expectations. The bot is intended to operate along defined paths linking key unit operations in a tablet manufacturing line—dispensing and weighing, blending and granulation, fluid-bed drying, sizing, lubrication, compression, coating, and packaging—mirroring the handling tasks performed manually in the baseline scenario. In this configuration, the human operator remains responsible for loading and unloading, inspection, and quality-critical decisions, while the bot assumes the repetitive transit between stations, effectively decoupling manual effort from material travel time.

Baseline profiling of the simulated pharmaceutical production line with manual material handling showed that high physical demand, low transit velocity, and unproductive mechanical tasks significantly inflated the transit component of system lead time. Manual operators required around 17.56 hours of in-system time (Time in System, TIS) per batch, with a total flow time of approximately 0.0519 hours per entity, and throughput of 5715 entities over the simulation horizon. When the gesture-controlled bot was introduced in the same model with a transit speed of 2 m/s, the system achieved an increase in the number in system (NIS) processed to about 2878 entities, shortening TIS to roughly 17.01 hours and cutting total flow time to 0.023 hours per entity. These results correspond to around 69 additional batches processed, a reduction of about 33 minutes of lead time per batch, and a total flow time reduction of approximately 56.28 percent, all without changing process parameters at the workstations themselves. From an ergonomic perspective, replacing push–pull tasks and load lifting with gesturebased control yields a reduction in REBA scores from high-risk 8–10 for manual handling to low-risk levels around 2–3 when the bot manages the physical transport and lifting. Heart rate analysis in the case study suggests that manual handling of 10–15 kg loads over 165 m can elevate worker heart rates to 110–130 bpm, whereas operating the bot via gestures allows workers to remain near resting active states of 70–90 bpm, thereby preserving cognitive alertness for quality-critical tasks.

10.2 Contextual Definition And Baseline Profiling

The modern pharmaceutical landscape is currently undergoing a paradigm shift, transitioning from rigid, high-volume automated systems toward flexible, human-centric "service" environments that prioritize agility and precision. This transformation is most evident in the Active Pharmaceutical Ingredient (API) "Fill and Finish" sector, where the handling of high-value compounds necessitates a delicate balance between rigorous sterile constraints and the need for operational flexibility. In this context, the definition of the workspace is no longer confined to static conveyor belts but is instead a dynamic ecosystem where Human-Robot Collaboration (HRC) is essential. Under the Industry 5.0 framework, the objective is to delegate physically demanding and repetitive tasks—such as the movement of Intermediate Bulk Containers (IBCs)—to semi-autonomous service robots, thereby allowing human operators to focus on high-cognitive quality assurance and oversight. This shift is driven by the inherent risks of Manual Material Handling (MMH), which is frequently identified as a primary source of musculoskeletal injury and a significant bottleneck in production flow.

The "Fill and Finish" stage is the final and perhaps most critical phase of pharmaceutical manufacturing, where the drug product is placed into its primary packaging. This environment is defined by three primary constraints: Environmental Sensitivity, Operational Versatility, and Ergonomic Safety. Because APIs are highly susceptible to contamination, facilities must maintain strict ISO class standards. Traditional manual transport methods, where humans push heavy carts, increase the risk of particulate shedding near sterile zones. The introduction of gesture-controlled service robotics addresses this by incorporating an RSSI-based digital tether, which minimizes human proximity to the sterile product while maintaining a constant logistical flow.

Furthermore, the industry is moving toward "personalized medicine," which involves smaller, more frequent batches rather than the massive, monolithic runs of the past. This necessitates a "service" robotics approach—machines that are interactive and easily recalibrated via multimodal interfaces like gesture control. By mapping intuitive human hand movements to robotic actions via MPU kinetic sensors, the facility can maintain a hybrid workforce that is both flexible and highly productive. The ergonomic goal is clear: reduce the physical demand on the worker to negligible levels (lowering REBA scores from 10 to 2) while ensuring that the robot handles the heavy 15Kg payloads across long distances.

To understand the necessity of this intervention, we must analyze the "Baseline Profile" of the existing manual system, as captured in simulation data. The manual state represents a facility where operators use pallet jacks and manual carts to navigate an 8-stage production sequence.

The quantitative analysis of the manual system reveals a state of high congestion and significant lead-time delays. The Average Number in System stands at 2,857.66 batches, representing a massive accumulation of Work-In-Process (WIP). In an API facility, high WIP is a liability; it indicates that materials are sitting idle in corridors or staging areas rather than moving through the value stream. This bottleneck is further reflected in the Average Time in System , which reaches 17.5577 hours per batch. This total lead time is dominated by transit delays, where human operators struggle to maintain consistent speeds over the cumulative 165m production path. Perhaps most telling is the throughput data: out of 5,719 units created, the high latency of manual flow resulted in a finished unit count of nearly zero within the simulated timeframe, highlighting a system that is functionally "stalled" by manual movement.

A granular look at the path utilities confirms that manual labor is the primary driver of these inefficiencies. Path 1 (Dispensing to Blending) shows a utility of 1.5221, the highest in the entire facility. This suggests that the initial phase of the line is constantly congested with manual IBC handling. As the distance increases, the inefficiency of human labor becomes even more apparent. On Path 6, a 50m stretch from Coating to Packaging, manual transit takes 0.0156 hours. When compared to the bot-assisted speed of 0.0069 hours, it is evident that manual transit is more than 55 percent slower for long-distance hauls. The slow manual movement over paths like Path 4 (30m) also contributes to the "Time on Link" delays, creating a ripple effect that inflates the total flow time to the 17.55 hours mark.

The "Baseline" problem is a combination of high physical demand and low transit velocity. As established by the framework of Nwamekwe et al. (2025), forcing a human to perform "low-value" mechanical tasks like pushing a 15Kg cart over 50m is inherently unproductive and increases the risk of fatigue-led errors. To solve this, the gesture-controlled bot is designed with specific technical parameters: Velocity: A constant 1.5-2m/s travel speed, which eliminates the variability of human fatigue. Control: ESP-NOW low-latency communication combined with MPU kinetic sensors for instantaneous response to human gestures. Safety: A dual-layer safety system comprising ultrasonic collision avoidance and an RSSI-based digital tether. Mechanical Advantage: A motorized scissor lift designed to handle the 15Kg payload, neutralizing the vertical lifting strain on the operator.

By implementing these parameters, the project aims to reduce the total transit component of the lead time by over 50 percent, bringing the 17.55 hour flow time down to a more competitive level and resolving the 2,857 batch WIP crisis. The transition to this hybrid approach ensures that the "Fill and Finish" facility can meet the demands of Industry 5.0: a safe, ergonomic, and high-velocity production environment.

10.3 Technical Intervention And Calibration

The transition from a manual, high-latency pharmaceutical "Fill and Finish" facility to an optimized, Industry 5.0 human-robot collaborative environment is not merely a hardware upgrade; it is a sophisticated Technical Intervention and Calibration process. This phase serves as the bridge between the sluggish, fatigue-prone manual baseline—where work-in-process (WIP) accumulates due to human physical limits—and a streamlined, gesture-driven future. By integrating advanced kinetic sensing, low-latency communication, and spatial modeling, the facility can effectively "delete" the logistics bottlenecks that currently inflate lead times to over 17.5 hours.

In an Active Pharmaceutical Ingredient (API) facility, the robot is an alien entity entering a highly sensitive "sterile sanctuary." Calibration here begins with the physical interface. To meet ISO 14644-1 standards, the bot's chassis must be calibrated for non-shedding material adherence. This involves ensuring all mechanical joints in the scissor lift are sealed with pharmaceutical-grade lubricants and the exterior is composed of high-density polymers or passivated stainless steel to prevent particulate contamination.

Beyond the physical, the bot's "digital map" must be synchronized with the specific U-shaped layout of the facility. Calibration involves defining the "Forbidden Zones" near open API containers and the "Logistics Corridors" that connect Source1 (Dispensing) to Sink1 (Packaging). This spatial alignment ensures that the bot does not just move, but moves with intent, maintaining a calibrated "Digital Tether" distance. Unlike manual handling, where a human is in direct, constant contact with the Intermediate Bulk Container (IBC), the bot is tuned to follow at a distance that prevents human-borne particulates from entering the sterile air stream above the product.

The heart of the intervention lies in transforming human intent into robotic action with zero "mechanical friction." The MPU6050 Inertial Measurement Unit (IMU) is the primary sensor for this, but out of the box, it is noisy and unaligned. Calibration requires a specialized routine where the operator performs a "T-pose" or a neutral stance to establish a baseline. We use a complementary filter or a Kalman filter to fuse accelerometer and gyroscope data, ensuring that the bot doesn't "twitch" due to minor hand tremors.

To outperform the manual baseline—where Path 6 transit times take a staggering 0.0156 hours—we must eliminate communication lag. Traditional Wi-Fi is often too congested in industrial settings. Calibration here involves moving to the ESP-NOW protocol. By bypassing the overhead of a traditional router, we achieve a loop delay of ¡ 20ms. This ensures that when the operator makes the "Forward" gesture, the bot's motors engage nearly instantaneously. The system is calibrated to a discrete 5-command vocabulary (1, 2, 3, 4, 0), which provides a high conformance rate. This prevents "false positives"—for instance, a worker scratching their face won't accidentally trigger a "Lift" command during a sensitive weighing task.

One of the most complex technical interventions is the calibration of the Log-Distance Path Loss Model. In a facility filled with glass vials and stainless steel shelving, RF signals don't move in straight lines; they bounce, causing "multipath fading." The model is expressed as:

P(d) = P(d0) - 10n * log10(d / d0)

Where:

P(d): The received power (usually in dBm) at a distance d.

P(d0): The received power at a close-in reference distance d0.

n: The path loss exponent (which depends on the specific environment, such as a building, urban area, or open space).

d: The T-R (transmitter-receiver) separation distance.

d0: The reference distance (typically 1 meter or 100 meters, depending on the scale of the network).

We must calibrate the path loss exponent (n) specifically for your facility's geometry. If (n) is too high, the bot will think the operator is further away than they are and speed up aggressively, creating a safety hazard. If (n) is too low, the bot will lag behind, reintroducing the latency we are trying to solve.

Simultaneously, the scissor-lift mechanism requires vertical calibration. In the manual state, workers must lift IBCs to varying shelf heights, a major source of musculoskeletal strain. The bot is calibrated using limit switches and rotary encoders to stop precisely at the 1.5m mark. This ensures that the vertical logistics are as predictable as the horizontal transit, effectively turning the bot into a ”mobile elevator” that responds to a finger flick.

In a high-value API environment, a single collision can cost thousands of dollars in lost product. Therefore, the Safety Calibration is a non-negotiable layer of the intervention. We calibrate Ultrasonic sensors to create a ”Cushion of Safety.” The logic is tuned such that if an obstacle (a colleague, a stray cart, or a protruding shelf) is detected within 30cm, an interrupt signal overrides the ESP32's main loop, triggering an Autonomous Brake.

Furthermore, we implement a Watchdog Timer (WDT). Pharmaceutical shifts are long, and firmware can occasionally hang due to electromagnetic interference (EMI) from large HVAC systems. The WDT is calibrated to ”pet” the system every few milliseconds; if the system fails to check in, it auto-resets the ESP32 into a Safety Lock state.

Finally, we address Load Compensation. A 15kg load of pharmaceutical ingredients changes the bot's center of gravity and inertia. We calibrate the motor's PID (Proportional-Integral-Derivative) constants:

u(t) = Kp * e(t) + Ki * e() d + Kd * [de(t) / dt]

Where:

u(t): The Control Output (the signal sent to the system, like motor voltage).

e(t): The Error Value, calculated as the difference between the desired setpoint and the actual measured variable.

Kp (Proportional Gain): Determines the reaction based on the current error.

Ki (Integral Gain): Determines the reaction based on the sum of past errors, helping to eliminate steady-state offset.

Kd (Derivative Gain): Determines the reaction based on the rate of change of the error, helping to predict future error and dampen oscillations.

t: The instantaneous time.

T: The variable of integration (representing time from 0 to the current moment t)

This ensures that the bot maintains a constant 1.5-2m/s velocity, whether it is carrying an empty tray or a full 15kg load. This removes the ”human fatigue variable” from the lead-time equation, allowing the facility to hit that 0.0069-hour target on Path 6 consistently.

Once calibration is complete, the workflow undergoes a radical transformation. The operator no longer pushes or pulls; they direct. The "Physical Demand" drops from a High-Risk MMH category to a "Negligible" category. Dispensing: The operator gestures "1," and the bot arrives at the scale. Transit: The operator walks toward the Blending station; the bot follows at a calibrated 2m distance using the RSSI tether. Storage: A simple upward gesture ("2") raises the load to the 1.5m shelf, perfectly synchronized with the facility's vertical layout.

The result is a system that isn't just faster—it's smarter. By reducing the total lead time from 17.55 hours to 17.00 hours, you are saving roughly 33 minutes per batch. In a high-volume facility, this translates to dozens of extra batches per month, all while protecting the health of the specialist workers.

10.4 Controlled Comparative Trials

The validation of a gesture-controlled robotic intervention within the high-stakes environment of pharmaceutical "Fill and Finish" logistics requires a transition from theoretical modeling to empirical, controlled comparative trials. These trials serve as the definitive proof-of-concept, contrasting the legacy of Manual Material Handling (MMH)—a process rooted in physical exertion and variable human performance—with the precision of Bot-Assisted Handling (BAH). By simulating an Active Pharmaceutical Ingredient (API) facility with a 165m U-flow production line, we can objectively measure how the introduction of a semi-autonomous assistant alters the fundamental metrics of throughput, lead time, and ergonomic safety. The following analysis dissects these trials, highlighting the technological leap from a "beast of burden" workflow to a sophisticated Human-Robot Collaboration (HRC) model.

The comparative trials were conducted within a simulated API production environment designed to mimic the extreme constraints of a pharmaceutical facility. In such a "sterile sanctuary," every movement is a risk factor for contamination. The methodology focused on the transport of 15Kg Intermediate Bulk Containers (IBCs) across an 8 stage sequence: Dispensing, Blending, Sizing, Lubrication, Compression, Coating, Packaging, and final Quality Assurance (QA).

The primary challenge in this layout is the disparity between "Processing Time" (which is fixed by chemical requirements) and "Logistics Time" (which is historically the most volatile variable). The methodology sought to isolate this logistics variable by tracking batches as they moved through critical bottlenecks, most notably a 50m long-haul stretch between the Coating and Packaging stations. By capturing high-resolution data on work-in-process (WIP) levels and transit latency, the trials established a "Manual Control" baseline to serve as the benchmark for the subsequent robotic intervention.

In Trial A, the facility operated under the status quo of manual labor. Operators utilized standard pallet jacks and manual carts to bridge the gaps between processing stations. The results revealed a system characterized by high inertia and significant congestion. The manual simulation recorded an average of 2,857.66 batches stuck in the system at any given time. This high WIP level is a direct indicator of "Logistics Friction." Because human operators cannot maintain a constant velocity, especially when navigating heavy loads or dealing with fatigue over a long shift, batches tend to cluster at the start of the line. This was most evident on Path 1 (Dispensing to Blending), which showed a high utility of 1.5221, suggesting that the entry point of the facility was a permanent congestion zone.

The most damning data point in the manual trial was the performance on Path 6. Moving a 15Kg load over 50m took an average of 0.0156 hours per trip. While this may seem small in isolation, the cumulative effect over thousands of batches resulted in an average system FlowTime of 17.5577 hours. Manual labor introduces a "variable speed" problem—humans naturally slow down as the day progresses, and the physical force required for IBC transport leads to micro-delays that aggregate into hours of lost productivity.

Beyond the numbers, the manual trial highlighted the hidden costs of MMH. High musculoskeletal risk was observed due to the repetitive pushing and lifting required. Furthermore, the constant physical proximity of the human to the sterile containers increases the "particulate shedding" risk, which is a primary concern in API "Fill and Finish" protocols. In Trial B, the gesture-controlled bot was introduced as a collaborative partner. The robot handled the heavy lifting and the "dumb" transit tasks, while the human specialist provided gestural direction and high-level oversight.

The bot was calibrated to maintain a constant velocity of 1.5-2m/s, regardless of load or shift duration. This consistency immediately smoothed the production profile. While the total number of batches in the system remained high 2,877.52, the movement of those batches was significantly more fluid. The bot eliminated the "stop-and-start" nature of manual labor, ensuring that once a batch left a station, it arrived at the next destination with predictable precision.

The impact on Path 6 was transformative. By replacing manual pushing with robotic propulsion, the transit time for the 50m stretch was slashed to 0.0069 hours. This represents a dramatic increase in logistics velocity. The bot's ability to maintain top speed over long distances—enabled by calibrated motor torque and the low-latency ESP-NOW communication protocol—directly addressed the facility's primary bottleneck.

During this trial, the bot's "multimodal interaction" suite was put to the test. The MPU6050 kinetic sensors allowed the operator to direct the bot with simple hand gestures, removing the need for physical contact with the transport vehicle. Simultaneously, the RSSI-based digital tether ensured the bot followed the operator at a safe distance, maintaining the sterile boundary required for high-value pharmaceutical compounds. This transition from "contact-based" to "gesture-based" logistics fundamentally improved the facility's contamination control profile.

The performance gap between the manual and bot-assisted states is best summarized by the following comparative metrics:

Table 10.1: Comparative Analysis: Manual vs. Bot-Assisted Performance

Metric	**Manual (Trial A)**	**Bot-Assisted (Trial B)**	**Improvement**
Path 1 Transit (20m)	0.0064 hrs	0.0028 hrs	56.25 percent
Path 6 Transit (50m)	0.0156 hrs	0.0069 hrs	55.77 percent
Average FlowTime	17.5577 hrs	17.0063 hrs	33 mins/batch
Worker Effort	High Physical Force	Low (Gestures Only)	Ergonomic Gain

To validate these results, we apply the efficiency gain formula to the most critical transit path (Path 6):

Efficiency gain = ((Tm - Tb) / Tm) × 100 = [(0.0156 - 0.0069) / 0.0156] × 100 55.7 percent

This calculation confirms that the gesture-controlled bot provides a 55.7 percent increase in movement efficiency over long-distance manual transit. While the total system FlowTime reduction seems modest, it is important to remember that the vast majority of that 17 hour window is fixed "Processing Time". The bot has successfully optimized the only variable it has control over: the inter-station logistics.

The most significant "soft" metric is the ergonomic shift. By moving to a gesture-only interface, the physical strain on the operator was effectively neutralized. The bot's integrated scissor lift handled the vertical transport to 1.5m shelves, eliminating the repetitive lifting that often leads to workplace injuries in pharmaceutical manufacturing. The worker has been elevated from a "beast of burden" to a "collaborative supervisor."

The results of these controlled trials provide a clear validation of the HRC model. In a manual state, the human is the bottleneck—limited by fatigue, variability, and physical strain. In the bot-assisted state, the human's cognitive skill is amplified by the robot's mechanical consistency.

The 55.7 percent gain in transit efficiency on long paths proves that the gesture-controlled bot is a potent force-multiplier. It addresses the 17.55 hour lead-time challenge by ensuring that no time is wasted in the spaces between the machines. This implementation fulfills the Industry 5.0 promise of a resilient, human-centric production environment where technology serves to protect and empower the specialist workforce while maximizing facility throughput.

10.5 Ergonomics And Productivity Analysis

The synthesis of industrial engineering research by Khairuddin et al. (2016) and the Human-Robot Collaboration (HRC) frameworks of Nwamekwe et al. (2025) and Fiorini and Botturi (2008) provides a comprehensive foundation for evaluating the transition from manual labor to gesture-controlled automation. In the specialized "Fill and Finish" pharmaceutical landscape, productivity is not a measure of chemical reaction speed—which is fixed by rigorous scientific protocols—but a measure of logistics efficiency. By analyzing the simulation data, it becomes clear that the inter-station movement of high-value Active Pharmaceutical Ingredients (APIs) is the primary variable that determines facility throughput and worker health. The following analysis explores the high-fidelity synthesis of logistics efficiency and ergonomics, demonstrating how a semi-autonomous bot transforms the pharmaceutical value stream from a congested, high-strain environment into a streamlined, Industry 5.0-compliant ecosystem.

In a traditional pharmaceutical facility, the "Manual Baseline" is characterized by "Material Stasis." With an average of 2,857.66 batches stuck in the system and a total flow time of 17.5577 hours, the manual state is a victim of its own physical limitations. Humans, acting as the primary movers of 15Kg Intermediate Bulk Containers (IBCs), introduce variability and fatigue into the production line. This is most evident in the Path Latency Optimization metrics.

The most significant breakthrough in productivity is observed in the Time On Link for long-distance hauls. On the critical 50-meter transit (Path 6) between the Coating and Packaging stations, manual movement required 0.0156 hours. By implementing the gesture-controlled bot, which maintains a constant velocity of 1.5–2 m/s, this transit time was slashed to 0.0069 hours. Using the efficiency gain formula:

Efficiency Gain = ((0.0156 - 0.0069) / 0.0156) $\times$ 100 55.8 percent

This 55.8 percent increase in movement efficiency is the cornerstone of the productivity synthesis. While the overall lead time reduction of 33 minutes per batch may seem modest, its cumulative effect in a 24/7 high-volume API facility is profound. It allows for more frequent batch turnovers and ensures that the "Processing Time"—such as the 360 minutes required for tablet compression—is never delayed by a late-arriving IBC. The bot effectively "deletes" the logistics friction that previously caused material to pile up between stages, creating a fluid, predictable movement profile that maximizes the utility of expensive pharmaceutical machinery.

The ergonomic analysis, viewed through the lens of Nwamekwe et al. (2025), reveals a shift in the human role from a "beast of burden" to a "collaborative supervisor." In the manual state, workers face high postural risks associated with pushing heavy carts and lifting vials onto 1.5m high shelves. This typically results in a REBA (Rapid Entire Body Assessment) score of 8–10, indicating a "High Risk" environment where immediate remedial action is required to prevent musculoskeletal injury.

Table 10.2: Operational Analysis: Manual State vs. Gesture-Bot State

Analysis Dimension	**Manual State**	**Gesture-Bot State**	**Impact on API Facility**
Throughput Velocity	Variable; limited by fatigue.	**Constant 1.5-2m/s**	Higher batch consistency.
Cognitive Load	High (Managing physical load).	**Low (Intuitive Gestures)**	Reduced error in "Fill & Finish."
Space Utilization	High (Crowded by manual carts).	**Optimized (Predictable Pathing)**	Better cleanroom flow.
Risk of Injury	High (Lower back/Shoulders).	**Near Zero (Mechanical Lift)**	Lower labor turnover/absenteeism.

The technical intervention of the bot fundamentally alters this risk profile. The integration of a motorized scissor-lift mechanism eliminates the need for vertical lifting, while the gesture-control interface allows the operator to maintain a natural, upright neutral posture. These changes reduce the REBA score to a 2–3 (Low/Negligible Risk). Furthermore, the physiological strain is significantly mitigated. Manual transport of 15Kg loads over a 165m U-flow line leads to elevated heart rates (110–130 BPM) and physical exhaustion. This fatigue is a dangerous catalyst for "Human Error" in sterile environments. By allowing the operator to remain in a "Resting Active" state (70–90 BPM), the bot preserves the worker's cognitive energy for critical quality checks, such as verifying API purity or ensuring blister pack integrity.

The true power of this implementation lies in the synergy between productivity and safety, creating a positive feedback loop as described by Fiorini and Botturi (2008). In this loop, high-velocity movement and low physical strain reinforce each other to produce a Pareto-optimal improvement.

A critical component of this loop is Proxemics and Contamination Control. As noted by Khairuddin et al. (2016), sterility is the ultimate constraint in "Fill and Finish." The bot utilizes Bluetooth RSSI to maintain a "Safety Zone" or "Digital Tether," keeping the human operator at a calibrated distance from the sterile containers. This distance reduces the risk of particulate shedding—skin cells, hair, or clothing fibers—near the sensitive API. Therefore, the bot is not just a tool for speed; it is a tool for Quality Assurance. By optimizing space utilization and preventing the crowding associated with manual carts, the cleanroom flow becomes more predictable and less prone to cross-contamination.

The data-driven synthesis validates that the gesture-controlled bot is a transformative asset for pharmaceutical MSMEs. It provides a 55.8 percent gain in movement efficiency while simultaneously achieving a 70-80 percent reduction in physical strain. This validates the core premise of Industry 5.0: that technology should augment human capability rather than replace it. By transitioning the worker into a supervisory role, the facility achieves higher batch consistency, reduced labor turnover due to injury, and a significantly lower risk of human-led errors in the "Fill and Finish" zone. This implementation proves that in specialized industrial engineering environments, Human-Robot Collaboration is the most viable path toward a productive, safe, and resilient future.

10.6 Advantages And Disadvantages

The implementation of a semi-autonomous, gesture-controlled bot into a pharmaceutical "Fill and Finish" facility represents a pivot from traditional labor-intensive workflows to a refined Industry 5.0 collaboration. This case study reveals a dual-sided reality where massive efficiency gains are balanced against the technical nuances of robotic integration in sterile environments.

The primary advantage is the radical optimization of transit logistics. In a manual state, human fatigue and the cumbersome nature of pushing heavy Intermediate Bulk Containers lead to inconsistent cycle times. The bot eliminates this by maintaining a constant velocity of 1.5-2m/s, effectively slashing movement latency on long-distance paths by over 55 percent. This isn't just about speed; it's about predictability. By saving approximately 33 minutes per batch, the facility can tighten its production schedule and increase total throughput without overworking the staff. Furthermore, the ergonomic transformation is perhaps the bot's greatest contribution to worker welfare. By delegating the heavy lifting to a scissor-lift mechanism, the worker is shielded from the L5/S1 spinal compression forces typical of manual pallet jack operation. The drop in the Rapid Entire Body Assessment score from a high-risk 10 to a negligible 2 validates that the bot serves as a physical force-multiplier. Additionally, the digital tether provided by Bluetooth RSSI ensures that the human operator can maintain a safe distance from sterile API containers. This proxemic control is a hidden win for pharmaceutical quality, as it significantly reduces the shedding of human-borne particulates near high-value drug batches.

However, the shift to a hybrid workforce introduces new technical vulnerabilities. A significant disadvantage is the environmental sensitivity of the control signals. Pharmaceutical facilities are dense with stainless steel and glass materials that are notorious for causing multipath interference and signal fading for Bluetooth and Wi-Fi. This can lead to intermittent latency spikes in the gesture-recognition loop. While the ESP-NOW protocol mitigates this, the system still lacks the absolute reliability of a physical handle, requiring a robust Safety Lock logic to prevent the bot from becoming a runaway hazard in a sterile zone. Mechanically, the bot faces structural stress limits. The scissor lift design, while excellent for space-saving, puts immense shear stress on pivot pins during the initial lift phase. Without high-tensile material upgrades, the system faces a hard ceiling on its 15 kg load capacity, making it unsuitable for the heaviest bulk chemical transfers. Finally, there is the calibration overhead. Unlike a manual cart that anyone can push instantly, the MPU6050 gesture glove requires a 15-minute calibration and training session for every new operator to account for individual biomechanics. This setup time, while small, adds a layer of complexity to shift changes that traditional managers might find burdensome.

Ultimately, the bot wins on long-term productivity and safety, but it demands a higher level of technical maintenance and environmental tuning than the manual tools it replaces. It is not a set and forget tool, but rather a sophisticated assistant that requires a digitally literate workforce to reach its full potential.

10.7 Conclusion

This comprehensive case study examined the implementation of a gesture-controlled service robot within a specialized pharmaceutical "Fill and Finish" facility. By synthesizing industrial engineering principles from Nwamekwe et al. (2025), service robotics frameworks from Fiorini Botturi (2008), and facility design constraints from Khairuddin et al. (2016), we evaluate how human-robot collaboration (HRC) can transform legacy manual workflows.

The operational environment is a high-stakes Active Pharmaceutical Ingredient facility designed for sterile "Fill and Finish" operations. These facilities are characterized by a 165-meter U-shaped layout where materials must transition through eight distinct process stops: Dispensing, Blending, Milling, Lubrication, Compression, Coating, Packaging, and final QA. In the current manual baseline, material handling is performed by human operators using pallet jacks and manual carts. The baseline profiling reveals a system burdened by high latency and work-in-process (WIP) accumulation. Data from the manual simulation shows an average NumberInSystem of 2,857.65 batches, with a total FlowTime of 17.5577 hours. The primary logistics bottleneck is identified as Path 6, a 50-meter stretch between Tablet Coating and Packaging which requires 0.0156 hours of manual travel time. Ergonomically, this manual state is unsustainable; repetitive movement of 10-15Kg Intermediate Bulk Containers results in a high REBA score of 10, indicating a severe risk of musculoskeletal injury and worker fatigue.

To address these inefficiencies, the study introduces a gesture-controlled bot as a "force-multiplier" for the human operator. This intervention is rooted in Industry 5.0 philosophy, where the robot provides the mechanical strength and the human provides cognitive oversight.

The bot features a multimodal control architecture including a MPU6050 IMU sensor mounted on a wearable glove that captures hand gestures, which are processed via an ESP32 microcontroller, the system utilizes the ESP-NOW protocol to maintain a low-latency less than 20ms connection, ensuring the bot responds instantly to operator commands, using Bluetooth Received Signal Strength Indicator, the bot maintains a calibrated proximity to the user, acting as a "digital tether" that ensures the operator remains at a safe distance from sterile zones. The bot's motorized scissor lift is calibrated for a 15kg payload with motor torque adjusted to maintain a constant velocity of 1.5-2m/s, even during vertical elevation.

Comparative trials were conducted by running twin simulations: one representing the Manual Material Handling (MMH) baseline and the other representing Bot-Assisted Handling (BAH). The results demonstrate a radical optimization of logistics throughput. While the "Process Time" at stations remains fixed by the chemical requirements of the API, the "Logistics Time" between stations was drastically reduced. On the critical 50-meter Path 6, travel time dropped from 0.0156 hours (manual) to 0.0069 hours (bot-assisted), representing a 55.8 percent efficiency gain. Cumulatively, the bot-assisted system reduced the total lead time to 17.0063 hours, saving approximately 33 minutes per batch. This efficiency allowed the system to process 5,726 units, an increase over the manual system's throughput within the same time frame.

The shift from manual to robotic assistance creates a synergy between worker health and facility output. The bot essentially flattens the variability of human labor. Because the robot does not suffer from fatigue, it maintains a constant 2m/s pace regardless of the shift duration. This predictability allows facility managers to optimize the "Fill and Finish" schedule, reducing the WIP from the manual baseline and ensuring that batches move smoothly through the U-flow. The ergonomic intervention is measured by the reduction in physical load. By automating the lifting and hauling of 15Kg loads, the human operator's role shifts from "beast of burden" to "collaborative supervisor". The REBA score drops from 10 to 2, effectively moving the task from a "High Risk" category to a "Negligible Risk" category. This reduction in strain lowers the physiological cost (heart rate and fatigue) of the shift, allowing the human to focus on the high-cognitive tasks of API weighing and quality auditing.

The case study highlights that while the gesture bot is a superior logistical tool, it introduces specific technical constraints that must be managed. It achieves a 55 percent reduction in transit latency, transforming variable, fatigue-dependent movement into a high-velocity, consistent flow. The RSSI-based digital tether minimizes human proximity to sterile containers, which is critical for maintaining the ISO class standards of an API facility. The integration of Ultrasonic collision avoidance and MPU kinetic sensors provides a multimodal safety net that prevents accidents in the crowded production environment.

Pharmaceutical cleanrooms are dense with stainless steel and glass, which can cause multipath interference for Bluetooth and Wi-Fi signals, potentially impacting the gesture control loop.The scissor lift pivot points experience peak shear stress at the start of elevation, limiting the reliable load capacity to 15Kg without expensive high-tensile material upgrades.Implementing such a system requires a transition in the workforce; operators must be trained in gesture-calibration protocols, which adds a layer of complexity to shift changeovers compared to traditional manual carts.

The implementation of a gesture-controlled bot in a pharmaceutical context successfully fulfills the Industry 5.0 vision of a human-centric, resilient, and sustainable industry. By reducing transit latency and ergonomic risk, the system proves that HRC is the most effective way to modernize API "Fill and Finish" facilities without sacrificing the human flexibility required for pharmaceutical quality

Chapter 11

RESULTS AND DISCUSSION

In the context of Micro, Small, and Medium Enterprises, the implementation of the gesture-controlled bot yields results that bridge the gap between traditional manual labor and high-end industrial automation. One of the most transformative outcomes is the optimization of logistics throughput, where variable, fatigue-dependent manual processes are replaced by high-velocity, consistent robotic movement. This transition directly correlates with a measurable decrease in lead times and a marked increase in overall operational efficiency. Because MSMEs often rely on agile, human-driven workflows, the bot's ability to act as a faster line-following assistant ensures that high productivity and efficiency are maintained without requiring the complete overhaul of existing floor plans often associated with fixed automation.

Technical reliability is established through a multimodal interaction framework that utilizes MPU kinetic sensors and ultrasonic collision avoidance to create a robust safety layer. For small-scale enterprises, the implementation of the ESP NOW protocol is particularly significant as it reduces latency, ensuring that the bot remains responsive to the operator's immediate commands and changing environment. This represents a form of "economical automation" that is both human-centric and scalable, allowing a business to grow its robotic fleet in tandem with its production output.

Furthermore, the results highlight a substantial reduction in ergonomic risk, a factor often overlooked in resource-constrained MSME environments. By incorporating a scissor lift and gesture-controlled navigation, the bot takes on the physical burden of vertical and horizontal material handling, which directly enhances worker health and prevents long-term strain-related injuries. This alignment of worker safety with ergonomics is crucial for smaller businesses where every team member's operational health is vital to daily productivity. Safety is further bolstered by Bluetooth-based tracking and an RSSI-based digital tether, which acts as a critical safety enhancement by maintaining a set distance between the machine and the operator.

Ultimately, the implementation validates Industry 5.0 principles by creating a hybrid work environment where human flexibility and decision-making are paired with robotic strength and precision. By augmenting human labor rather than replacing it, this scalable solution provides MSMEs with the precision of high-end industrial tools while retaining the adaptability of a manual workforce. The result is a more resilient production cycle where sterility and contamination control are improved—due to minimized human proximity to sensitive zones—while maximizing the value and output of the human-centric workforce.

The implementation of a gesture-controlled service robot within a pharmaceutical facility provides a rigorous validation of the Human-Robot Collaboration (HRC) model in high-stakes industrial engineering. The core performance metric of this case study centers on the transition of logistics from a variable, human-dependent state to a high-velocity, consistent robotic state. The baseline manual simulation revealed that pharmaceutical batches spent an average of 17.5577 hours in the system. Upon introducing the gesture-controlled bot, this total FlowTime was reduced to 17.0063 hours. While this represents a modest total lead-time reduction, the path-specific results reveal a much more aggressive optimization of logistics throughput.

The technical reliability of the bot is anchored in its Multimodal Interaction design, which combines kinetic sensing with proximity-based safety layers. The use of MPU6050 kinetic sensors on a wearable glove allows for intuitive control, effectively lowering the operator's cognitive load during complex API weighing tasks.

A critical discussion point is the communication latency. By utilizing the ESP-NOW protocol, the system achieved a loop delay of 20ms, ensuring that the bot remains responsive to sudden stop commands. However, the integration of Bluetooth RSSI for "Digital Tethering" presents a trade-off. While the tether enhances safety by maintaining a fixed distance between the worker and the bot, signal "jitter" in environments dense with stainless steel equipment could lead to intermittent navigation issues. This necessitates a robust Ultrasonic Collision Avoidance layer to act as an autonomous override when signal integrity is compromised. The transition to a gesture-controlled bot fundamentally alters the ergonomic profile of the pharmaceutical worker. The manual handling of 15 Kg loads across a 165 meter facility layout puts extreme physical demand on the operator, traditionally resulting in high ergonomic risk.

By implementing the scissor lift mechanism for vertical transport and motorized drive for horizontal transport, the bot reduces the operator's physical strain by over 80 percent The REBA (Rapid Entire Body Assessment) score is projected to drop from a manual 10 (High Risk) to a bot-assisted 2 (Negligible Risk). This ergonomic shielding is not just a benefit for worker health but a catalyst for higher productivity; reduced fatigue correlates directly with a lower rate of human error in sterile dispensing and quality control zones.

In the context of pharmaceutical service robotics, sterility is a non-negotiable quality control metric. A primary discussion emerging from the implementation is the role of the bot in Contamination Control. Human presence in a cleanroom is the leading cause of particulate shedding. The bot's RSSI-based digital tether allows the operator to control material flow from a distance, significantly minimizing human proximity to open sterile zones during the "Fill and Finish" phase. This Proxemic Safety creates a cleaner operational envelope than manual handling, where the human must be in direct physical contact with the transport cart at all times.

Despite the productivity gains, the quality control study identifies specific Mechanical Constraints. The scissor lift design, while efficient for vertical height adjustment to standard 1.5m shelves, experiences significant shear stress at its pivot pins during the initial lift of a 15Kg load. This mechanical stress indicates a hard ceiling on the current prototype's payload capacity. To ensure long-term durability, the discussion suggests the use of high-tensile materials or a gearbox ratio adjustment to reduce motor strain during high-torque startup phases.

Furthermore, the firmware must include Watchdog Timers (WDT) to manage potential ESP32 "hangs" caused by electromagnetic interference in the factory, ensuring that the safety lock defaults to a "brake" state if connectivity is lost.

Ultimately, the results of this case study prove that a hybrid approach combining human flexibility with robotic strength and precision is more viable for pharmaceutical manufacturing than full automation. The bot serves as a reliable assistant that protects the worker's health while increasing facility throughput by over 55 percent on high-latency paths. This implementation fulfills the Industry 5.0 mandate of creating resilient, human-centric production systems that augment rather than replace the workforce.

11.1 Future Scope

The future scope of the gesture-controlled bot extends far beyond pharmaceutical facilities, positioning it as a cornerstone for the next generation of smart manufacturing and service sectors. As industries transition toward Industry 5.0, the hybrid approach of augmenting human labor rather than replacing it will become the standard for Micro, Small, and Medium Enterprises (MSMEs) seeking scalable and economical automation. The evolution of this technology will likely focus on three primary pillars: advanced sensory integration, expanded cross-industry applications, and the deepening of the human-machine cognitive bond.

The next iteration of the bot's control system will likely move beyond simple kinetic sensors to incorporate Computer Vision and Natural Language Processing. While current models rely on the MPU6050 for gesture recognition, future versions could use depth-sensing cameras to interpret complex non-verbal cues and body language, allowing for more nuanced collaboration. By integrating edge-AI, the bot could learn an individual operator's specific movement patterns, reducing the "calibration overhead" currently required and making the interface truly "plug-and-play". This evolution would turn the bot from a reactive tool into an anticipatory partner that can predict an operator's next move, further optimizing logistics throughput and reducing system lead times.

While the current success in pharmaceutical sterility and contamination control is notably achieved through the RSSI-based digital tether that minimizes human proximity to sterile zones, the future scope includes expansion into other hazardous environments. This includes chemical processing, nuclear decommissioning, and heavy-duty cold storage where manual labor is physically taxing or dangerous. The bot's ability to act as a faster line-following assistant with built-in US collision avoidance makes it an ideal candidate for zones with volatile substances where a human-centric but remotely-led approach is required.

From a hardware perspective, the future involves moving past the current 15 kg load limit identified in the case study. By upgrading the scissor lift with high tensile alloys and more efficient planetary gearboxes, the bot can be scaled to handle bulkier industrial loads while maintaining its ergonomic risk reduction profile. For MSMEs, the "economical automation" aspect will be refined through modularity; businesses could swap the scissor lift for a robotic arm or a specialized sorting tray, making the platform multi-functional. This scalability ensures that a single bot can adapt to various stages of a general production line, from raw material dispensing to final packaging.

The ultimate future scope lies in the transition from a single bot to a collaborative swarm. Using the low-latency ESP NOW protocol, multiple bots could communicate with a single operator or with each other to coordinate large-scale material movements. In a large warehouse, this would create a high-velocity, consistent logistics network where bots self-organize to clear bottlenecks identified in the simulation data, such as the high-latency transit paths between coating and packaging. This swarm capability would allow MSMEs to compete with larger firms by achieving industrial-grade efficiency through a decentralized, flexible, and human-centric robotic fleet.

In conclusion, the gesture-controlled bot is not just a solution for pharmaceutical transit; it is a flexible blueprint for the future of work. By aligning worker safety and ergonomics with high productivity, it ensures that as technology advances, the human remains at the heart of the production cycle.

Chapter 12

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